

A Simple Algorithm Breaking the $1/3$ Barrier for Approximate Nash Equilibria in Bimatrix Games

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Abstract

The PPAD-completeness of computing Nash equilibria in bimatrix games has driven extensive research into polynomial-time algorithms for ϵ -approximate Nash equilibria. Determining the strongest approximation guarantee achievable in polynomial time is a fundamental question in the complexity of approximate Nash equilibria. Despite sustained efforts, breaking the $1/3$ barrier has long been regarded as a difficult open problem. In this paper, we give an affirmative answer by proposing a simple, symmetric deterministic polynomial-time algorithm that has no tunable hyperparameters and achieves an approximation guarantee of approximately $0.30954 + \delta$. The main idea is to enrich the optimization framework of Tsaknakis and Spirakis with building blocks drawn from our work on strategy mixing and automated approximation analysis, yielding a new algorithmic structure that provides a conceptual explanation for the improved guarantee.

1 Introduction

Nash equilibrium (NE), introduced by Nash [1], is a foundational solution concept in game theory. It describes a strategy profile in which no player can increase their payoff by changing their strategy unilaterally. Nash’s existence theorem guarantees an equilibrium in mixed strategies for every finite normal-form game, but computing one has remained a persistent algorithmic challenge. A major breakthrough came with the complexity results of Daskalakis, Goldberg, and Papadimitriou [2] and Chen, Deng, and Teng [3], which established PPAD-completeness even for two-player games, providing strong evidence against polynomial-time algorithms for computing exact Nash equilibria.

Approximate Nash equilibria offer a way to relax this computational problem. For payoffs normalized to $[0, 1]$, an ϵ -approximate Nash equilibrium (ϵ -NE) allows each player to gain at most ϵ by a unilateral deviation. Thus an exact NE is the special case $\epsilon = 0$. Lipton, Markakis, and Mehta [4] established quasi-polynomial-time computation for every fixed $\epsilon > 0$, with improved small-support bounds subsequently given by Babichenko, Barman, and Peretz [5]. Under the exponential-time hypothesis for PPAD, Rubinstein [6] showed that quasi-polynomial time is necessary for sufficiently small constant errors. Thus, determining the smallest error achievable in polynomial time remains a fundamental open question in algorithmic game theory.

Most algorithmic work on this question has focused on two-player games, or bimatrix games [7]. Early polynomial-time algorithms achieved guarantees of 0.75 by Kontogiannis, Panagopoulou, and Spirakis [8] and 0.5 by Daskalakis, Mehta, and Papadimitriou [9]. The guarantee was further improved to $0.382 + \delta$ by Daskalakis, Mehta, and Papadimitriou [10] and 0.364 by Bosse, Byrka,

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and Markakis [11], before Tsaknakis and Spirakis [12] achieved $0.3393 + \delta$ through an optimization approach. This guarantee remained the state of the art for fifteen years, until Deligkas, Fasoulakis, and Markakis [13] reached $1/3 + \delta$ by refining the Tsaknakis–Spirakis optimization framework. Here, $\delta > 0$ denotes an arbitrarily small accuracy parameter.

Despite these advances, going below $1/3$ has long been viewed as a difficult methodological challenge [7]. Tsaknakis and Spirakis suggested that a radically different, possibly randomized, approach might be needed [12]. Moreover, the guarantees of the Tsaknakis–Spirakis and Deligkas–Fasoulakis–Markakis algorithms are tight [14], so sharper analysis alone cannot improve them. Achieving a polynomial-time approximation guarantee strictly below $1/3$ has remained a central open problem [7].

In this paper, we resolve this open problem with a deterministic polynomial-time algorithm that improves the approximation guarantee from $1/3 + \delta$ to approximately $0.30954 + \delta$.

Theorem 1. *There is a deterministic algorithm that, given a bimatrix game with rational payoffs in $[0, 1]$ and a rational $\delta > 0$, computes an $(r_* + \delta)$ -approximate Nash equilibrium, where $r_* \approx 0.30954$ is an absolute constant. The running time is polynomial in the input encoding length and $1/\delta$.*

Our methodology follows LegoNE [16, 15], a framework for systematically understanding the construction and analysis of approximate NE algorithms. LegoNE organizes algorithmic and proof insights into composable building blocks and combines domain knowledge, large language models (LLMs), and computer-aided analysis of worst-case guarantees. It improved the three-player guarantee from $0.6 + \delta$ to $0.5 + \delta$ [16, 15], using a construction beyond the previously established recursive extension paradigm [11, 17].

Following this perspective, we used OpenAI Codex to explore new algorithms for bimatrix games. Codex proposed and refined candidate algorithms, formulated their building-block properties, and developed the proof of the approximation guarantee. In doing so, Codex used `SimpleMixing` [18] and `OptimalMixing` [19] as building blocks for algorithm design. These operations were originally formalized to understand mixing and automate approximation analysis. Combining them with the framework of Tsaknakis and Spirakis led to a fixed, symmetric construction. The analysis extends the LegoNE methodology to incorporate the properties of these optimization operations.

Historical note. This research was initiated after an email from Davit Gondauri and subsequent correspondence concerning his preprint [20], which reports an approximation guarantee of $1/3 - 7.5 \times 10^{-45}$. At the level of algorithmic construction, it combines mixing ideas from Bosse, Byrka, and Markakis [11] with the optimization framework of Deligkas, Fasoulakis, and Markakis [13]. The correspondence and our examination of the manuscript with Codex motivated us to seek a conceptually new construction yielding a substantial improvement, beyond the very small reported separation from $1/3$. Our algorithm and approximation analysis were developed independently.

2 Preliminaries

We first specify the game and the notation used throughout the paper. The objective is the maximum regret of the two players: minimizing this quantity gives an approximate Nash equilibrium. We also record the best-response and convexity properties used in the construction and its analysis.

For a positive integer n , write $[n] = \{1, \dots, n\}$.

A bimatrix game is specified by two *payoff matrices* $R, C \in [0, 1]^{n_1 \times n_2}$, with n_1 rows and n_2 columns. Player 1 is the *row player* and player 2 is the *column player*. A *pure strategy* chooses

one row or column; if the players choose $k \in [n_1]$ and $\ell \in [n_2]$, their payoffs are $R_{k\ell}$ and $C_{k\ell}$. Thus player $i \in [2]$ has n_i pure strategies. A mixed strategy is a probability distribution over pure strategies. We write $(x^i)_k$ for the probability assigned to pure strategy k by a mixed strategy x^i . For any positive integer n , the *probability simplex* is

$$\Delta_n = \left\{ z \in \mathbb{R}_{\geq 0}^n : \sum_{k \in [n]} z_k = 1 \right\}.$$

Thus the strategy set of player i is Δ_{n_i} . The *support* of a mixed strategy $z \in \Delta_n$ is

$$\text{supp}(z) = \{k \in [n] : z_k > 0\}.$$

The k th *standard basis vector* $e_k^i \in \mathbb{R}^{n_i}$ represents the pure strategy that chooses k with probability one: $(e_k^i)_k = 1$ and $(e_k^i)_\ell = 0$ for $\ell \neq k$. A strategy profile is $\mathbf{x} = (x^1, x^2)$, where $x^i \in \Delta_{n_i}$. Its *expected payoffs* are

$$u_1(\mathbf{x}) = (x^1)^\top R x^2, \quad u_2(\mathbf{x}) = (x^1)^\top C x^2.$$

We also write $u_i(x^1, x^2)$ for $u_i(\mathbf{x})$. Superscripts on strategies identify the player. Subscripts on named strategies distinguish outputs of different operations or rounds; coordinates are written with the subscript outside parentheses, as in $(x^i)_k$.

For a vector $v \in \mathbb{R}^n$, $\text{suppmax}(v)$ denotes the index set of its maximum entries [14]. For $\delta \geq 0$, we also write $\text{suppmax}_\delta(v)$ for the indices whose entries are within δ of the maximum:

$$\begin{aligned} \text{suppmax}(v) &= \{k \in [n] : v_k = \max_{\ell \in [n]} v_\ell\}, \\ \text{suppmax}_\delta(v) &= \{k \in [n] : v_k \geq \max_{\ell \in [n]} v_\ell - \delta\}. \end{aligned}$$

Thus $\text{suppmax}_0(v) = \text{suppmax}(v)$.

We write $x^{i'}$ for an arbitrary comparison strategy of player i , and $\mathbf{x}' = (x^{1'}, x^{2'})$ for a comparison profile. The prime marks a comparison strategy, while the superscript identifies the player.

Following the notation of LegoNE [16], the *regret* of each player and the *maximum regret* are

$$\begin{aligned} f_1(\mathbf{x}) &= \max_{x^{1'} \in \Delta_{n_1}} u_1(x^{1'}, x^2) - u_1(\mathbf{x}), \\ f_2(\mathbf{x}) &= \max_{x^{2'} \in \Delta_{n_2}} u_2(x^1, x^{2'}) - u_2(\mathbf{x}), \\ f(\mathbf{x}) &= \max\{f_1(\mathbf{x}), f_2(\mathbf{x})\}. \end{aligned} \tag{1}$$

For $\epsilon \geq 0$, the definition of an ϵ -approximate Nash equilibrium (ϵ -NE) can therefore be written as

$$\mathbf{x} \text{ is an } \epsilon\text{-NE} \iff f(\mathbf{x}) \leq \epsilon.$$

In particular, $\mathbf{x}$ is an exact NE if and only if $f_1(\mathbf{x}) = f_2(\mathbf{x}) = 0$. A *best response* attains the corresponding maximum in (1); it can be found by comparing pure-strategy payoffs. The vectors Rx^2 and $C^\top x^1$ give the pure-strategy payoffs of players 1 and 2, respectively. Hence $x^{1'} \in \Delta_{n_1}$ and $x^{2'} \in \Delta_{n_2}$ are best responses to x^2 and x^1 , respectively, if and only if

$$\text{supp}(x^{1'}) \subseteq \text{suppmax}(Rx^2), \quad \text{supp}(x^{2'}) \subseteq \text{suppmax}(C^\top x^1).$$

For a fixed strategy of one player, both regrets, and hence f , are convex functions of the other player's strategy. They need not be jointly convex in the two strategies.

A *convex combination* of strategies is a weighted sum with nonnegative coefficients summing to one. We write $\text{conv } S$ for the *convex hull* of a finite set $S = \{s_j : j \in [m]\}$, that is,

$$\text{conv } S = \left\{ \sum_{j \in [m]} p_j s_j : p \in \Delta_m \right\}.$$

For two strategies of the same player, these are the mixtures $(1 - p)x^i + px^{i'}$ with $p \in [0, 1]$.

3 The LegoNE framework

LegoNE provides a symbolic language for composing approximate Nash equilibrium algorithms and an analyzer for deriving their approximation guarantees [16, 15]. We review these two components, following [16]. The Daskalakis–Mehta–Papadimitriou (DMP) 1/2-NE algorithm [9] illustrates both the construction of a program and its analysis. This viewpoint will also be used to present and analyze our algorithm in Sections 4 and 5.

3.1 Language for algorithm design

The LegoNE language specifies algorithms for games with a fixed number of players by composing high-level building blocks. These blocks encapsulate operations from the literature, including best responses, mixing, linear programming, and stationary-point computation. An algorithm is a sequence of assignments: each call uses previously available variables as inputs and introduces new variables for its outputs. The program ends by returning one strategy for each player. This gives a structured design space in which candidate algorithms are formed by choosing blocks, their order, and their inputs.

We fix the game throughout a construction. The blocks used in this paper have strategy inputs and outputs. Their accuracy parameters are fixed throughout a construction and specified when the blocks are introduced. Their computational guarantees are stated when the blocks are introduced. A block also has a mathematical specification of the properties guaranteed by its outputs. The analyzer uses these specifications to evaluate a composition of blocks.

For the DMP example, **Random1** supplies an initial row strategy, **BestResponse** returns a best response to its input strategy for the other player, and **UniformMixing** returns the equal mixture of its input strategies. In particular, $\text{UniformMixing}(x^i, x^{i'}) = (x^i + x^{i'})/2$ for $x^i, x^{i'} \in \Delta_{n_i}$. Algorithm 1 first constructs a column best response, then a row best response, and finally mixes the two row strategies. All the calls in this example take polynomial time.

Algorithm 1 The DMP 1/2-NE algorithm

Input: A bimatrix game.

- 1: $x^1 = \text{Random1}()$
 - 2: $y^2 = \text{BestResponse}(x^1)$
 - 3: $y^1 = \text{BestResponse}(y^2)$
 - 4: $z^1 = \text{UniformMixing}(x^1, y^1)$
 - 5: **return** (z^1, y^2)
-

Write $z = (z^1, y^2)$ for its output. The algorithm uses only the stated operations; its approximation guarantee is derived from their properties below. More generally, this modular language allows new compositions to be proposed and evaluated using the same library of building blocks.

3.2 Automated approximation analysis

The analysis must establish a bound on the output regret for every game, with payoff matrices of arbitrary size and a continuum of mixed strategies. LegoNE first translates the program into logical properties. It then applies two abstraction principles, *instantiation* and *forgetting*, to turn this proof obligation into a fixed-size constrained optimization problem. We describe the translation before explaining these two principles.

Logic encoding. Under Floyd–Hoare semantics, each line is represented by the properties it guarantees. For example, the row best-response call in Algorithm 1 has the *logic encoding*

$$\phi[y^1 = \text{BestResponse}(y^2)] : \quad \forall x^{1'} \in \Delta_{n_1}, u_1(x^{1'}, y^2) \leq u_1(y^1, y^2). \quad (2)$$

The column best-response call has the analogous encoding. The **UniformMixing** call is encoded by its behavior on payoff and regret terms: for every $x^{2'} \in \Delta_{n_2}$, it guarantees

$$\begin{aligned} U(z^1, x^{2'}) &= \frac{1}{2}U(x^1, x^{2'}) + \frac{1}{2}U(y^1, x^{2'}), \\ f_2(z^1, x^{2'}) &\leq \frac{1}{2}f_2(x^1, x^{2'}) + \frac{1}{2}f_2(y^1, x^{2'}). \end{aligned}$$

Here U ranges over the payoff functions and their linear combinations, as in the LegoNE encoding; the second line is the convexity property of the other player’s regret. Thus the mixing step supplies the properties used in its analysis, in addition to defining a strategy. The initialization imposes no further property on x^1 .

The encoding of the whole program Γ combines these line encodings with *inherent formulas*, denoted by ϕ_0 :

$$\phi[\Gamma] = \phi_0 \wedge \phi[\text{line 1}] \wedge \cdots \wedge \phi[\text{line } k],$$

where k is the number of building-block calls. The inherent formulas include payoff and regret bounds in $[0, 1]$, the definition of regret, and the bounds relating a best-response payoff to the payoff of any strategy and to 1. The proof goal is

$$\phi[\Gamma] \implies f(\mathbf{x}_{\text{out}}) \leq b$$

for all payoff functions and all values of the strategy variables in the program, where $\mathbf{x}_{\text{out}}$ is its returned profile and b is the proposed approximation bound. At this stage the formulas still quantify over strategies and payoff functions, so they are not a finite real-variable optimization problem.

Instantiation. The analyzer substitutes each universally quantified strategy variable with all strategy variables of the same player appearing in the program. For the row best-response property in (2), these are x^1, y^1, z^1 . Besides the trivial substitution $x^{1'} = y^1$, this gives

$$u_1(x^1, y^2) \leq u_1(y^1, y^2), \quad u_1(z^1, y^2) \leq u_1(y^1, y^2).$$

Universally quantified payoff variables are instantiated with the payoff functions appearing in the program.

There is also an instantiation using the maximum-payoff operator. When a quantified strategy occurs only once, in a payoff term, that term can be replaced by its maximum over the strategy. For the same best-response property, this gives

$$\max_{x^{1'} \in \Delta_{n_1}} u_1(x^{1'}, y^2) \leq u_1(y^1, y^2).$$

The inherent maximum-payoff bound supplies the reverse inequality, so the two terms are equal. Applying the corresponding procedures to the other line encodings and to ϕ_0 produces finitely many constraints involving payoff, regret, and maximum-payoff terms. If a block property includes an auxiliary real variable, one value of that variable is shared across all instantiations of the property.

Forgetting. The instantiated constraints still involve unknown payoff functions. Forgetting renames every distinct payoff, regret, and maximum-payoff term as a symbolic real variable, consistently using the same name for every occurrence of the same term. For example,

$$u_1(x^1, y^2) \mapsto v_{1,x^1,y^2}, \quad u_1(y^1, y^2) \mapsto v_{1,y^1,y^2},$$

so their comparison becomes $v_{1,x^1,y^2} \leq v_{1,y^1,y^2}$. Each instantiated constraint is rewritten using these new symbols. Quantification over payoff functions and strategies is replaced by quantification over these real variables. The functional origin of the terms is forgotten; their encoded arithmetic relations remain.

Auxiliary existential real variables in the encodings appear as variables of the constrained program, with their ranges and relations unchanged. Let $\mathbf{v}$ collect these variables and the renamed terms, and write $\phi'[\Gamma]$ for the resulting real constraints. The same renaming turns the output maximum regret into an arithmetic expression $g_\Gamma(\mathbf{v})$. The smallest bound valid for all assignments satisfying the constraints is the optimal value of

$$\begin{aligned} & \underset{\mathbf{v}}{\text{maximize}} && g_\Gamma(\mathbf{v}) \\ & \text{subject to} && \phi'[\Gamma]. \end{aligned}$$

The analyzer compiles the constraints and objective into this problem for an external optimizer. Its size is fixed once the program, encodings, and instantiation tactic are fixed; it does not grow with the number of pure strategies. Every execution satisfies the encoded and instantiated properties, and therefore gives a feasible assignment of the real variables. An upper bound on this optimization problem consequently certifies an approximation guarantee. It is the best bound obtainable from this constraint system, which may also admit assignments that are not realized by a game.

For DMP, the instantiated best-response properties and regret definitions give $f_1(y^1, y^2) = f_2(x^1, y^2) = 0$. The mixing encoding, instantiated at $x^{2'} = y^2$, and the same regret definitions give

$$\begin{aligned} f_1(\mathbf{z}) &= \tfrac{1}{2}f_1(x^1, y^2), \\ f_2(\mathbf{z}) &\leq \tfrac{1}{2}f_2(y^1, y^2). \end{aligned}$$

These are relations among terms in the instantiated system, so they remain arithmetic constraints after forgetting. The inherent regret bounds then imply $g_\Gamma(\mathbf{v}) \leq 1/2$. This illustrates how the analysis derives the guarantee from a program's encodings. For our construction, Section 5 gives the necessary properties and their arithmetic consequences; the complete encoding, exact proof checkers, and replay instructions are available in the accompanying repository¹.

4 The algorithm

We now use the building-block viewpoint of Section 3 to describe our algorithm. We first recall the framework of Tsaknakis and Spirakis, the improvement of Deligkas, Fasoulakis, and Markakis, and the motivation for a further strategy construction. We then define the mixing operations, state their properties and computational guarantees, and present the complete algorithm.

¹<https://github.com/lhydave/new-approx-ne-algo>.

4.1 The Tsaknakis–Spirakis framework

The Tsaknakis–Spirakis algorithm (TS) formulates approximate Nash equilibrium computation as the minimization of f over $\Delta_{n_1} \times \Delta_{n_2}$. Its idea is to use a descent method to find a stationary point and then construct a profile with lower regret [12, 14]. Accordingly, the algorithm has a descent phase and a strategy-construction phase.

The descent phase computes a profile $\mathbf{x}_s = (x_s^1, x_s^2)$ and two dual strategies y_s^1, y_s^2 . The strategy-construction phase uses these strategies to choose the output profile.

We first describe the descent outputs and their properties, following the minimax formulation in [14]. This is a classical technique for approximate Nash equilibrium computation, so we state the formulation and its output properties without repeating the derivation; see [12, 14] for details. We then explain how TS and the algorithm of Deligkas, Fasoulakis, and Markakis (DFM) use these outputs, and why a further improvement requires a different strategy construction.

Let $\mathbf{x}$ be the current profile and $\mathbf{x}'$ a comparison profile, both in $\Delta_{n_1} \times \Delta_{n_2}$. The *Dini directional derivative* of f at $\mathbf{x}$ in the direction $\mathbf{x}' - \mathbf{x}$ is

$$Df(\mathbf{x}, \mathbf{x}') = \lim_{\theta \downarrow 0} \frac{f((1 - \theta)\mathbf{x} + \theta\mathbf{x}') - f(\mathbf{x})}{\theta}.$$

A profile $\mathbf{x}$ is a *stationary point* if $Df(\mathbf{x}, \mathbf{x}') \geq 0$ for every $\mathbf{x}' \in \Delta_{n_1} \times \Delta_{n_2}$. The descent procedure equalizes the two players' regrets without increasing f . In the following description, we therefore assume $f_1(\mathbf{x}) = f_2(\mathbf{x})$.

We use $\mathbf{y} = (y^1, y^2)$ for the best-response strategies in the minimax problem. Let $\mathbf{y} \in \Delta_{n_1} \times \Delta_{n_2}$ satisfy

$$\text{supp}(y^1) \subseteq \text{suppmax}(Rx^2), \quad \text{supp}(y^2) \subseteq \text{suppmax}(C^\top x^1).$$

Thus y^1, y^2 are best responses to x^2, x^1 , respectively. For $\rho \in [0, 1]$, define

$$\begin{aligned} T(\mathbf{x}, \mathbf{x}', \rho, \mathbf{y}) &= \rho[u_1(y^1, x^{2'}) - u_1(x^1, x^{2'}) - u_1(x^{1'}, x^2) + u_1(\mathbf{x})] \\ &\quad + (1 - \rho)[u_2(x^{1'}, y^2) - u_2(x^{1'}, x^2) - u_2(x^1, x^{2'}) + u_2(\mathbf{x})]. \end{aligned} \tag{3}$$

The corresponding *minimax value* is

$$V(\mathbf{x}) = \min_{\mathbf{x}'} \max_{\rho, \mathbf{y}} T(\mathbf{x}, \mathbf{x}', \rho, \mathbf{y}),$$

where $\mathbf{x}' \in \Delta_{n_1} \times \Delta_{n_2}$, $\rho \in [0, 1]$, and $\mathbf{y} \in \Delta_{n_1} \times \Delta_{n_2}$ satisfies the support conditions above. The minimum of $Df(\mathbf{x}, \mathbf{x}')$ over $\mathbf{x}'$ equals $V(\mathbf{x}) - f(\mathbf{x})$. Consequently, $\mathbf{x}$ is stationary if and only if $V(\mathbf{x}) = f(\mathbf{x})$.

The polynomial-time descent procedure uses a rational accuracy parameter $\delta_f > 0$. Its minimax problem replaces the exact best-response sets by

$$\text{supp}(y^1) \subseteq \text{suppmax}_{\delta_f}(Rx^2), \quad \text{supp}(y^2) \subseteq \text{suppmax}_{\delta_f}(C^\top x^1).$$

Let $V_{\delta_f}(\mathbf{x})$ denote the value of this enlarged problem. The procedure terminates at a profile $\mathbf{x}_s$ satisfying

$$f_1(\mathbf{x}_s) = f_2(\mathbf{x}_s) = f(\mathbf{x}_s), \quad V_{\delta_f}(\mathbf{x}_s) \geq f(\mathbf{x}_s) - \delta_f. \tag{4}$$

This minimax value can be computed by a linear program (LP). An optimal dual solution at $\mathbf{x}_s$ determines

$$(\rho, y_s^1, y_s^2) \in [0, 1] \times \Delta_{n_1} \times \Delta_{n_2}.$$

Write $\mathbf{y}_s = (y_s^1, y_s^2)$ and $\sigma = 1 - \rho$. The weights ρ, σ correspond to the row and column players, respectively. In the enlarged LP, a positive row weight normalizes dual weights on pure strategies whose payoffs are within δ_f of the best-response payoff. Averaging these payoffs therefore gives, when $\rho > 0$,

$$\begin{aligned} u_1(y_s^1, x_s^2) &= \sum_{k \in [n_1]} (y_s^1)_k u_1(e_k^1, x_s^2) \\ &\geq \max_{x^{1'} \in \Delta_{n_1}} u_1(x^{1'}, x_s^2) - \delta_f. \end{aligned}$$

The column-player argument is the same. The output properties used in our analysis can thus be stated directly as payoff comparisons:

$$\begin{aligned} \forall x^{1'} \in \Delta_{n_1}, \quad \rho > 0 &\implies u_1(x^{1'}, x_s^2) \leq u_1(y_s^1, x_s^2) + \delta_f, \\ \forall x^{2'} \in \Delta_{n_2}, \quad \sigma > 0 &\implies u_2(x_s^1, x^{2'}) \leq u_2(x_s^1, y_s^2) + \delta_f. \end{aligned} \tag{5}$$

When a weight is zero, its corresponding strategy can be chosen arbitrarily. Moreover, the same dual solution satisfies

$$f(\mathbf{x}_s) - \delta_f \leq T(\mathbf{x}_s, \mathbf{x}', \rho, \mathbf{y}_s) \quad \forall \mathbf{x}' \in \Delta_{n_1} \times \Delta_{n_2}. \tag{6}$$

We regard this procedure as the **StationaryPoint** building block of LegoNE [16], with accuracy input δ_f and four strategy outputs:

$$(x_s^1, x_s^2, y_s^1, y_s^2) = \text{StationaryPoint}(\delta_f).$$

For a fixed bimatrix game, this block has no strategy input; δ_f is its only explicit input. It computes the four strategies in time polynomial in the input encoding length and $1/\delta_f$ [12, 13].

We call $\mathbf{x}_s$ the stationary profile below, meaning the approximate output characterized by (4) and (6). It need not have small regret. Even when y_s^1, y_s^2 are best responses to x_s^2, x_s^1 , respectively, the profile $\mathbf{y}_s$ can have large regret. In its strategy-construction phase, TS compares $\mathbf{x}_s$ with $\mathbf{y}_s$ and profiles in which one player mixes her stationary and dual strategies while the other uses her dual strategy. The mixing coefficients are chosen using payoff differences between the stationary and dual strategies. These comparisons give the guarantee $0.3393 + \delta$ [12], but in the bottleneck cases they do not guarantee $1/3 + \delta$ [13].

DFM uses the same descent procedure and replaces the strategy construction in these bottleneck cases. Its main addition is a best response to an equal mixture of the opponent's stationary and dual strategies, such as $(x_s^2 + y_s^2)/2$ for the row player. DFM uses this additional candidate in further convex combinations and uses the resulting payoff relations to bound both players' regrets. The column-player construction is symmetric. These modifications improve the guarantee to $1/3 + \delta$ [13].

The central role of **StationaryPoint** can be understood through its logical encoding. Following the viewpoint of [15], consider the strategy variables that can be instantiated in a block property. The inequality (6) has *two* universally quantified strategy variables, one for each player:

$$\forall x^{1'} \in \Delta_{n_1}, \forall x^{2'} \in \Delta_{n_2}, \quad f(\mathbf{x}_s) - \delta_f \leq T(\mathbf{x}_s, (x^{1'}, x^{2'}), \rho, \mathbf{y}_s).$$

Both players' payoff terms occur in this single inequality, and the same dual weight ρ is shared by all its instantiations. If the construction provides t_1 row strategies and t_2 column strategies, substituting every pair gives $t_1 t_2$ inequalities. Thus, the stationary and dual strategies can be related to strategies from different calls through a family of joint payoff constraints.

For a fixed $x^2 \in \Delta_{n_2}$, the output $y^1 = \text{BestResponse}(x^2)$ instead satisfies

$$\forall x^{1'} \in \Delta_{n_1}, \quad u_1(x^{1'}, x^2) \leq u_1(y^1, x^2).$$

This property has *one* universally quantified strategy variable. Substituting the t_1 row strategies gives t_1 comparisons of row payoffs, all against the same opponent strategy x^2 . DFM's additional best response supplies comparisons of this form, with x^2 equal to a prescribed mixture. They improve the bound, but provide limited additional relations between the two players' strategies. The tightness results in [14] show that the resulting construction still admits instances attaining $1/3$, or approaching $1/3$ even in the cases using the additional best response.

This comparison suggests seeking further polynomial-time building blocks whose properties relate both players' payoffs and regrets, and composing them so that outputs of one call can be substituted into the properties of another. Additional strategies also provide further instantiations of the two strategy variables in (6). The number of quantified variables indicates the available comparisons; their content and interaction determine the approximation bound. This motivates our construction while using the same descent procedure. The following subsections introduce the operations and then explain their composition.

4.2 OptimalMixing

Given $x_s^1, y_s^1, x_s^2, y_s^2$, the objective of **OptimalMixing** is to minimize the maximum regret over mixtures of each player's stationary and dual strategies [19, 16]. The two players may use different mixing coefficients. Formally, write

$$x^1(p) = (1-p)x_s^1 + py_s^1, \quad x^2(q) = (1-q)x_s^2 + qy_s^2,$$

and define the corresponding profile by

$$\mathbf{x}(p, q) = (x^1(p), x^2(q)), \quad (p, q) \in [0, 1]^2.$$

The optimization problem is

$$\min_{(p, q) \in [0, 1]^2} f(\mathbf{x}(p, q)).$$

For a fixed accuracy parameter $\tau > 0$, **OptimalMixing** returns a profile whose maximum regret is at most τ above this minimum. We separate the four strategy inputs from the accuracy input by a semicolon:

$$(x_o^1, x_o^2) = \text{OptimalMixing}(x_s^1, y_s^1, x_s^2, y_s^2; \tau).$$

Formally, the profile $\mathbf{x}_o = (x_o^1, x_o^2)$ satisfies

$$\begin{aligned} \mathbf{x}_o &\in \{\mathbf{x}(p, q) : (p, q) \in [0, 1]^2\}, \\ f(\mathbf{x}_o) &\leq \min_{(p, q) \in [0, 1]^2} f(\mathbf{x}(p, q)) + \tau. \end{aligned} \tag{7}$$

We also require the output to have regret no greater than that of any of the four profiles formed by choosing one input strategy for each player; namely,

$$f(\mathbf{x}_o) \leq \min_{(p, q) \in \{0, 1\}^2} f(\mathbf{x}(p, q)).$$

For rational input and rational $\tau > 0$, **OptimalMixing** can be computed in time polynomial in the input encoding length and $1/\tau$ [19]. When the accuracies are fixed and clear from context, we omit them from the notation and write **StationaryPoint()** and **OptimalMixing**($x_s^1, y_s^1, x_s^2, y_s^2$). We display them explicitly in Algorithm 2, where their values are inputs.

To analyze this operation in LegoNE, its optimality comparisons must be combined with relations between the payoffs and regrets of mixed strategies [16]. First, fixing one player's strategy

makes each payoff linear in the other player's strategy. Thus the following identities hold against every comparison strategy:

$$\begin{aligned} \forall i \in [2], \forall p \in [0, 1], \forall x^{2'} \in \Delta_{n_2}, \quad u_i(x^1(p), x^{2'}) &= (1-p)u_i(x_s^1, x^{2'}) + pu_i(y_s^1, x^{2'}), \\ \forall i \in [2], \forall q \in [0, 1], \forall x^{1'} \in \Delta_{n_1}, \quad u_i(x^{1'}, x^2(q)) &= (1-q)u_i(x^{1'}, x_s^2) + qu_i(x^{1'}, y_s^2). \end{aligned} \quad (8)$$

The best-response payoff is convex in the opponent's strategy. Write these two payoff functions as

$$\begin{aligned} \forall x^2 \in \Delta_{n_2}, \quad B_1(x^2) &= \max_{x^{1'} \in \Delta_{n_1}} u_1(x^{1'}, x^2), \\ \forall x^1 \in \Delta_{n_1}, \quad B_2(x^1) &= \max_{x^{2'} \in \Delta_{n_2}} u_2(x^1, x^{2'}). \end{aligned}$$

Jensen's inequality bounds a best-response payoff at a mixture by the mixture of the endpoint best-response payoffs. The nonnegative differences are the *Jensen gaps*:

$$\begin{aligned} \forall q \in [0, 1], \quad J_1(q) &= (1-q)B_1(x_s^2) + qB_1(y_s^2) - B_1(x^2(q)) \geq 0, \\ \forall p \in [0, 1], \quad J_2(p) &= (1-p)B_2(x_s^1) + pB_2(y_s^1) - B_2(x^1(p)) \geq 0. \end{aligned} \quad (9)$$

By payoff linearity, each regret is exactly the bilinear interpolation of its four endpoint values minus its Jensen gap. Formally,

$$\begin{aligned} \forall p, q \in [0, 1], \quad f_1(\mathbf{x}(p, q)) &= (1-p)(1-q)f_1(\mathbf{x}_s) + p(1-q)f_1(y_s^1, x_s^2) \\ &\quad + (1-p)qf_1(x_s^1, y_s^2) + pqf_1(\mathbf{y}_s) - J_1(q), \\ \forall p, q \in [0, 1], \quad f_2(\mathbf{x}(p, q)) &= (1-p)(1-q)f_2(\mathbf{x}_s) + p(1-q)f_2(y_s^1, x_s^2) \\ &\quad + (1-p)qf_2(x_s^1, y_s^2) + pqf_2(\mathbf{y}_s) - J_2(p). \end{aligned} \quad (10)$$

Each bracket below is the regret of one comparison strategy against an endpoint opponent strategy. Minimizing the weighted sum of these two regrets gives the corresponding Jensen gap:

$$\begin{aligned} \forall q \in [0, 1], \quad J_1(q) &= \min_{x^{1'} \in \Delta_{n_1}} \{ (1-q)[B_1(x_s^2) - u_1(x^{1'}, x_s^2)] \\ &\quad + q[B_1(y_s^2) - u_1(x^{1'}, y_s^2)] \}, \\ \forall p \in [0, 1], \quad J_2(p) &= \min_{x^{2'} \in \Delta_{n_2}} \{ (1-p)[B_2(x_s^1) - u_2(x_s^1, x^{2'})] \\ &\quad + p[B_2(y_s^1) - u_2(y_s^1, x^{2'})] \}. \end{aligned} \quad (11)$$

The minimum in each formula is attained by a best response to the corresponding mixture. A lower bound valid for every comparison strategy therefore bounds the Jensen gap from below. Joint dual properties can supply such bounds, giving stronger regret comparisons than using only $J_1, J_2 \geq 0$.

Remark. For two strategies per player, the encoding used in LegoNE [16] bounds the regret by considering only the boundary of $[0, 1]^2$. Our analysis of **OptimalMixing** is more detailed: it bounds the Jensen gaps using shared dual constraints and also applies the **OptimalMixing** comparison to interior points $(p, q) \in (0, 1)^2$.

4.3 SimpleMixing and its dual strategies

For a fixed opponent's strategy, a best response maximizes one player's payoff. Minimizing f with one player's strategy fixed provides another way to construct a strategy, taking both players' regrets into account. This is the **SimpleMixing** operation [18]. Given a row strategy $x^1 \in \Delta_{n_1}$, the operation

fixes x^1 and chooses a column strategy x^2 minimizing the maximum regret of the resulting profile. Formally,

$$x^2 \in \arg \min_{x^{2'} \in \Delta_{n_2}} f(x^1, x^{2'}). \quad (12)$$

This problem can be solved by linear programming [18]. To write it as an LP, introduce a real variable h that bounds both players' regrets. The row regret is the largest gain from a pure row deviation, so we impose one constraint for each $k \in [n_1]$. The column player's best-response payoff depends only on the fixed input x^1 . Thus the LP is

$$\begin{aligned} & \underset{x^{2'}, h}{\text{minimize}} && h \\ & \text{subject to} && h \geq u_1(e_k^1, x^{2'}) - u_1(x^1, x^{2'}), \quad k \in [n_1], \\ & && h \geq \max_{\ell \in [n_2]} u_2(x^1, e_\ell^2) - u_2(x^1, x^{2'}), \\ & && x^{2'} \in \Delta_{n_2}, \quad h \in \mathbb{R}. \end{aligned} \quad (13)$$

The maximum in the second constraint is a constant, so all constraints are linear in $x^{2'}$ and h . An optimal column strategy x^2 is the primal solution, and the optimal value is $h = f(x^1, x^2)$.

An optimal dual solution certifies that no column strategy can achieve a smaller maximum regret. Let $\lambda_k \geq 0$ be the dual variable for the k th row-regret constraint in (13), and let $1 - \omega$ be the dual variable for the column-regret constraint, where ω is defined by the sum below. These variables satisfy

$$\omega = \sum_{k \in [n_1]} \lambda_k \in [0, 1], \quad \sum_{k \in [n_1]} \lambda_k + (1 - \omega) = 1.$$

They therefore assign weights to the pure row-deviation gains and the column regret. The total row weight is ω , and the column weight is $1 - \omega$. When $\omega > 0$, normalizing the row weights gives the row dual strategy

$$y^1 = \sum_{k \in [n_1]} \frac{\lambda_k}{\omega} e_k^1 \in \Delta_{n_1}.$$

When $\omega = 0$, there is no row weight to normalize, and y^1 can be chosen arbitrarily.

The operation therefore has one strategy input, the fixed row strategy x^1 , and two strategy outputs: the minimizing column strategy x^2 and the row dual strategy y^1 . We write

$$(x^2, y^1) = \text{SimpleMixing}(x^1).$$

Both strategy outputs can be computed exactly in polynomial time for rational input [18]. The dual weight ω is an auxiliary scalar associated with these outputs. It appears in their properties below.

The dual strategy y^1 combines the row-deviation gains into the payoff difference $u_1(y^1, x^{2'}) - u_1(x^1, x^{2'})$. The dual solution combines this difference with the column regret, using weights ω and $1 - \omega$, respectively. LP duality guarantees that the resulting expression is at least the optimal maximum regret $f(x^1, x^2)$ for every comparison column strategy $x^{2'} \in \Delta_{n_2}$. Formally,

$$\forall x^{2'} \in \Delta_{n_2}, \quad \omega[u_1(y^1, x^{2'}) - u_1(x^1, x^{2'})] + (1 - \omega)f_2(x^1, x^{2'}) \geq f(x^1, x^{2'}). \quad (14)$$

Substituting $x^{2'} = x^2$ into (14) gives

$$\begin{aligned} f(x^1, x^2) &\leq \omega[u_1(y^1, x^2) - u_1(x^1, x^2)] + (1 - \omega)f_2(x^1, x^2) \\ &\leq \omega f_1(x^1, x^2) + (1 - \omega)f_2(x^1, x^2) \leq f(x^1, x^2). \end{aligned}$$

The second inequality follows from the definition of f_1 , and the last from $f = \max\{f_1, f_2\}$. All inequalities are therefore equalities. If $\omega > 0$, equality in the second inequality implies

$$u_1(y^1, x^2) = \max_{x^{1'} \in \Delta_{n_1}} u_1(x^{1'}, x^2).$$

Thus y^1 is a best response to x^2 , or equivalently,

$$\forall x^{1'} \in \Delta_{n_1}, \quad \omega > 0 \quad \implies \quad u_1(x^{1'}, x^2) \leq u_1(y^1, x^2). \quad (15)$$

Both (14) and (15) refer to this same optimal dual solution. In particular, y^1 and ω do not depend on the strategy $x^{2'}$ used in (14). The LP formulation and proofs of these properties are given in [18].

These output properties motivate using **SimpleMixing** in strategy construction. In DFM, an additional strategy is obtained by taking a best response to a prescribed mixture of the opponent's stationary and dual strategies [13]. This step maximizes one player's payoff against that mixture. The **SimpleMixing** operation instead minimizes the maximum regret with one player's strategy fixed, and also supplies a dual strategy. The additional information is the dual inequality (14), which relates the candidate regret to both players' payoff and regret terms. This suggests forming a candidate profile from the primal output and using the dual output to construct further strategies. The associated dual relations may help such a construction improve on DFM's approximation guarantee.

The operation **SimpleMixing**(x^2) is defined symmetrically: it fixes a column strategy $x^2 \in \Delta_{n_2}$ and chooses a row strategy $x^1 \in \Delta_{n_1}$ minimizing the maximum regret of the resulting profile. It also returns a dual column strategy $y^2 \in \Delta_{n_2}$. Namely,

$$\begin{aligned} (x^1, y^2) &= \text{SimpleMixing}(x^2), \\ x^1 &\in \arg \min_{x^{1'} \in \Delta_{n_1}} f(x^{1'}, x^2). \end{aligned}$$

It has the same polynomial-time guarantee. In this symmetric call, ω is the total dual weight on the pure column-deviation constraints; $1 - \omega$ is the dual multiplier on the single row-regret constraint. Here ω belongs to **SimpleMixing**(x^2), independently of the weight in **SimpleMixing**(x^1). In particular,

$$\forall x^{2'} \in \Delta_{n_2}, \quad \omega > 0 \quad \implies \quad u_2(x^1, x^{2'}) \leq u_2(x^1, y^2).$$

4.4 Complete algorithm

With these building blocks, we obtain the algorithm for Theorem 1, presented in Algorithm 2. Starting from the output of **StationaryPoint**, it introduces further strategies through **SimpleMixing** and uses **OptimalMixing** to form another candidate profile. The same operations are applied to both players, giving a simple, symmetric construction. Figure 1 gives a graphical illustration.

The **SimpleMixing** calls are organized into two rounds. The first round uses the stationary dual strategies y_s^1, y_s^2 as inputs; the second uses the dual strategies y_1^1, y_1^2 returned by the first. In x_j^i, y_j^i , the subscript $j \in [2]$ indicates the round and the superscript $i \in [2]$ identifies the player who uses the strategy.

Algorithm 2 makes one **StationaryPoint** call, four **SimpleMixing** calls, and one **OptimalMixing** call. The algorithm displays the two accuracy inputs at the calls that use them. Its running time is therefore polynomial in the input encoding length, $1/\delta_f$, and $1/\tau$.

Algorithm 2 Our algorithm for bimatrix games

Input: A bimatrix game with rational payoffs and rational $\delta_f, \tau > 0$.

- 1: $(x_s^1, x_s^2, y_s^1, y_s^2) = \text{StationaryPoint}(\delta_f)$
 - 2: $(x_1^2, y_1^1) = \text{SimpleMixing}(y_s^1)$
 - 3: $(x_1^1, y_1^2) = \text{SimpleMixing}(y_s^2)$
 - 4: $(x_2^2, y_2^1) = \text{SimpleMixing}(y_1^1)$
 - 5: $(x_2^1, y_2^2) = \text{SimpleMixing}(y_1^2)$
 - 6: $(x_o^1, x_o^2) = \text{OptimalMixing}(x_s^1, y_s^1, x_s^2, y_s^2; \tau)$
 - 7: $\mathcal{S} = \{(y_s^1, x_1^2), (x_1^1, y_1^2), (y_1^1, x_2^2), (x_2^1, y_2^2), (x_o^1, x_o^2)\}$
 - 8: **return** $\mathbf{x}_{\text{out}} \in \arg \min_{\mathbf{x} \in \mathcal{S}} f(\mathbf{x})$
-

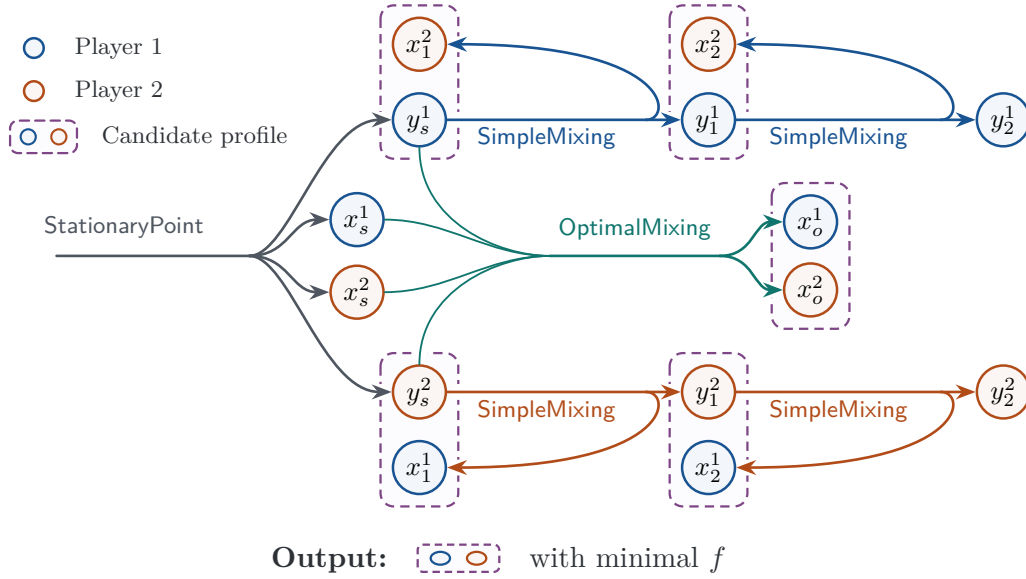


Figure 1: Construction of the candidate profiles. Vertices represent strategies, and labelled edges indicate building-block calls; the labels show operation names, while Algorithm 2 shows their accuracy inputs. **StationaryPoint** has no strategy input and produces four strategies. Each purple dashed frame pairs the two players' strategies in one candidate profile. The output minimizes $f = \max\{f_1, f_2\}$ over the five framed profiles. The strategies y_2^1, y_2^2 are used only in the analysis.

5 Approximation analysis

We first consider the formal zero-error versions of the two operations' output properties, written $\delta_f = \tau = 0$, and prove their guarantee by contradiction. The implemented calls use positive accuracy inputs. Suppose the selected output has regret above a proposed threshold r . Because it selects the best of five candidates, all five must then exceed r . We will show that this cannot happen when r is above the claimed bound.

The argument proceeds from the game to arithmetic. The properties of the algorithm's calls first impose restrictions on any such bad output. We combine and simplify those restrictions until they are a finite list of inequalities in a few numbers, rather than in the game's strategies and payoffs. Numerical exploration suggests the threshold where these inequalities become critical; an exact calculation defines that threshold, and a global argument shows that the inequalities cannot hold above it. Finally, we account for the errors δ_f and τ in the actual calls. The main text explains

these transitions; Appendices A–F give the detailed calculations and exact-checker paths.

5.1 Properties needed for the contradiction

This subsection gathers the comparisons used later in the proof. **StationaryPoint** relates the two players' payoffs at its output; each **SimpleMixing** call chooses an opponent strategy and compares the result with other choices; and **OptimalMixing** compares its output with profiles formed by mixing the stationary and best-response strategies. We state the precise inequalities below and explain what each will contribute. They follow from the stationary-point results [12, 14] and the mixing operations [18, 19].

StationaryPoint: one comparison for both players. The operation returns a stationary profile $\mathbf{x}_s$ and one auxiliary strategy for each player; write the pair as $\mathbf{y}_s = (y_s^1, y_s^2)$. It makes the two players' regrets at $\mathbf{x}_s$ equal; call their common value g :

$$g = f_1(\mathbf{x}_s) = f_2(\mathbf{x}_s) = f(\mathbf{x}_s). \quad (16)$$

The operation also returns a number $\rho \in [0, 1]$ that weighs the row player's part of a payoff comparison; $1 - \rho$ weighs the column player's part. The same ρ and $\mathbf{y}_s$ work against every pair of comparison strategies:

$$\exists \rho \in [0, 1], \forall x^{1'} \in \Delta_{n_1}, \forall x^{2'} \in \Delta_{n_2}, \quad g - \delta_f \leq T(\mathbf{x}_s, (x^{1'}, x^{2'}), \rho, \mathbf{y}_s). \quad (17)$$

Here T is the payoff expression in (3). The returned strategies also nearly maximize each player's payoff against the other player's stationary strategy. For the row player,

$$\forall x^{1'} \in \Delta_{n_1}, \quad \rho > 0 \implies u_1(x^{1'}, x_s^2) \leq u_1(y_s^1, x_s^2) + \delta_f. \quad (18)$$

The column-player property is symmetric. Later we will choose comparison strategies produced by both **SimpleMixing** rounds. The same ρ and $\mathbf{y}_s$ must satisfy every resulting inequality.

SimpleMixing: comparing a fixed-side choice with alternatives. For a fixed row strategy x^1 , the operation chooses the column strategy x^2 that minimizes the two players' maximum regret. Its linear program also returns a row strategy y^1 derived from the constraints that certify this choice. Write the outputs as

$$(x^2, y^1) = \text{SimpleMixing}(x^1).$$

The chosen x^2 minimizes maximum regret with x^1 fixed, so changing the column strategy alone cannot reduce the candidate's maximum regret:

$$\forall x^{2'} \in \Delta_{n_2}, \quad f(x^1, x^2) \leq f(x^1, x^{2'}). \quad (19)$$

The same LP assigns a number $\omega \in [0, 1]$ to the row player's possible payoff gain and $1 - \omega$ to the column player's regret, as explained in Section 4.3. Unlike ρ , this ω belongs to the particular **SimpleMixing** call. It gives a second comparison that holds for every column strategy:

$$\exists \omega \in [0, 1], \forall x^{2'} \in \Delta_{n_2}, \quad f(x^1, x^2) \leq \omega[u_1(y^1, x^{2'}) - u_1(x^1, x^{2'})] + (1 - \omega)f_2(x^1, x^{2'}). \quad (20)$$

The same y^1 and ω work for every $x^{2'}$ in this inequality. This is what lets us compare the output of one call with strategies from another. At the chosen x^2 , the inequality is an equality:

$$f(x^1, x^2) = \omega[u_1(y^1, x^2) - u_1(x^1, x^2)] + (1 - \omega)f_2(x^1, x^2). \quad (21)$$

If a player's regret has positive total weight, that player's regret equals the candidate's maximum regret. The LP's complementary-slackness conditions state this as

$$\begin{aligned}\omega[f(x^1, x^2) - f_1(x^1, x^2)] &= 0, \\ (1 - \omega)[f(x^1, x^2) - f_2(x^1, x^2)] &= 0.\end{aligned}\tag{22}$$

It also makes every positively weighted row-deviation constraint tight. Consequently, when $\omega > 0$, the returned y^1 is a best response to x^2 :

$$\forall x^{1'} \in \Delta_{n_1}, \quad \omega > 0 \implies u_1(x^{1'}, x^2) \leq u_1(y^1, x^2).\tag{23}$$

These are properties of the same LP solution [18, 21]; the properties of **SimpleMixing**(x^2) are symmetric. We apply them to all four calls in Algorithm 2. The first round's y strategies become the second round's inputs, so the comparisons from the two rounds constrain some of the same payoffs.

OptimalMixing: comparing all pairs of mixtures. As defined in Section 4.2, $\mathbf{x}(p, q)$ uses probability p to mix the row player's two given strategies and probability q for the column player's two. The two probabilities vary independently over $[0, 1]$. The output is one such profile:

$$\exists p_o, q_o \in [0, 1], \quad \mathbf{x}_o = \mathbf{x}(p_o, q_o).$$

The operation minimizes the maximum regret over these mixtures up to τ , so its output regret is at most τ above that of any mixed profile:

$$\forall p, q \in [0, 1], \quad f(\mathbf{x}_o) \leq f(\mathbf{x}(p, q)) + \tau.\tag{24}$$

To bound regret at these mixtures, we use two identities from Section 4.2. Equation (10) starts with the four unmixed profiles and interpolates their regrets. It then subtracts a nonnegative correction, called the Jensen gap. The correction arises because a best response to a mixture must use one strategy against both endpoint opponents, whereas the endpoints may have different best responses. Equation (11) expresses the correction as a minimum over that one response strategy. Later comparisons will bound it from below. The operation also compares with the four unmixed profiles without an additive error:

$$\forall p, q \in \{0, 1\}, \quad f(\mathbf{x}_o) \leq f(\mathbf{x}(p, q)).$$

In particular, choosing $p = q = 0$ gives the stationary profile and therefore yields

$$f(\mathbf{x}_o) \leq g.\tag{25}$$

Output selection. Finally, the algorithm chooses the candidate of smallest maximum regret. With $\mathcal{S}$ as in Algorithm 2, this gives

$$\mathbf{x}_{\text{out}} \in \mathcal{S}, \quad \forall \mathbf{z} \in \mathcal{S}, \quad f(\mathbf{x}_{\text{out}}) \leq f(\mathbf{z}).\tag{26}$$

Thus a regret bound for any one candidate suffices. We now assume all five candidates exceed a proposed threshold and combine these comparisons to rule out that possibility; the accompanying materials¹ record the full logical description.

5.2 From a failed output to inequalities it must satisfy

Fix $0 < r \leq 1/3$ and any normalized bimatrix game. In the zero-error analysis, suppose for contradiction that the selected output has regret greater than r :

$$f(\mathbf{x}_{\text{out}}) > r. \quad (27)$$

We derive inequalities that any such run must satisfy. We then eliminate the strategies and payoff values, leaving conditions on a few real numbers. If these conditions have no solution, the assumed failure is impossible.

We obtain these conditions by applying one call's inequality to strategies produced by another call, then keeping only the consequences needed later. This is the instantiation-and-forgetting step of LegoNE [15, 16]. For each call, its returned strategies and comparison coefficients stay fixed while the strategies being compared vary. The resulting inequalities may also admit numbers that do not describe any game; ruling out this larger set still rules out every actual failure. Throughout this reduction we use the zero-error versions of their comparison properties:

$$\delta_f = \tau = 0. \quad (28)$$

Section 5.5 restores the approximation errors. For the fixed game, fix one optimal solution of each LP, including the strategies and comparison coefficients it returns. Denote the four SimpleMixing candidates and their regrets by

$$\begin{aligned} \mathbf{z}_1^1 &= (y_s^1, x_1^2), & \mathbf{z}_1^2 &= (x_1^1, y_s^2), \\ \mathbf{z}_2^1 &= (y_1^1, x_2^2), & \mathbf{z}_2^2 &= (x_2^1, y_1^2), \\ \forall i, j \in [2], & F_j^i &= f(\mathbf{z}_j^i). \end{aligned} \quad (29)$$

Here the superscript on $\mathbf{z}_j^i, F_j^i$ identifies the player whose input is fixed, and the subscript identifies the round. This differs from the superscript on a single strategy x_j^i or y_j^i , which identifies the player using that strategy.

Each SimpleMixing call supplies a coefficient ω that balances the two players' regret terms in (20). We give the four calls' coefficients different names:

$$\begin{aligned} \text{SimpleMixing}(y_s^1) : \quad \omega &= \eta, & \text{SimpleMixing}(y_s^2) : \quad \omega &= \xi, \\ \text{SimpleMixing}(y_1^1) : \quad \omega &= \alpha, & \text{SimpleMixing}(y_1^2) : \quad \omega &= \gamma. \end{aligned} \quad (30)$$

The algorithm chooses the least-regret candidate. Therefore its failure means that all four fixed-side candidates are bad; the stationary regret g also exceeds r , and every allowable mixture by OptimalMixing must exceed r . Formally,

$$\forall i, j \in [2], F_j^i > r, \quad g \geq f(\mathbf{x}_o) > r, \quad \forall p, q \in [0, 1], f(\mathbf{x}(p, q)) > r. \quad (31)$$

Here $g = f(\mathbf{x}_s)$ and $\mathbf{x}(p, q)$ mixes the stationary strategies with the auxiliary strategies returned by StationaryPoint, as defined in Section 4.2. The last condition lets us choose the mixing probabilities p, q later to test the supposed failure. Restricting r to $1/3$ loses no counterexample to a smaller bound, since any output above $1/3$ also exceeds a threshold strictly below $1/3$.

We now simplify what a failed run would require. First we show that the comparison coefficients lie strictly between zero and one; this makes the returned auxiliary strategies best responses. Next we bound the payoffs used in later comparisons, combine the two SimpleMixing rounds, and test

selected mixtures. Finally we remove the stationary regret g from the conditions. The remaining five numbers are $\rho, \eta, \xi, \alpha, \gamma$.

First we need positive coefficients; otherwise the best-response conclusions in (18) and (23) need not apply. The next lemma derives positivity from (31) and uses complementary slackness to identify both regrets at each SimpleMixing candidate.

Lemma 2 (Positive comparison coefficients). *For an execution satisfying the properties of Section 5.1, suppose that there is no approximation error as in (28) and that the selected output has regret above r as in (27). Then all five comparison coefficients satisfy*

$$r < \rho, \eta, \xi, \alpha, \gamma < 1 - r.$$

The auxiliary strategies from StationaryPoint are best responses to the opponents' stationary strategies, and every auxiliary strategy from SimpleMixing is a best response to that call's chosen strategy:

$$\begin{aligned} B_1(x_s^2) &= u_1(y_s^1, x_s^2), & B_2(x_s^1) &= u_2(x_s^1, y_s^2), \\ \forall j \in [2], & B_1(x_j^2) &= u_1(y_j^1, x_j^2), & B_2(x_j^1) &= u_2(x_j^1, y_j^2). \end{aligned}$$

Both player regrets at each candidate equal its maximum regret:

$$\forall i, j \in [2], \quad f_1(z_j^i) = f_2(z_j^i) = F_j^i.$$

Proof. Recall

$$\sigma = 1 - \rho.$$

Substitute (y_s^1, x_s^2) and (x_s^1, y_s^2) for the comparison profile in (17), with $\delta_f = 0$. One bracket vanishes in each substitution, leaving

$$\begin{aligned} g &\leq \sigma[u_2(y_s^1, y_s^2) - u_2(y_s^1, x_s^2)], \\ g &\leq \rho[u_1(y_s^1, y_s^2) - u_1(x_s^1, y_s^2)]. \end{aligned}$$

Each payoff difference is at most one, so $g > r$ from (31) implies $r < \rho, \sigma$. This justifies using the stationary auxiliary strategies as best responses. Nonnegative payoffs also give

$$u_1(y_s^1, y_s^2) \geq g/\rho, \quad u_2(y_s^1, y_s^2) \geq g/\sigma. \quad (32)$$

For the first-round call SimpleMixing(y_s^1), apply (20), with $\omega = \eta$, first at a column best response to y_s^1 , where the column regret is zero, and then at x_s^2 , to which y_s^1 is a row best response. Payoff differences are at most one, so these comparisons yield

$$F_1^1 \leq \eta, \quad F_1^1 \leq 1 - \eta.$$

Since $F_1^1 > r$ in (31), both coefficients are positive. Complementary slackness (22) gives

$$\eta(F_1^1 - f_1(z_1^1)) = 0, \quad (1 - \eta)(F_1^1 - f_2(z_1^1)) = 0,$$

and y_1^1 is a best response to x_1^2 . For the second-round call SimpleMixing(y_1^1), apply the same comparison inequality, now with $\omega = \alpha$, at x_1^2 and at a column best response to y_1^1 . The best-response property just established gives $F_2^1 \leq 1 - \alpha$ at x_1^2 , and zero column regret at the other comparison gives $F_2^1 \leq \alpha$. The corresponding applications for SimpleMixing(y_s^2) and SimpleMixing(y_1^2) are symmetric. The complete substitutions are in Appendix A. $\square$

Bounding the cross-profile payoffs. The best-response identities from Lemma 2 let us bound the payoffs appearing when we compare strategies from different calls. Because all payoffs lie in $[0, 1]$, each player's regret satisfies $f_i(\mathbf{x}) \leq 1 - u_i(\mathbf{x})$. We therefore bound these payoff shortfalls at three profiles made from first-round strategies. The six bounds below follow by applying (17) and (20); later comparisons use them in place of game-specific payoffs. For compactness, collect the threshold, stationary regret, and first-round comparison coefficients in

$$\boldsymbol{\lambda} = (r, g, \rho, \eta, \xi). \quad (33)$$

For fixed r , the other entries of $\boldsymbol{\lambda}$ can vary. Each entry in the next table bounds the payoff shortfall $1 - u_i$ for the player named by its column, and therefore also that player's regret. The six symbols s_1, s_2, h_1, h_2, S, T depend only on $\boldsymbol{\lambda}$; their formulas are given in (98), (102), and (105); Appendix A gives the substitutions.

profile	upper bound on $1 - u_1$	upper bound on $1 - u_2$
(y_1^1, y_s^2)	$s_1(\boldsymbol{\lambda})$	$h_2(\boldsymbol{\lambda})$
(y_s^1, y_1^2)	$h_1(\boldsymbol{\lambda})$	$s_2(\boldsymbol{\lambda})$
(y_1^1, y_1^2)	$S(\boldsymbol{\lambda})$	$T(\boldsymbol{\lambda})$

(34)

Here the numerical bound $T(\boldsymbol{\lambda})$ is distinct from the stationary expression $T(\mathbf{x}, \mathbf{x}', \rho, \mathbf{y})$; the number and types of their arguments distinguish them. To obtain the last row, we abbreviate two ratios of the first-round comparison coefficients as

$$u = \frac{1 - \eta}{\eta}, \quad v = \frac{1 - \xi}{\xi}. \quad (35)$$

Applying (20) for $\text{SimpleMixing}(y_s^1)$ at y_1^2 and its column-player counterpart for $\text{SimpleMixing}(y_s^2)$ at y_1^1 gives

$$S(\boldsymbol{\lambda}) = h_1(\boldsymbol{\lambda}) - r + u(s_2(\boldsymbol{\lambda}) - r), \quad T(\boldsymbol{\lambda}) = h_2(\boldsymbol{\lambda}) - r + v(s_1(\boldsymbol{\lambda}) - r), \quad (36)$$

We have now bounded the payoff shortfalls needed below in terms of r, g and the first-round coefficients ρ, η, ξ . The second-round coefficients enter when we compare the two rounds.

Using both SimpleMixing rounds together. The auxiliary strategy returned by the first round becomes the input of the second. Applying both calls' inequalities to the same comparison strategy gives a lower and an upper bound on one payoff. Eliminating that payoff will link the rounds. To do so, apply the comparison inequalities for the first-round call $\text{SimpleMixing}(y_s^2)$ and the second-round call $\text{SimpleMixing}(y_1^2)$ at the same arbitrary row strategy $x^{1'} \in \Delta_{n_1}$. Their coefficients are ξ and γ , respectively, by (30). The column-player version of (20), together with (31) and payoff bounds in $[0, 1]$, gives

$$\begin{aligned} \forall x^{1'} \in \Delta_{n_1}, \quad \xi u_2(x^{1'}, y_1^2) &\geq r - (1 - \xi) + (1 - \xi)u_1(x^{1'}, y_s^2), \\ \forall x^{1'} \in \Delta_{n_1}, \quad \gamma u_2(x^{1'}, y_1^2) &\leq 1 - r - (1 - \gamma)u_1(x^{1'}, y_1^2). \end{aligned} \quad (37)$$

Multiply the first inequality by $\gamma/\xi > 0$ and substitute its lower bound for the common payoff into the second inequality. Using v from (35), this eliminates $u_2(x^{1'}, y_1^2)$ and gives

$$\forall x^{1'} \in \Delta_{n_1}, \quad (1 - \gamma)u_1(x^{1'}, y_1^2) + \gamma v u_1(x^{1'}, y_s^2) \leq 1 - r - \frac{\gamma}{\xi}[r - (1 - \xi)]. \quad (38)$$

This inequality holds for every row strategy, so it bounds the row best-response payoff against a mixture of y_1^2 and y_s^2 . Define a mixed column strategy to use as a comparison in the proof:

$$D_E = 1 - \gamma + \gamma v > 0, \quad y_E^2 = \frac{(1 - \gamma)y_1^2 + \gamma v y_s^2}{D_E} \in \Delta_{n_2}. \quad (39)$$

By payoff linearity and the definition of B_1 , inequality (38) implies

$$B_1(y_E^2) \leq \frac{1 - r - \frac{\gamma}{\xi}[r - (1 - \xi)]}{D_E}. \quad (40)$$

We can therefore apply the comparison inequality (20) for the second-round call **SimpleMixing**(y_1^1) at y_E^2 , since that inequality holds throughout the column simplex. This call has coefficient α . Applying its inequality at y_s^2 and y_1^2 , and that of **SimpleMixing**(y_1^2) with coefficient γ at y_s^1 and y_1^1 , using (34), gives four upper bounds on second-round candidate regret. Since all those candidates exceed r under the failure assumption, these bounds must be at least r :

$$\begin{aligned} U_g &= \alpha s_1(\boldsymbol{\lambda}) + (1 - \alpha)h_2(\boldsymbol{\lambda}) \geq r, & H_g &= \alpha S(\boldsymbol{\lambda}) + (1 - \alpha)T(\boldsymbol{\lambda}) \geq r, \\ V_g &= (1 - \gamma)h_1(\boldsymbol{\lambda}) + \gamma s_2(\boldsymbol{\lambda}) \geq r, & K_g &= (1 - \gamma)S(\boldsymbol{\lambda}) + \gamma T(\boldsymbol{\lambda}) \geq r. \end{aligned} \quad (41)$$

These four expressions depend on $\boldsymbol{\lambda}$ and the second-round coefficients α, γ . At y_E^2 , the row player's best-response term in the second-round comparison is bounded by (40). Because y_E^2 mixes y_1^2 and y_s^2 , its other payoff terms are weighted averages of terms already bounded in H_g and U_g . Collecting these bounds gives

$$F_2^1 \leq \frac{(1 - \gamma)H_g + \gamma v U_g + \alpha[\gamma(1 - r - rv) - r]}{D_E}. \quad (42)$$

Since $F_2^1 > r$, multiplication by D_E yields the further necessary inequality

$$E_{1,g} = (1 - \gamma)(H_g - r) + \gamma v(U_g - r) + \alpha\gamma(1 - r - rv) - \alpha r \geq 0. \quad (43)$$

The shared inequality $E_{1,g} \geq 0$ is therefore a new necessary condition for failure: it links both players' **SimpleMixing** calls across the two rounds. Its remaining variables are the threshold, stationary regret, and all five comparison coefficients. Appendix A gives the substitution before regrouping terms and its symmetric counterpart.

Using OptimalMixing to test selected mixtures. The preceding inequalities connect the four **SimpleMixing** calls. We now use **OptimalMixing**: every profile $\mathbf{x}(p, q)$ must have regret above r under (31). We derive upper bounds on both players' regrets at these profiles. If both bounds are at most r for even one choice of p, q , the assumed failure is impossible. The bounds have simple linear pieces, so we will test four selected segments rather than carry p, q through the rest of the proof.

By (10), each mixed regret equals an average of endpoint regrets minus the Jensen gap explained in Section 5.1. The gap reflects the fact that one response strategy must serve against both endpoint opponents. To bound the column player's gap from below, consider the regrets of the same column strategy $x^{2'}$ against the two row strategies x_s^1 and y_s^1 :

$$A = f_2(x_s^1, x^{2'}), \quad B = f_2(y_s^1, x^{2'}). \quad (44)$$

Both numbers are nonnegative, and the same strategy is used at both endpoints. Apply stationarity (17) at $(x_s^1, x^{2'})$, then substitute the resulting payoff lower bound into the comparison inequality (20) for the first-round call **SimpleMixing**(y_s^1) at $x^{2'}$, with $\omega = \eta$. This gives

$$\forall x^{2'} \in \Delta_{n_2}, \quad \frac{\eta\sigma}{\rho}A + (1 - \eta)B \geq \ell_c(\boldsymbol{\lambda}), \quad \ell_c(\boldsymbol{\lambda}) = r - \eta(1 - g/\rho). \quad (45)$$

The function ℓ_c is the lower bound obtained by these two comparisons; the index identifies the column player. If it is positive, every column strategy has positive regret against at least one of the two row strategies. The Jensen-gap identity (11) takes the weighted minimum of these same regrets:

$$J_2(p) = \min_{x^{2'} \in \Delta_{n_2}} \{(1-p)A + pB\}.$$

The following lemma therefore converts the joint regret comparison into the correction needed in the mixed profile's regret bound.

Lemma 3 (A lower bound on the gain from mixing). *Consider an execution satisfying the building-block properties of Section 5.1 with zero errors (28), and suppose that its selected output has regret above r as in (27). Then, for every $p \in [0, 1]$, the column Jensen gap satisfies*

$$J_2(p) \geq \ell_c(\boldsymbol{\lambda}) \min \left\{ \frac{(1-p)\rho}{\eta\sigma}, \frac{p}{1-\eta} \right\}.$$

There is a symmetric bound for the row Jensen gap $J_1(q)$.

Proof. Set $c_1 = \eta\sigma/\rho$, $c_2 = 1 - \eta$, and $M = \min\{(1-p)/c_1, p/c_2\}$. Both c_1, c_2 are positive. For each comparison strategy, nonnegativity of A, B gives

$$(1-p)A + pB \geq M(c_1A + c_2B) \geq M\ell_c(\boldsymbol{\lambda}).$$

Minimizing the left side over $x^{2'}$ gives $J_2(p)$ by its regret identity in Section 4.2. Thus a joint restriction on endpoint best responses becomes a reduction in the regret of the mixed profile. The inequality remains valid if $\ell_c(\boldsymbol{\lambda})$ is nonpositive. $\square$

Subtracting these Jensen-gap bounds from the endpoint interpolation gives regret upper bounds $R_g(p, q)$ and $C_g(p, q)$ for the row and column players. They depend on $(\boldsymbol{\lambda}, p, q)$. For the row player, define the symmetric Jensen-gap lower bound by

$$\ell_r(\boldsymbol{\lambda}) = r - \xi(1 - g/\sigma), \quad j_r(\boldsymbol{\lambda}, q) = \ell_r(\boldsymbol{\lambda}) \min \left\{ \frac{(1-q)\sigma}{\xi\rho}, \frac{q}{1-\xi} \right\}.$$

Using $J_1(q) \geq j_r(\boldsymbol{\lambda}, q)$ gives

$$R_g(p, q) = (1-p)(1-q)g + q(1 - pg/\rho) - j_r(\boldsymbol{\lambda}, q).$$

The first two terms interpolate the four row-regret bounds $g, 0, 1, 1 - g/\rho$; the last term is the improvement from the Jensen gap. For the column player, the corresponding formulas are

$$\begin{aligned} j_c(\boldsymbol{\lambda}, p) &= \ell_c(\boldsymbol{\lambda}) \min \left\{ \frac{(1-p)\rho}{\eta\sigma}, \frac{p}{1-\eta} \right\}, \\ C_g(p, q) &= (1-p)(1-q)g + p(1 - qg/\sigma) - j_c(\boldsymbol{\lambda}, p). \end{aligned}$$

Both formulas and the two Jensen-gap bounds are also stated in (129) and (127); Appendix B gives the derivation. Since both actual regrets are bounded above by these functions, failure requires

$$\forall p, q \in [0, 1], \quad \max\{R_g(p, q), C_g(p, q)\} > r.$$

Each Jensen-gap bound uses the minimum of two affine expressions. We choose comparison coefficients where these expressions agree:

$$\frac{(1-p)\rho}{\eta\sigma} = \frac{p}{1-\eta}, \quad \frac{(1-q)\sigma}{\xi\rho} = \frac{q}{1-\xi}.$$

Solving these equations gives

$$p_{\dagger} = \frac{\rho(1-\eta)}{\rho(1-\eta) + \sigma\eta}, \quad q_{\dagger} = \frac{\sigma(1-\xi)}{\sigma(1-\xi) + \rho\xi}.$$

It is enough for our contradiction to keep four necessary *segment conditions* along these cuts:

$$\begin{aligned} \min_{0 \leq q \leq q_{\dagger}} \max\{R_g(p_{\dagger}, q), C_g(p_{\dagger}, q)\} &\geq r, \\ \min_{q_{\dagger} \leq q \leq 1} \max\{R_g(p_{\dagger}, q), C_g(p_{\dagger}, q)\} &\geq r, \\ \min_{0 \leq p \leq p_{\dagger}} \max\{R_g(p, q_{\dagger}), C_g(p, q_{\dagger})\} &\geq r, \\ \min_{p_{\dagger} \leq p \leq 1} \max\{R_g(p, q_{\dagger}), C_g(p, q_{\dagger})\} &\geq r. \end{aligned} \tag{46}$$

Both regret bounds are affine on each of these segments, so each minimum occurs at an endpoint or an intersection of the two bounds. Symbolic calculation with SymPy [29] constructs the endpoint and intersection inequalities, and an exact machine check reconstructs their identities. This turns (46) into finite inequalities and eliminates the mixing parameters. Appendix B expands these inequalities.

Removing the stationary regret. The payoff bounds have removed the individual payoff variables, and the segment tests have removed the mixing parameters p, q . For fixed r , only g and the five comparison coefficients $\rho, \eta, \xi, \alpha, \gamma$ remain. We now remove g as well. Every assumed failure has $g > r$. The next lemma shows that its necessary inequalities still hold if we substitute $g = r$ while keeping those coefficients fixed. This is an algebraic substitution in the inequalities; it does not require another game or algorithm execution.

Lemma 4 (Eliminating the stationary regret). *Suppose an execution satisfies the properties of Section 5.1, with zero errors (28) and output regret above r as in (27), where $3/10 \leq r \leq 1/3$. Hold $r, \rho, \eta, \xi, \alpha, \gamma$ fixed and replace g by r in the numerical bounds. The six payoff upper bounds remain nonnegative; the second-round inequalities (41), the shared inequality (43), and its symmetric counterpart remain valid, as do the four segment conditions (46).*

Proof idea. We present only the main proof ideas here; Lemma 12 in Appendix C provides the complete proof of Lemma 4. Applying the comparison inequalities for **SimpleMixing**(y_s^1) at y_s^2 and for **SimpleMixing**(y_s^2) at y_s^1 , and using the stationary payoff bounds (32), gives

$$\begin{aligned} r &\leq \eta(1 - g/\rho) + (1 - \eta)(1 - g/\sigma), \\ r &\leq \xi(1 - g/\sigma) + (1 - \xi)(1 - g/\rho). \end{aligned} \tag{47}$$

The right sides decrease with g , so these inequalities bound the admissible interval for g from above. Together with $g \geq r$, they specify the interval on which we prove preservation.

Along this interval, each of the six payoff upper bounds increases or stays unchanged as g decreases. For example,

$$\frac{\partial s_1(\lambda)}{\partial g} = -\frac{1}{\rho} - \frac{u}{\sigma} < 0.$$

The second-round and shared inequalities combine these bounds with nonnegative coefficients, so they continue to hold.

For the four mixing conditions, we use their finite endpoint and intersection cases. On the two segments ending at a mixing coefficient equal to one, both regret upper bounds increase as g decreases.

For the segment $p = p_{\dagger}$, $0 \leq q \leq q_{\dagger}$, the first-round comparisons (47) give

$$C_g(p_{\dagger}, 0) \geq R_g(p_{\dagger}, 0).$$

The column bound at this outer endpoint and both bounds at the central endpoint increase as g decreases, so the endpoint conditions are preserved. Along the segment, the row bound increases with q and the column bound decreases. If they intersect inside the segment, clearing the positive difference of their slopes shows that the common value is at least r exactly when $\mathcal{D}(g) \geq 0$, where

$$\begin{aligned} \mathcal{D}(g) = \frac{1}{q_{\dagger}} \{ & [C_g(p_{\dagger}, 0) - r][R_g(p_{\dagger}, q_{\dagger}) - r] \\ & - [R_g(p_{\dagger}, 0) - r][C_g(p_{\dagger}, q_{\dagger}) - r] \}. \end{aligned} \quad (48)$$

We use an exact machine check to establish

$$\frac{\partial \mathcal{D}(g)}{\partial g} < 0$$

on the admissible parameter ranges. After clearing positive denominators, this becomes a polynomial sign condition, verified exactly as detailed in Appendix C [31]. That appendix gives the polynomial, domain, and checker. Thus $\mathcal{D}(g)$ increases as g decreases, preserving the intersection condition. An intersection enters or leaves the segment through an endpoint, whose condition is already preserved. Exchanging the two players gives the segment ending at $p = 0$. All four finite segment conditions therefore continue to hold down to $g = r$. $\square$

After substituting $g = r$, write the remaining parameters as

$$\lambda_r = (r, r, \rho, \eta, \xi). \quad (49)$$

We now simplify the first-round regret bounds to obtain explicit formulas. Applying the comparison inequalities of `SimpleMixing`(y_s^1) at x_s^2 and of `SimpleMixing`(y_s^2) at x_s^1 gives

$$f_1(y_1^1, x_s^2) \leq \min\{1, (1 - \eta - r)/\eta\}, \quad f_2(x_s^1, y_1^2) \leq \min\{1, (1 - \xi - r)/\xi\}. \quad (50)$$

Replacing the two right sides by $(1 - \eta - r)/\eta$ and $(1 - \xi - r)/\xi$ gives larger upper bounds. Combining them with stationarity (17) at (y_1^1, x_s^2) and (x_s^1, y_1^2) gives relaxed bounds for h_1, h_2 . From this point on, the bare symbols s_1, s_2, h_1, h_2, S, T denote the functions after substituting $g = r$ and making this relaxation:

$$\begin{aligned} s_1 &= 1 - r - r/\rho + u(1 - r - r/\sigma), \\ s_2 &= 1 - r - r/\sigma + v(1 - r - r/\rho), \\ h_1 &= \min\{1, 1 - r(1 + \sigma)/\rho + \sigma(1 - r)v/\rho\}, \\ h_2 &= \min\{1, 1 - r(1 + \rho)/\sigma + \rho(1 - r)u/\sigma\}, \\ S &= h_1 - r + u(s_2 - r), \quad T = h_2 - r + v(s_1 - r). \end{aligned} \quad (51)$$

These functions depend on r, ρ, η, ξ . Using them in (41) and (43) defines the corresponding expressions U, V, H, K, E_1 after the relaxation. The mixing bounds become $R_r(p, q) = R_g(p, q)|_{g=r}$ and $C_r(p, q) = C_g(p, q)|_{g=r}$.

The next proposition states the implication we need. A failed run would assign values to $\rho, \eta, \xi, \alpha, \gamma$ that satisfy all the remaining necessary inequalities. Call this finite collection Σ_r . Proving that no five values satisfy it will rule out the assumed failure. The complete list is (162) in Appendix C.

Proposition 5 (Necessary inequalities for a failed run). *There is an explicit collection of inequalities $\Sigma_r(\boldsymbol{\theta})$ in the five real variables*

$$\boldsymbol{\theta} = (\rho, \eta, \xi, \alpha, \gamma)$$

such that, for an execution with the properties of Section 5.1 and zero errors (28),

$$\forall r \in [3/10, 1/3], \quad f(\mathbf{x}_{\text{out}}) > r \implies \exists \boldsymbol{\theta}, \Sigma_r(\boldsymbol{\theta}).$$

The collection consists of coefficient bounds, the payoff bounds and shared comparisons derived above, and the four finite segment conditions.

5.3 Finding and defining the candidate threshold

Proposition 5 has reduced a failed run to five real numbers $\boldsymbol{\theta} = (\rho, \eta, \xi, \alpha, \gamma)$ that must satisfy Σ_r ; strategies, payoffs, the mixing probabilities p, q , and g no longer appear. We now search for where these inequalities come closest to holding at a large r . Numerical exploration suggests the comparisons responsible; exact equations then define a candidate threshold r_* . This subsection defines that candidate. The next one proves that no other assignment satisfies Σ_r above it. The exploratory problem is

$$\begin{aligned} & \underset{r, \boldsymbol{\theta}}{\text{maximize}} \quad r \\ & \text{subject to} \quad 3/10 \leq r \leq 1/3, \quad \Sigma_r(\boldsymbol{\theta}). \end{aligned} \tag{52}$$

We explore (52) from several initial points using SciPy’s SLSQP routine [23, 24]. The search suggests a threshold near 0.30954 and identifies comparisons close to equality; Appendix D records the script and these observations. They guide the following algebraic definition, but the numerical search is not used to prove the bound.

Near the numerical candidate, two second-round comparisons equal r , the payoff bound h_1 reaches its upper limit of one, and both a selected mixing comparison and the comparison linking the two SimpleMixing rounds become equalities. The first three observations let us express α, γ, ξ using fewer variables. The remaining two equalities, plus a condition identifying the top of their local solution curve, define r_* exactly.

First consider the two second-round comparisons at the profile (y_1^1, y_1^2) formed by the auxiliary strategies returned in the first round. They are (41) after the projection and relaxation:

$$H = \alpha S + (1 - \alpha)T \geq r, \quad K = (1 - \gamma)S + \gamma T \geq r. \tag{53}$$

Exchanging the players if necessary, we describe the numerical candidate in the orientation $S > r > T$. Both $H - r$ and $K - r$ are close to zero. We therefore impose $H = r$ and $K = r$ in the local algebraic calculation. Solving these two equations determines α and γ . Denote the resulting value of α by A_0 :

$$A_0 = \frac{r - T}{S - T}, \quad \alpha = A_0, \quad \gamma = 1 - A_0. \tag{54}$$

The denominator is positive because $S > r > T$. Thus A_0 is the coefficient that makes the weighted average of S and T equal to r .

Another observed equality determines the first-round weight ξ . Recall the upper-bound function from (51):

$$h_1 = \min \left\{ 1, 1 - \frac{r(1 + \sigma)}{\rho} + \frac{\sigma(1 - r)}{\rho} v \right\}, \quad \sigma = 1 - \rho, \quad v = \frac{1 - \xi}{\xi}. \tag{55}$$

The numerical points approach the place where this minimum changes which formula it uses. Equating the second expression to one and solving for v gives that formula-change value, denoted by v_0 :

$$v_0 = \frac{r(1+\sigma)}{\sigma(1-r)}, \quad v = v_0, \quad \xi = \frac{\sigma(1-r)}{\sigma+r}. \quad (56)$$

On the line $v = v_0$, $h_1 = 1$, and ξ is determined by r, ρ . The same numerical points have $h_2 < 1$. For the local calculation we therefore use the second expression in the minimum defining h_2 :

$$h_2 = 1 - \frac{r(1+\rho)}{\sigma} + \frac{\rho(1-r)}{\sigma}u, \quad u = \frac{1-\eta}{\eta}. \quad (57)$$

The formulas for s_1, s_2, S, T remain those in (51), with these substitutions.

We have now expressed ξ, α, γ in terms of (r, ρ, η) . Two further equalities appear in the numerical outputs. One is $E_1 = 0$ in the shared comparison (43), after the projection and relaxation. The other comes from the segment $p = p_\dagger$, $0 \leq q \leq q_\dagger$ in (46): the two affine regret upper bounds intersect at value r . By (48), this intersection equality is $\mathcal{D}(r) = 0$. Substituting (54)–(57) into these two equalities leaves rational equations in (r, ρ, η) . Clearing their positive denominators gives polynomial equations $N_1(r, \rho, \eta) = 0$ and $N_2(r, \rho, \eta) = 0$: N_1 comes from the segment intersection, and N_2 from $E_1 = 0$. Equation (173) in Appendix D specifies the positive factors that fix their exact normalization.

The equations $N_1 = N_2 = 0$ define a curve near the candidate in coordinates (r, ρ, η) , with linearly independent constraint gradients. We want its locally largest value of r . At such a point, the Lagrange multiplier condition [25] gives real multipliers ν_1, ν_2 such that

$$\nabla r = (1, 0, 0) = \nu_1 \nabla N_1 + \nu_2 \nabla N_2. \quad (58)$$

The vector $\nabla N_1 \times \nabla N_2$ is orthogonal to both constraint gradients. Taking its dot product with both sides of (58) sets the vector's r -coordinate to zero, giving the third equation

$$N_1 = N_2 = 0, \quad H_{\text{root}} = (N_1)_\rho (N_2)_\eta - (N_1)_\eta (N_2)_\rho = 0, \quad (59)$$

where subscripts indicate partial derivatives. We define (r_*, ρ_*, η_*) as the unique solution of (59) in the rational isolating box (176). Appendix D gives the polynomials and the checker for this root. Here an exact machine check uses rational interval arithmetic and the Krawczyk method [26, 27] to prove existence and uniqueness in that box, and gives

$$(r_*, \rho_*, \eta_*) \approx (0.3095399654, 0.4799488361, 0.5461144474), \quad 0.30953996 < r_* < 0.30953997.$$

This certifies the local algebraic root suggested by the equalities seen in the search. It does not yet bound the algorithm: the inequalities Σ_r could have a solution far from this root. We must exclude every such solution for $r > r_*$.

5.4 Ruling out every failure above the threshold

Fix $r_* < r \leq 1/3$. The previous subsection identified r_* from equalities near one candidate assignment. We must now rule out *every* assignment satisfying Σ_r , including ones far from that candidate. First, we show that any such assignment would still satisfy the inequalities needed below after setting the two second-round coefficients to the values that balance S and T . That leaves ρ, η, ξ . Next, we move their values, while preserving the needed inequalities, to the line $v = v_0$ where the formula

for h_1 changes. On that line, the comparison linking the two SimpleMixing rounds conflicts with one of two mixing tests. Thus

$$\Sigma_r \text{ has a solution} \implies \text{the necessary inequalities hold on } v = v_0 \implies \text{contradiction.}$$

The algebraic root r_* enters in the final exclusion on that line; the preceding reductions cover the entire domain.

Recall $\sigma = 1 - \rho$ and the relative weights $u = (1 - \eta)/\eta$, $v = (1 - \xi)/\xi$ from (35). The bounds on η, ξ in Σ_r become

$$k = \frac{r}{1-r}, \quad k \leq u, v \leq 1/k. \quad (60)$$

The moves below change numerical values satisfying necessary inequalities; they need not correspond to new games. Each move preserves the inequalities used at its destination. If no destination is possible, no starting assignment is possible.

Removing the two second-round coefficients. The comparisons $H, K \geq r$ in (53) are convex combinations of S, T , so at least one of S, T is at least r . The payoff and segment conditions exclude $S, T \geq r$ simultaneously. If $S = r > T$, then $H < r$ because $\alpha < 1$; the symmetric case contradicts $K \geq r$. Hence, exchanging both players' labels if necessary, we can arrange $S > r > T$. This exchange also exchanges ρ, σ , u, v , and α, γ ; it leaves ρ free to range over $[r, 1 - r]$. The exclusion of $S, T \geq r$ uses exact machine checks of polynomial signs together with the payoff comparisons. Appendix E.1 gives the complete argument, including the bound $s_1, s_2 < r$, and identifies the checkers at the sign conditions they verify.

In this orientation, $H, K \geq r$ restrict the two second-round weights to opposite sides of the balancing value A_0 defined in (54). Define the positive distances from the threshold by

$$m = S - r > 0, \quad n = r - T > 0, \quad A_0 = \frac{n}{m+n} = \frac{r-T}{S-T}. \quad (61)$$

Then

$$H - r = \alpha(m+n) - n, \quad K - r = m - \gamma(m+n), \quad \alpha \geq A_0, \quad \gamma \leq 1 - A_0. \quad (62)$$

The numerical discovery used equality in these two comparisons. For the global proof, we must justify replacing arbitrary allowed α, γ by those balancing values while retaining the inequalities needed for the contradiction. The next lemma does so.

Lemma 6 (Eliminating the second-round coefficients). *For $r_* < r \leq 1/3$, any solution of Σ_r can, after exchanging both players' labels if necessary, be changed to an assignment with*

$$S > r > T, \quad \alpha = A_0, \quad \gamma = 1 - A_0, \quad 0 < A_0 \leq 1 - r.$$

Changing only α, γ preserves the comparison $U \geq r$ from (41) and the shared comparison $E_1 \geq 0$ from (43), with the projected functions of (51). The six payoff bounds in (51) and the four segment conditions in (46) are unchanged.

Proof idea. We present only the main proof ideas here; the complete proof of Lemma 6 is given in Appendix E.2. We first lower α to A_0 . Differentiating the projected expression in (43) gives

$$\partial_\alpha E_1 = (1 - \gamma)(S - T) + \gamma(1 + T - h_2/\xi) - r < 0 \quad (r \leq \gamma \leq 1 - A_0). \quad (63)$$

To prove the sign, it suffices to check the two endpoints in γ , since this derivative is affine in γ . Here exact machine checks establish the endpoint signs and the bound $h_2 > \xi \geq r$, using the original nonnegative payoff bounds, $H \geq r$, and $\alpha \leq 1 - r$. Appendix E.2 specifies the domains and code paths for these checks. Thus lowering α increases E_1 . It also increases $U = \alpha s_1 + (1 - \alpha)h_2$, since $s_1 < r < h_2$. The inequality $A_0 \leq \alpha \leq 1 - r$ gives the stated upper bound on A_0 .

With $\alpha = A_0$, we have $H = r$ and

$$E_1(A_0, 0) = -A_0 r < 0.$$

The function $E_1(A_0, \gamma)$ is affine in γ and nonnegative at the current positive γ . Its slope is therefore positive. Raising γ to $1 - A_0$ preserves $E_1 \geq 0$ and leaves U unchanged. The payoff functions and mixing comparisons are independent of these two weights. $\square$

Lemma 6 has moved any putative solution to the balancing values of α, γ . Only three comparisons will be needed in the remaining reduction: $U \geq r$, the bound $A_0 \leq 1 - r$, and the shared comparison $E_1 \geq 0$. We rewrite them so their signs can be tracked as the other variables move. Let $z = n/m > 0$ be the ratio of the two distances from r , and set $c = 1 - 2r > 0$. Multiply $U(A_0) - r \geq 0$ and $1 - r - A_0 \geq 0$ by $m + n > 0$, and divide $E_1(A_0, 1 - A_0) \geq 0$ by $(1 - A_0)^2 > 0$. Define the resulting nonnegative expressions by

$$\begin{aligned} G &= (m + n)[U(A_0) - r] = m(h_2 - r) - n(r - s_1) \geq 0, \\ B_{\text{mass}} &= (m + n)[1 - r - A_0] = (1 - r)m - rn \geq 0, \\ E &= \frac{E_1(A_0, 1 - A_0)}{(1 - A_0)^2} = v(h_2 - r) + z[c + v(s_1 - 2r)] - rz^2 \geq 0. \end{aligned} \tag{64}$$

Thus $G \geq 0$ is $U \geq r$ rewritten, $B_{\text{mass}} \geq 0$ is $A_0 \leq 1 - r$ rewritten, and $E \geq 0$ is the shared comparison rewritten. We use these three inequalities in the paths below; the other weighted comparisons are not needed there.

Selecting the mixing tests for the boundary paths. After eliminating α, γ , we need mixing conditions that remain useful as u, v change. The four segment conditions (46) are still available. We use the two segments joining $(p_\dagger, q_\dagger)$ to $(p_\dagger, 0)$ and $(0, q_\dagger)$. At the common endpoint $(p_\dagger, q_\dagger)$, we first need to know which player's regret upper bound can exceed r . The factor D_{mix} below is positive, so multiplying by it does not change signs; N_R, N_C are the resulting differences from r :

$$\begin{aligned} D_{\text{mix}} &= (\rho u + \sigma)(\sigma v + \rho) > 0, \\ N_R &= D_{\text{mix}}[R_r(p_\dagger, q_\dagger) - r], \quad N_C = D_{\text{mix}}[C_r(p_\dagger, q_\dagger) - r]. \end{aligned} \tag{65}$$

To keep the two line-segment tests short, write $\chi_1 = 1 - r - r/\rho$ and $\chi_2 = 1 - r - r/\sigma$, the coefficients in $s_1 = \chi_1 + \chi_2 u$ and $s_2 = \chi_2 + \chi_1 v$. Along either selected segment, the two regret upper bounds are affine and may cross. After fixing the endpoint signs, the next expressions test whether such a crossing can stay at or above r ; positive factors have been removed without changing their signs:

$$L = (\chi_1 + cu)N_R + ruN_C, \quad B_{\text{mix}} = (\chi_2 + cv)N_C + rvN_R. \tag{66}$$

Here L corresponds to the segment toward $(p_\dagger, 0)$ and B_{mix} to the segment toward $(0, q_\dagger)$. The endpoint and intersection conditions imply

$$(N_R \geq 0 \wedge L \geq 0) \quad \text{or} \quad (N_R < 0 \leq N_C \wedge B_{\text{mix}} \geq 0). \tag{67}$$

If $N_R \geq 0$, the first segment supplies $L \geq 0$; if $N_R < 0$, the central comparison requires $N_C \geq 0$, and the second segment supplies $B_{\text{mix}} \geq 0$. Appendix E.1 expands these substitutions and checks the endpoint signs. The paths use the weaker first alternative $L \geq 0$ alone. The second alternative continues to require $N_R < 0 \leq N_C$.

We next want one fixed formula for h_2 , so that only the change in h_1 separates the cases in the path argument. Exact machine checks combine the coefficient bound $B_{\text{mass}} \geq 0$ with the applicable mixing condition to exclude $u \geq 1$ on the full weight range $r \leq \rho \leq 1 - r$. Appendix E.2 gives the corresponding polynomial signs and their checkers. This also determines the formula for h_2 . By (51), reaching its cap would require

$$u \geq \frac{r(1 + \rho)}{\rho(1 - r)} > 1.$$

Hence h_2 always uses the second expression in its minimum in the reduced problem. The only remaining change of formula is that of h_1 : it uses the affine expression in (55) for $v < v_0$, and equals one for $v \geq v_0$.

Any solution of Σ_r would therefore give values (r, ρ, u, v) satisfying the following shorter list. We keep only the payoff signs needed by the paths below; dropping other conditions admits more assignments, so excluding this list still excludes every failed run.

$$\begin{aligned} r_* < r \leq 1/3, \quad r \leq \rho \leq 1 - r, \quad k \leq u < 1, \quad k \leq v \leq 1/k, \quad S > r > T, \\ T, s_2 \geq 0, \quad (\rho \leq 1/2 \implies s_1 \geq 0), \quad G, B_{\text{mass}}, E \geq 0, \\ L \geq 0 \quad \text{or} \quad (N_R < 0 \leq N_C \wedge B_{\text{mix}} \geq 0). \end{aligned} \tag{68}$$

The functions are those defined in (51), (64), and (65)–(66). For $\rho > 1/2$, allowing s_1 to become negative enlarges the list of necessary conditions and permits the paths used next.

Moving solutions to where h_1 changes formula. The reduced conditions now involve (r, ρ, u, v) . For each fixed pair (r, ρ) , the remaining freedom is in u, v . The two formulas for h_1 suggest two paths. From $v \geq v_0$, we can try decreasing v directly to the switch. From $v < v_0$, we first increase u until a mixing condition becomes an equality, then use that equality to guide a continuation toward v_0 . To justify these moves, we must preserve the inequalities in (68) and exclude every possible stop before reaching the destination. The next proposition records the resulting reduction.

Proposition 7 (Boundary reduction). *Any point satisfying (68) produces a point satisfying the same conditions on $v = v_0$, with r, ρ unchanged.*

Proof idea. We present only the main proof ideas here; the complete proof of Proposition 7 is given in Appendix E.3. Suppose first that $v \geq v_0$. Keep u fixed and decrease v . Exact machine checks establish the derivative signs that make $z = (r - T)/(S - r)$ decrease during this move. Since s_1, h_2 are fixed, the identities

$$G = m[(h_2 - r) - z(r - s_1)], \quad B_{\text{mass}} = m[(1 - r) - rz]$$

show that $G, B_{\text{mass}} \geq 0$ persist while $m > 0$. The same machine check establishes the sign needed at a zero of the shared comparison:

$$E = 0 \implies -\partial_v E > 0.$$

This excludes a crossing from $E \geq 0$ to $E < 0$ as v decreases. Further exact machine checks verify that the applicable determinant increases and that $T < r$. Together with the decrease of z , these

signs exclude the possible stops $S = r$ and $T = r$. The payoff signs are preserved along this move. Appendix E.3 gives these checks, their domains, and the remaining stopping arguments. Thus this path reaches v_0 .

Suppose now that $v < v_0$. Keep v fixed and increase u . Here exact machine checks of the derivatives show that z decreases and $(h_2 - r)/(r - s_1)$ increases. These signs preserve $G, B_{\text{mass}} \geq 0$ by the same two identities. The same derivative checks give

$$E = 0 \implies \partial_u E > 0,$$

so it also remains nonnegative. The applicable mixing inequality can become tight during this move. The other stopping events, including $u = 1$, are excluded by the selected comparisons. The path therefore reaches $L = 0$ or $B_{\text{mix}} = 0$.

If $\rho \leq 1/2$, the payoff conditions force $N_C < 0$ before the switch, so the contact is on $L = 0$. At the contact used for continuation, an exact subdivision check verifies $L_u < 0$, $L_v < 0$, and the zero-crossing sign in the following display. The implicit-function theorem [33] then gives a boundary curve with

$$\frac{du}{dv} = -\frac{L_v}{L_u} < 0, \quad E = 0 \implies \frac{dE}{dv} = E_v - E_u \frac{L_v}{L_u} > 0. \quad (69)$$

Increasing v along this curve decreases u and preserves $E \geq 0$. The same subdivision check, combined with the stopping arguments, allows the curve to continue to v_0 . If $\rho \geq 1/2$, two separate exact subdivision checks exclude the contact systems $L = 0$ and $B_{\text{mix}} = 0$ under (68). Consequently, no starting point with $v < v_0$ and $\rho \geq 1/2$ can satisfy these reduced conditions.

Each of these three subdivision checks verifies exact rational interval or Bernstein bounds [26, 31] on every terminal box and checks that the boxes cover its stated domain. Appendix E.3 gives the path selection, stopping arguments, and the code paths for each check. $\square$

Ruling out the switch boundary. The path reduction leaves $v = v_0$, so ξ is determined by r, ρ as in (56). We return to $\eta = 1/(1+u) > 1/2$ and work in (r, ρ, η) . Here $h_1 = 1$, and h_2 uses precisely the formula (57) used to define the local root. This is why the switch was chosen: the polynomials from Section 5.3 now describe necessary inequalities. Clearing their positive denominators gives

$$L \geq 0 \iff N_1(r, \rho, \eta) \geq 0, \quad E \geq 0 \iff N_2(r, \rho, \eta) \geq 0. \quad (70)$$

For the first mixing alternative, we must exclude simultaneous $N_1, N_2 \geq 0$; the second alternative replaces N_1 by the polynomial for its other segment. In each case, we find the first value of η allowed by the mixing condition, show that the shared condition is already negative there, and prove it cannot recover as η increases. Thus the first branch reuses the local equations defining r_* ; the second uses the same shared comparison with a different mixing test.

Lemma 8 (Exclusion on the switch boundary). *The necessary conditions (68) have no point on $v = v_0$ with $r_* < r \leq 1/3$.*

Proof idea. We present only the main proof ideas here; the complete proof of Lemma 8 is given in Appendix E.4. First suppose $L \geq 0$, so $N_1, N_2 \geq 0$ are necessary. The polynomial N_1 is a strictly concave quadratic in η . Exact signs exclude parts of the (r, ρ) domain where this quadratic is negative throughout $1/2 < \eta \leq 1 - r$. Whenever $N_1 \geq 0$ is possible, the quadratic has a smaller real root $\eta_-(r, \rho) > 1/2$, and $N_1 \geq 0$ requires $\eta \geq \eta_-$. We will prove $N_2 < 0$ at this mixing boundary and propagate that sign to any proposed feasible value of η .

To prove $N_2 < 0$ at η_- , we first rule out a zero there. The resultant of N_1 and N_2 with respect to η is a polynomial in r, ρ that must vanish if they have a common η -root [28]. Define its relevant factor by

$$\text{Res}_\eta(N_1, N_2) = \sigma^6 \rho^2 r^2 R_{\text{resultant}}(r, \rho). \quad (71)$$

The prefactor is positive. A zero of N_2 at η_- would be a common root of N_1, N_2 and would force $R_{\text{resultant}} = 0$. We therefore seek a strict sign for this factor above r_* . Once common zeros are excluded, one negative reference value and continuity will fix the sign of N_2 along the whole boundary.

The extremum equation (59), together with the regularity check at the isolated root, yields a double factor at $r = r_*$:

$$R_{\text{resultant}}(r_*, \rho) = (\rho - \rho_*)^2 W(\rho), \quad W(\rho) < 0 \quad (r_* \leq \rho \leq 1 - r_*). \quad (72)$$

This identity defines the univariate polynomial W . An exact machine check verifies the regularity condition used for this factorization and certifies $W < 0$ using its Bernstein coefficients and the rational isolating intervals for r_*, ρ_* . The same checker establishes the preliminary exclusions restricting possible points with $N_1 \geq 0$ to $r < 8/25$ and $\rho > 21/50$. On the remaining parameter range, it verifies $\partial_r R_{\text{resultant}} < 0$. Hence increasing r from r_* gives $R_{\text{resultant}}(r, \rho) < 0$ at every required pair with $r > r_*$, and excludes a common root.

Because the resultant is nonzero on the relevant region, N_2 cannot cross zero along the smaller-root boundary $\eta_-(r, \rho)$. An exact check gives $N_2(r_*, 1/2, \eta_-(r_*, 1/2)) < 0$. The connection paths in Appendix E.4 link every relevant smaller-root point to this reference within the real-root domain, including cases of coincident quadratic roots. Continuity therefore gives $N_2 < 0$ at every such η_- .

We must still exclude the inequalities at larger η . At fixed (r, ρ, v_0) , consider the shared residual $E_1(A_0, 1 - A_0) = (1 - A_0)^2 E$, with all payoff functions and A_0 evaluated at the varying η . Exact machine checks provide the derivative signs that keep $S > r > T$, $0 < A_0 \leq 1 - r$, and $h_2 > r$ on the interval from η_- to a proposed feasible η . At a zero of the shared residual, the chain rule and the derivative of its quadratic expression then give

$$E_1(A_0, 1 - A_0) = 0 \implies \frac{d}{d\eta} E_1(A_0, 1 - A_0) < 0. \quad (73)$$

By (70), its starting value at η_- is negative. The derivative sign prevents it from crossing to a nonnegative value as η increases. This excludes $N_1, N_2 \geq 0$ simultaneously.

Finally suppose $N_R < 0 \leq N_C$ and $B_{\text{mix}} \geq 0$. On the switch, this alternative can occur only for $\rho > 1/2$; the payoff condition $T < r$ excludes $N_C \geq 0$ on the other half. Define the polynomial for this segment by substituting $u = (1 - \eta)/\eta$ and $v = v_0$ into

$$N_B(r, \rho, \eta) = \sigma^2 \eta (1 - r)^2 B_{\text{mix}}. \quad (74)$$

The multiplier is positive, so this alternative requires $N_B, N_2 \geq 0$. The polynomial N_B is affine and strictly increasing in η on $1/2 \leq \rho \leq 1 - r$. Its unique zero, denoted $\eta_B(r, \rho)$, is above $1/2$, so $N_B \geq 0$ implies $\eta \geq \eta_B$. Substitution at this zero gives $N_2(r, \rho, \eta_B) < 0$ throughout the stated domain. Here an exact machine check verifies the strictly negative Bernstein coefficients after substitution. The interval conditions and the same zero derivative (73) then exclude a nonnegative shared residual at every larger feasible η . Both mixing alternatives are therefore excluded. Appendix E.4 specifies the sign domains, the reference connection paths, and the checker paths beside the resultant and N_B calculations. $\square$

Proposition 5 maps an output above r to a solution of Σ_r . Lemma 6 reduces such a solution to (68), Proposition 7 moves it to $v = v_0$, and Lemma 8 excludes that boundary. These implications cover every point with $r_* < r \leq 1/3$ and give the following conclusion.

Proposition 9 (Global arithmetic bound). *For every r with $r_* < r \leq 1/3$, no five values satisfy the necessary inequalities:*

$$\nexists \theta, \quad \Sigma_r(\theta).$$

Consequently, the ideal building-block properties imply $f(\mathbf{x}_{\text{out}}) \leq r_*$.

The last conclusion follows by choosing a comparison threshold strictly between r_* and a hypothetical larger output regret, with the threshold below $1/3$. The exact checks used in these reductions verify specified arithmetic identities, signs, and coverage; the implication from an algorithm failure to the numerical inequalities follows from the comparisons in Section 5.2. All proof computations are offline and are absent from Algorithm 2. The constant is an upper bound for the algorithm. An exact machine check also verifies that the necessary inequalities can hold with equality at r_* ; Appendix D identifies that check. This alone does not show the algorithm attains regret r_* : that would require a game realizing those numerical values. Section 5.5 now restores the approximation errors.

5.5 Restoring the approximation errors

The preceding subsection proved that Σ_r has no solution for $r_* < r \leq 1/3$ and hence established the ideal guarantee. The actual algorithm uses accuracy δ_f for **StationaryPoint** and τ for **OptimalMixing**, as specified in Section 4.4. To prove the actual guarantee, suppose the output exceeds $r + \delta_f + \tau$. The **OptimalMixing** guarantee then makes every tested mixture exceed $r + \delta_f$. We next account for the stationary comparison and its nearly best-response strategies using a single δ_f margin. This recovers the same necessary inequalities Σ_r , contradicting the preceding subsection. The four **SimpleMixing** LPs are solved exactly and add no error. The next lemma states this transfer.

Lemma 10 (Transfer to approximate building blocks). *For the actual accuracy parameters $\delta_f, \tau > 0$, an execution with the building-block properties of Section 5.1 satisfies*

$$\forall r \in (r_*, 1/3), \quad f(\mathbf{x}_{\text{out}}) > r + \delta_f + \tau \implies \exists \theta, \Sigma_r(\theta).$$

Proof idea. We present only the main proof ideas here; the complete proof of Lemma 10 is given in Appendix F. Fix $r \in (r_*, 1/3)$ and assume the stated failed-output threshold. Output selection (26) makes every candidate exceed $r + \delta_f + \tau$. The comparison of **OptimalMixing** in (24) gives $f(\mathbf{x}_o) \leq f(\mathbf{x}(p, q)) + \tau$ for every $p, q \in [0, 1]$. Therefore

$$\forall p, q \in [0, 1], \quad f(\mathbf{x}(p, q)) > r + \delta_f. \tag{75}$$

This is the only subtraction of τ . All four **SimpleMixing** LPs are exact, so their comparison inequalities and complementary slackness introduce no additional accuracy term.

The stationary comparison guarantees only $g - \delta_f$ where the ideal proof used g . To reuse the earlier inequalities, call this conservative value G_s :

$$G_s = g - \delta_f > r. \tag{76}$$

We use this same value in every subsequent comparison. The comparisons at (x_s^1, y_s^2) and (y_s^1, x_s^2) give $G_s \leq \rho, \sigma$, so both stationary coefficients are above r . The near-best-response error also fits

within the same surplus. For the row player, let τ_1 measure how far y_s^1 is from being a best response to x_s^2 ; it is at most δ_f and is unrelated to the **OptimalMixing** accuracy τ . Also let d be the row regret after replacing y_s^1 by y_1^1 :

$$\tau_1 = f_1(y_s^1, x_s^2) = B_1(x_s^2) - u_1(y_s^1, x_s^2), \quad 0 \leq \tau_1 \leq \delta_f, \quad d = f_1(y_1^1, x_s^2) \geq 0. \quad (77)$$

Applying the comparison inequality (20) of **SimpleMixing**(y_s^1) at a column best response to y_s^1 gives $F_1^1 \leq \eta$, hence $\eta > r$. Applying it at x_s^2 instead gives

$$\begin{aligned} r + \delta_f &< F_1^1 \leq \eta(\tau_1 - d) + (1 - \eta), \\ \eta d &< 1 - \eta - r - \delta_f + \eta\tau_1 \leq 1 - \eta - r. \end{aligned} \quad (78)$$

Thus $\eta < 1 - r$ and $d \leq \min\{1, (1 - \eta - r)/\eta\}$, the same regret bound used in (50). The error has canceled before taking the minimum with one. An exact machine check reconstructs this cancellation identity and both cases of the minimum. The symmetric comparisons give $r < \xi < 1 - r$ and the corresponding column-regret bound. The auxiliary strategies from the first round are then exact best responses to those calls' chosen strategies; the second-round comparisons give $r < \alpha, \gamma < 1 - r$ as in the ideal proof.

Using the stationary value G_s and these regret bounds, the six payoff bounds are now evaluated at

$$\lambda_s = (r, G_s, \rho, \eta, \xi). \quad (79)$$

The same applications of the comparison inequalities from Section 5.2 give their nonnegativity and the weighted and shared comparisons (41) and (43), with $g = G_s$. The joint regret comparisons used for the Jensen-gap bounds have additional nonnegative terms proportional to τ_1 and its column counterpart. Dropping these terms and applying the weighted minimum argument of Lemma 3 gives the row lower bound $j_r(\lambda_s, q)$ and the symmetric column bound, both evaluated at $g = G_s$.

It remains to transfer the segment conditions. In the row-regret interpolation (10), the stationary corner contributes $g = G_s + \delta_f$ and the corner (y_s^1, x_s^2) contributes τ_1 . Their total error is bounded by

$$\delta_f(1 - p)(1 - q) + \tau_1 p(1 - q) \leq \delta_f(1 - q).$$

The column calculation is symmetric. With the Jensen-gap lower bounds just obtained, this gives

$$\begin{aligned} \forall p, q \in [0, 1], \quad f_1(\mathbf{x}(p, q)) &\leq R_{G_s}(p, q) + \delta_f(1 - q), \\ \forall p, q \in [0, 1], \quad f_2(\mathbf{x}(p, q)) &\leq C_{G_s}(p, q) + \delta_f(1 - p). \end{aligned} \quad (80)$$

Here R_{G_s}, C_{G_s} are R_g, C_g with g replaced by G_s . The same machine check reconstructs these error identities and the nonnegative combinations giving the payoff and regret bounds. If their maximum were at most r at any (p, q) , both actual regrets would be at most $r + \delta_f$, contradicting (75). The four segment conditions (46) therefore hold at this same adjusted parameter.

The first-round comparisons (47), with $g = G_s$, give the interval needed for the substitution in Lemma 4. Its preservation argument uses only the coefficient bounds, nonnegativity, weighted and shared comparisons, segment conditions, and first-round comparisons established above. It therefore applies to these adjusted numerical values. Substituting $G_s = r$ and then using the payoff bounds (51) produces Σ_r . Appendix F identifies the checker path for these error calculations. $\square$

For every $r \in (r_*, 1/3)$, Proposition 9 excludes the solution supplied by Lemma 10. If the output regret exceeded $r_* + \delta_f + \tau$, we could choose

$$r_* < r < \min\{f(\mathbf{x}_{\text{out}}) - \delta_f - \tau, 1/3\},$$

contradicting that exclusion. Hence

$$f(\mathbf{x}_{\text{out}}) \leq r_* + \delta_f + \tau. \quad (81)$$

Taking $\delta_f = \tau = \delta/2$ gives the claimed $(r_* + \delta)$ approximation and proves Theorem 1; the running time follows from the fixed composition in Section 4.4.

6 Discussion

The present work extends the role of LLM agents in the LegoNE methodology. In the reported LegoNE discovery runs, the programming language, building-block specifications, and compilation rules were fixed before the search [16, 15]; the LLM agent explored compositions within that space. Here, an LLM agent can also develop the search space: identify or propose building blocks, interleave instantiation and forgetting as the argument evolves, and choose proof methods and write suitable compilation or verification routines. The mathematical abstractions and verification methods can therefore evolve together with the algorithm.

Codex played a central role in discovering the algorithm and developing its proof, including the specialized routines and exact arithmetic checkers. In our experience, LLM agents appear indispensable for the present improvement and may remain so for future advances in approximate Nash equilibrium algorithms: the proof complexity already exceeds what we can reliably handle through unaided human reasoning.

An open problem is a polynomial-time $(c + \delta)$ -approximation algorithm for bimatrix games with a constant $c < 0.3$ and every $\delta > 0$. We explored this target with Codex, but prolonged work did not produce a certified bound: the growing complexity of the proof became the main obstacle. Progress may require new building blocks or mathematical abstractions that make the analysis simpler.

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References

- [1] John Nash. Non-cooperative games. *Annals of Mathematics*, 54(2):286–295, 1951. doi:10.2307/1969529.
- [2] Constantinos Daskalakis, Paul W. Goldberg, and Christos H. Papadimitriou. The complexity of computing a Nash equilibrium. *SIAM Journal on Computing*, 39(1):195–259, 2009. doi:10.1137/070699652.
- [3] Xi Chen, Xiaotie Deng, and Shang-Hua Teng. Settling the complexity of computing two-player Nash equilibria. *Journal of the ACM*, 56(3):14:1–14:57, 2009. doi:10.1145/1516512.1516516.

- [4] Richard J. Lipton, Evangelos Markakis, and Aranyak Mehta. Playing large games using simple strategies. In *Proceedings of the 4th ACM Conference on Electronic Commerce*, pages 36–41, 2003. doi:10.1145/779928.779933.
- [5] Yakov Babichenko, Siddharth Barman, and Ron Peretz. Empirical distribution of equilibrium play and its testing application. *Mathematics of Operations Research*, 42(1):15–29, 2017. doi:10.1287/moor.2016.0794.
- [6] Aviad Rubinstein. Settling the complexity of computing approximate two-player Nash equilibria. In *Proceedings of the 57th IEEE Symposium on Foundations of Computer Science*, pages 258–265, 2016. doi:10.1109/FOCS.2016.35.
- [7] Hanyu Li, Wenhan Huang, Zhijian Duan, David Henry Mguni, Kun Shao, Jun Wang, and Xiaotie Deng. A survey on algorithms for Nash equilibria in finite normal-form games. *Computer Science Review*, 51:100613, 2024. doi:10.1016/j.cosrev.2023.100613.
- [8] Spyros C. Kontogiannis, Panagiota N. Panagopoulou, and Paul G. Spirakis. Polynomial algorithms for approximating Nash equilibria of bimatrix games. *Theoretical Computer Science*, 410(17):1599–1606, 2009. doi:10.1016/j.tcs.2008.12.033.
- [9] Constantinos Daskalakis, Aranyak Mehta, and Christos H. Papadimitriou. A note on approximate Nash equilibria. *Theoretical Computer Science*, 410(17):1581–1588, 2009. doi:10.1016/j.tcs.2008.12.031.
- [10] Constantinos Daskalakis, Aranyak Mehta, and Christos H. Papadimitriou. Progress in approximate Nash equilibria. In *Proceedings of the 8th ACM Conference on Electronic Commerce*, pages 355–358, 2007. doi:10.1145/1250910.1250962.
- [11] Hartwig Bosse, Jaroslaw Byrka, and Evangelos Markakis. New algorithms for approximate Nash equilibria in bimatrix games. *Theoretical Computer Science*, 411(1):164–173, 2010. doi:10.1016/j.tcs.2009.09.023.
- [12] Haralampos Tsaknakis and Paul G. Spirakis. An optimization approach for approximate Nash equilibria. *Internet Mathematics*, 5(4):365–382, 2008. doi:10.1080/15427951.2008.10129172.
- [13] Argyrios Deligkas, Michail Fasoulakis, and Evangelos Markakis. A polynomial-time algorithm for 1/3-approximate Nash equilibria in bimatrix games. *ACM Transactions on Algorithms*, 19(4):31:1–31:17, 2023. doi:10.1145/3606697.
- [14] Zhaohua Chen, Xiaotie Deng, Wenhan Huang, Hanyu Li, and Yuhao Li. On tightness of Tsaknakis–Spirakis descent methods for approximate Nash equilibria. *Information and Computation*, 293:105046, 2023. doi:10.1016/j.ic.2023.105046.
- [15] Xiaotie Deng, Dongchen Li, and Hanyu Li. A computer-aided approach for approximate Nash equilibria. In *Proceedings of the 20th International Conference on Web and Internet Economics (WINE 2024)*, volume 15534 of *Lecture Notes in Computer Science*, pages 314–331, 2026. doi:10.1007/978-3-032-08560-3_18.
- [16] Hanyu Li, Dongchen Li, and Xiaotie Deng. Discovering expert-level Nash equilibrium algorithms with large language models. *Nature Communications*, 17:7168, 2026. doi:10.1038/s41467-026-74003-1.

- [17] Sébastien Hémon, Michel de Rougemont, and Miklos Santha. Approximate Nash equilibria for multi-player games. In *Proceedings of the 1st International Symposium on Algorithmic Game Theory*, volume 4997 of *Lecture Notes in Computer Science*, pages 267–278, 2008. doi:10.1007/978-3-540-79309-0_24.
- [18] Xiaotie Deng, Dongchen Li, and Hanyu Li. Automating approximation analysis for Nash equilibria algorithms in two-player games. *Information and Computation*, 307:105362, 2025. doi:10.1016/j.ic.2025.105362.
- [19] Xiaotie Deng, Dongchen Li, and Hanyu Li. On the optimal mixing problem of approximate Nash equilibria in bimatrix games. *Theoretical Computer Science*, 1031:115072, 2025. doi:10.1016/j.tcs.2025.115072.
- [20] Davit Gondauri. A deterministic polynomial-time approximation below one third for bimatrix games. Preprint, 2026. Zenodo record 23077766.
- [21] Stephen Boyd and Lieven Vandenbergh. *Convex Optimization*. Cambridge University Press, 2004. Author-hosted book and supplementary materials.
- [22] Saugata Basu, Richard Pollack, and Marie-Françoise Roy. *Algorithms in Real Algebraic Geometry*, second edition. Algorithms and Computation in Mathematics, volume 10. Springer, 2006. doi:10.1007/3-540-33099-2.
- [23] Pauli Virtanen, Ralf Gommers, Travis E. Oliphant, et al. SciPy 1.0: fundamental algorithms for scientific computing in Python. *Nature Methods*, 17:261–272, 2020. doi:10.1038/s41592-019-0686-2.
- [24] Dieter Kraft. A software package for sequential quadratic programming. Technical Report DFVLR-FB 88-28, Institut für Dynamik der Flugsysteme, DFVLR, Oberpfaffenhofen, 1988. Technical report.
- [25] Jorge Nocedal and Stephen J. Wright. *Numerical Optimization*, second edition. Springer, 2006. doi:10.1007/978-0-387-40065-5.
- [26] Ramon E. Moore, R. Baker Kearfott, and Michael J. Cloud. *Introduction to Interval Analysis*. Society for Industrial and Applied Mathematics, 2009. doi:10.1137/1.9780898717716.
- [27] R. Krawczyk. Newton-Algorithmen zur Bestimmung von Nullstellen mit Fehlerschranken. *Computing*, 4:187–201, 1969. doi:10.1007/BF02234767.
- [28] David A. Cox, John Little, and Donal O’Shea. *Ideals, Varieties, and Algorithms: An Introduction to Computational Algebraic Geometry and Commutative Algebra*, fourth edition. Springer, 2015. doi:10.1007/978-3-319-16721-3.
- [29] Aaron Meurer, Christopher P. Smith, Mateusz Paprocki, et al. SymPy: symbolic computing in Python. *PeerJ Computer Science*, 3:e103, 2017. doi:10.7717/peerj-cs.103.
- [30] Python Software Foundation. *Python 3 Documentation*. Official documentation; `fractions`—rational numbers. Accessed October 4, 2026.
- [31] Jürgen Garloff. Convergent bounds for the range of multivariate polynomials. In Karl Nickel, editor, *Interval Mathematics 1985*, volume 212 of *Lecture Notes in Computer Science*, pages 37–56. Springer, 1986. doi:10.1007/3-540-16437-5_5.

- [32] Charles R. Harris, K. Jarrod Millman, Stéfan J. van der Walt, et al. Array programming with NumPy. *Nature*, 585:357–362, 2020. doi:10.1038/s41586-020-2649-2.
- [33] Walter Rudin. *Principles of Mathematical Analysis*, third edition. McGraw-Hill, 1976. Publisher’s book page.

A Instantiating the building-block properties

This appendix supplies the strategy instantiations used in Lemma 2 and in the payoff elimination of Section 5.2. We analyze the building-block properties with zero approximation errors, as specified in (28). The accuracy transfer is proved in Appendix F. Fix one game with payoffs in $[0, 1]$, one execution, and one optimal dual solution for each of its five LPs, including the stationary LP. All comparisons below use this same game and these same dual solutions. The stationary-point and mixing encodings are those of [12, 14, 18, 19]; their instantiation follows the LegoNE framework [15, 16] and uses LP optimality and complementary slackness [21].

Use the four candidate profiles $\mathbf{z}_j^i$ and their maximum regrets F_j^i from (29). In these two symbols, the superscript identifies the player whose input is fixed and the subscript identifies the round. In a single strategy x_j^i or y_j^i , the superscript instead identifies the player using that strategy. The four weights $\eta, \xi, \alpha, \gamma$ are the call-specific dual weights defined in (30). For example, η weights the row-deviation gains in the first-round operation $\text{SimpleMixing}(y_s^1)$, and $1 - \eta$ weights its column regret.

To analyze a possible failure of the approximation guarantee, assume that the selected output has maximum regret above r , with $0 < r \leq 1/3$, as in (27). The resulting inequalities (31) say that all four F_j^i exceed r , that $g \geq f(\mathbf{x}_o) > r$, and that every profile $\mathbf{x}(p, q)$ in the original mixing rectangle has maximum regret above r . These assumptions allow comparisons at strategies chosen during the proof, as well as at the returned candidates.

The stationary LP comparison. The joint inequality (17) can be represented by two additive constants from the stationary LP. Write A_s, B_s for lower bounds on its two separate comparison parts. These constants are chosen for the fixed stationary dual solution, independently of the comparison strategies. With zero error, their conditions are

$$\begin{aligned} & \exists A_s, B_s \in \mathbb{R} \text{ such that} \\ & \forall x^{1'} \in \Delta_{n_1}, \quad A_s \leq -\rho u_1(x^{1'}, x_s^2) + \sigma[u_2(x^{1'}, y_s^2) - u_2(x^{1'}, x_s^2)], \\ & \forall x^{2'} \in \Delta_{n_2}, \quad B_s \leq -\sigma u_2(x_s^1, x^{2'}) + \rho[u_1(y_s^1, x^{2'}) - u_1(x_s^1, x^{2'})], \\ & g \leq \rho u_1(\mathbf{x}_s) + \sigma u_2(\mathbf{x}_s) + A_s + B_s. \end{aligned} \tag{82}$$

Such constants exist: minimize each comparison part over its own simplex. The sum of these two minima is the minimum of their sum, since the two comparison strategies are independent; the universal stationarity inequality gives the last line. Conversely, adding the three lines gives (17). The quantified registry records this LP representation. We use the combined inequality directly, eliminating A_s, B_s before forming the scalar necessary system. With approximation error, the last line has $g - \delta_f$ in place of g .

The four LP comparisons. We first record the dual inequalities for the first-round operations $\text{SimpleMixing}(y_s^1)$ and $\text{SimpleMixing}(y_s^2)$ and the second-round operations $\text{SimpleMixing}(y_1^1)$

and $\text{SimpleMixing}(y_1^2)$, so that each subsequent strategy instantiation has an explicit premise. Instantiating (20) with their inputs and outputs in Algorithm 2 and the weight assignments (30) gives

$$\forall x^{2'} \in \Delta_{n_2}, \quad F_1^1 \leq \eta[u_1(y_1^1, x^{2'}) - u_1(y_s^1, x^{2'})] + (1 - \eta)f_2(y_s^1, x^{2'}), \quad (83a)$$

$$\forall x^{1'} \in \Delta_{n_1}, \quad F_1^2 \leq \xi[u_2(x^{1'}, y_1^2) - u_2(x^{1'}, y_s^2)] + (1 - \xi)f_1(x^{1'}, y_s^2), \quad (83b)$$

$$\forall x^{2'} \in \Delta_{n_2}, \quad F_2^1 \leq \alpha[u_1(y_2^1, x^{2'}) - u_1(y_1^1, x^{2'})] + (1 - \alpha)f_2(y_1^1, x^{2'}), \quad (83c)$$

$$\forall x^{1'} \in \Delta_{n_1}, \quad F_2^2 \leq \gamma[u_2(x^{1'}, y_2^2) - u_2(x^{1'}, y_1^2)] + (1 - \gamma)f_1(x^{1'}, y_1^2). \quad (83d)$$

Each inequality holds throughout the stated comparison simplex, with one fixed dual strategy and weight for its specified operation.

Positive weights and best responses. The best-response conclusions (18) and (23) require positive dual weights. We establish that positivity before using the conclusions to compare payoffs for the second-round operations $\text{SimpleMixing}(y_1^1)$ and $\text{SimpleMixing}(y_1^2)$. Recall that ρ is the stationary row weight and $\sigma = 1 - \rho$ is the stationary column weight. Define the two stationary/dual payoff differences by

$$\lambda_{\text{payoff}} = u_1(y_s^1, y_s^2) - u_1(x_s^1, y_s^2), \quad \mu_{\text{payoff}} = u_2(y_s^1, y_s^2) - u_2(y_s^1, x_s^2). \quad (84)$$

Instantiate (17), with $\delta_f = 0$, at the comparison profiles (x_s^1, y_s^2) and (y_s^1, x_s^2) , respectively:

$$\begin{aligned} g &\leq \rho[u_1(y_s^1, y_s^2) - u_1(x_s^1, y_s^2)], \\ g &\leq \sigma[u_2(y_s^1, y_s^2) - u_2(y_s^1, x_s^2)]. \end{aligned} \quad (85)$$

The column bracket is zero in the first instantiation and the row bracket is zero in the second. Each displayed payoff difference is at most one. Since $g > r$ in (31), this gives

$$r < \rho < 1 - r, \quad \rho, \sigma > 0. \quad (86)$$

Thus (18), with zero error, implies

$$B_1(x_s^2) = u_1(y_s^1, x_s^2), \quad B_2(x_s^1) = u_2(x_s^1, y_s^2). \quad (87)$$

The functions B_1, B_2 are the best-response payoff functions defined in Section 4.2. Nonnegativity of $u_1(x_s^1, y_s^2)$ and $u_2(y_s^1, x_s^2)$ in (85) also gives

$$u_1(y_s^1, y_s^2) \geq g/\rho, \quad u_2(y_s^1, y_s^2) \geq g/\sigma. \quad (88)$$

Equivalently, the two payoff differences satisfy the lower bounds

$$\lambda_{\text{payoff}} \geq \lambda_0 = g/\rho, \quad \mu_{\text{payoff}} \geq \mu_0 = g/\sigma. \quad (89)$$

These lower bounds will bound regrets from above, since all best-response payoffs are at most one.

For the first-round operation $\text{SimpleMixing}(y_s^1)$, choose a column best response x_{br}^2 to y_s^1 . It exists because payoffs are continuous and the simplex is compact. Instantiate (83a) with $x^{2'} = x_{\text{br}}^2$. The column regret is zero, so

$$\begin{aligned} f_2(y_s^1, x_{\text{br}}^2) &= 0, \\ F_1^1 &\leq \eta[u_1(y_1^1, x_{\text{br}}^2) - u_1(y_s^1, x_{\text{br}}^2)] \leq \eta. \end{aligned} \quad (90)$$

Next instantiate the same inequality with $x^{2'} = x_s^2$. By (87), its row payoff difference is nonpositive; the column regret is at most one. Hence

$$F_1^1 \leq \eta[u_1(y_1^1, x_s^2) - u_1(y_s^1, x_s^2)] + (1 - \eta)f_2(y_s^1, x_s^2) \leq 1 - \eta. \quad (91)$$

Together with $F_1^1 > r$ in (31), these instantiations give $r < \eta < 1 - r$.

To transfer a best-response conclusion to the second round, instantiate the strong-duality equality (21) with the input y_s^1 , primal output x_1^2 , dual strategy y_1^1 , and weight η of the first-round operation **SimpleMixing**(y_s^1). The regret definition (1) then gives

$$\begin{aligned} F_1^1 &= \eta[u_1(y_1^1, x_1^2) - u_1(y_s^1, x_1^2)] + (1 - \eta)f_2(y_s^1, x_1^2) \\ &\leq \eta f_1(y_s^1, x_1^2) + (1 - \eta)f_2(y_s^1, x_1^2) \leq F_1^1. \end{aligned} \quad (92)$$

Every inequality is an equality. Since $\eta > 0$, y_1^1 attains the row best-response payoff against x_1^2 . Since η and $1 - \eta$ are both positive, both regrets equal the maximum regret:

$$B_1(x_1^2) = u_1(y_1^1, x_1^2), \quad f_1(z_1^1) = f_2(z_1^1) = F_1^1. \quad (93)$$

For the first-round operation **SimpleMixing**(y_s^2), instantiate (83b) at a row best response to y_s^2 ; the row regret vanishes and $F_1^2 \leq \xi$. Instantiate the same inequality with $x^{1'} = x_s^1$; the column payoff difference is nonpositive by (87), so $F_1^2 \leq 1 - \xi$. Apply the strong-duality equality and regret definition at the primal output x_1^1 , now with input y_s^2 , dual strategy y_1^2 , and weight ξ of **SimpleMixing**(y_s^2). This gives

$$r < \xi < 1 - r, \quad B_2(x_1^1) = u_2(x_1^1, y_1^2), \quad f_1(z_1^2) = f_2(z_1^2) = F_1^2. \quad (94)$$

For the second-round operation **SimpleMixing**(y_1^1), instantiate (83c) with $x^{2'} = x_1^2$. Its input y_1^1 is a best response to x_1^2 by (93), giving $F_2^1 \leq 1 - \alpha$. Instantiating the same inequality at a column best response to y_1^1 instead gives $F_2^1 \leq \alpha$. For the second-round operation **SimpleMixing**(y_1^2), instantiate (83d) first with $x^{1'} = x_1^1$ and then at a row best response to y_1^2 . These give $F_2^2 \leq 1 - \gamma$ and $F_2^2 \leq \gamma$, respectively. The failures in (31) therefore make all four weights and their complements positive. Instantiating (21) at the respective primal outputs of **SimpleMixing**(y_1^1) and **SimpleMixing**(y_1^2) and repeating the equality chain (92) gives

$$\begin{aligned} r &< \eta, \xi, \alpha, \gamma < 1 - r, \\ B_1(x_2^2) &= u_1(y_2^1, x_2^2), \quad B_2(x_1^2) = u_2(x_1^2, y_2^2), \\ f_1(z_2^1) &= f_2(z_2^1) = F_2^1, \quad f_1(z_2^2) = f_2(z_2^2) = F_2^2. \end{aligned} \quad (95)$$

Thus the first-round best-response conclusion is proved before it is used to instantiate the dual inequalities of the second-round operations **SimpleMixing**(y_1^1) and **SimpleMixing**(y_1^2).

Lower bounds on payoffs used by the second round. Recall the parameter tuple $\lambda = (r, g, \rho, \eta, \xi)$ from (33). All six payoff-bound functions below are evaluated at this full tuple. Dividing the dual inequalities of the first-round operations **SimpleMixing**(y_s^1) and **SimpleMixing**(y_s^2) by their positive weights repeatedly produces the same ratios. Recall the relative weights u, v from (35). The following display uses exactly these same u, v and records their common lower endpoint:

$$u = \frac{1 - \eta}{\eta}, \quad v = \frac{1 - \xi}{\xi}, \quad k = \frac{r}{1 - r}, \quad k < u, v < 1/k. \quad (96)$$

The inequalities follow from (95). These are scalar ratios; the payoff functions continue to be written as u_1, u_2 .

We seek lower bounds on the payoffs of the first-round dual strategies, because they are the inputs of $\text{SimpleMixing}(y_1^1)$ and $\text{SimpleMixing}(y_1^2)$. Instantiate the dual inequality (83a) of $\text{SimpleMixing}(y_s^1)$ with $x^{2'} = y_s^2$, and use (88):

$$\begin{aligned} r < F_1^1 &\leq \eta[u_1(y_1^1, y_s^2) - u_1(y_s^1, y_s^2)] + (1 - \eta)f_2(y_s^1, y_s^2) \\ &\leq \eta[u_1(y_1^1, y_s^2) - g/\rho] + (1 - \eta)(1 - g/\sigma). \end{aligned} \quad (97)$$

Solving for $u_1(y_1^1, y_s^2)$, and then subtracting from one, gives the first of the following upper-bound functions. The second follows by instantiating the dual inequality (83b) of $\text{SimpleMixing}(y_s^2)$ with $x^{1'} = y_s^1$:

$$\begin{aligned} s_1(\lambda) &= 1 - g/\rho - \frac{r - (1 - \eta)(1 - g/\sigma)}{\eta}, \\ s_2(\lambda) &= 1 - g/\sigma - \frac{r - (1 - \xi)(1 - g/\rho)}{\xi}. \end{aligned} \quad (98)$$

They satisfy

$$\begin{aligned} 1 - u_1(y_1^1, y_s^2) &\leq s_1(\lambda), \quad f_1(y_1^1, y_s^2) \leq s_1(\lambda), \\ 1 - u_2(y_s^1, y_1^2) &\leq s_2(\lambda), \quad f_2(y_s^1, y_1^2) \leq s_2(\lambda). \end{aligned} \quad (99)$$

The bounds on $1 - u_i$ give payoff information as well as regret information; later inequalities will use both.

We also need the row regret of y_1^1 against x_s^2 . Rearrange (91), using $F_1^1 > r$, $f_2 \leq 1$, and (87). This regret is at most $(1 - \eta - r)/\eta$ and also at most one. Instantiating the dual inequality (83b) of $\text{SimpleMixing}(y_s^2)$ with $x^{1'} = x_s^1$ gives the second bound:

$$\begin{aligned} d_1 &= \min\{1, (1 - \eta - r)/\eta\}, \quad B_1(x_s^2) - u_1(y_1^1, x_s^2) \leq d_1, \\ d_2 &= \min\{1, (1 - \xi - r)/\xi\}, \quad B_2(x_s^1) - u_2(x_s^1, y_1^2) \leq d_2. \end{aligned} \quad (100)$$

Both d_i are nonnegative by (95).

To turn these regrets into payoff bounds for the other player, instantiate (17) at (x_s^1, y_1^2) and (y_1^1, x_s^2) . The zero-error comparisons are

$$\begin{aligned} g &\leq \rho[u_1(y_s^1, y_1^2) - u_1(x_s^1, y_1^2)] + \sigma[B_2(x_s^1) - u_2(x_s^1, y_1^2)], \\ g &\leq \rho[B_1(x_s^2) - u_1(y_1^1, x_s^2)] + \sigma[u_2(y_1^1, y_s^2) - u_2(y_1^1, x_s^2)]. \end{aligned} \quad (101)$$

Use (100) and nonnegative payoffs to obtain $u_1(y_s^1, y_1^2) \geq g/\rho - (\sigma/\rho)d_2$ and $u_2(y_1^1, y_s^2) \geq g/\sigma - (\rho/\sigma)d_1$. Together with $u_i \geq 0$, these define

$$\begin{aligned} h_1(\lambda) &= \min\{1, 1 - g/\rho + (\sigma/\rho)d_2\}, \\ h_2(\lambda) &= \min\{1, 1 - g/\sigma + (\rho/\sigma)d_1\}. \end{aligned} \quad (102)$$

Their meanings are

$$1 - u_1(y_s^1, y_1^2) \leq h_1(\lambda), \quad 1 - u_2(y_1^1, y_s^2) \leq h_2(\lambda). \quad (103)$$

Thus $(s_1(\lambda), h_2(\lambda))$ bounds the two regrets at (y_1^1, y_s^2) , and $(h_1(\lambda), s_2(\lambda))$ bounds them at (y_s^1, y_1^2) .

We next compare the dual strategies returned by $\text{SimpleMixing}(y_s^1)$ and $\text{SimpleMixing}(y_s^2)$. Instantiate the dual inequality (83a) of $\text{SimpleMixing}(y_s^1)$ with $x^{2'} = y_1^2$ and the dual inequality (83b) of $\text{SimpleMixing}(y_s^2)$ with $x^{1'} = y_1^1$. Using (31), then the four payoff bounds (99) and (103), gives

$$\begin{aligned} r &\leq \eta[u_1(y_1^1, y_1^2) - u_1(y_s^1, y_1^2)] + (1 - \eta)f_2(y_s^1, y_1^2), \\ r &\leq \xi[u_2(y_1^1, y_1^2) - u_2(y_1^1, y_s^2)] + (1 - \xi)f_1(y_1^1, y_s^2). \end{aligned} \quad (104)$$

Solving for the actual payoffs at (y_1^1, y_1^2) defines

$$S(\boldsymbol{\lambda}) = h_1(\boldsymbol{\lambda}) - r + u(s_2(\boldsymbol{\lambda}) - r), \quad T(\boldsymbol{\lambda}) = h_2(\boldsymbol{\lambda}) - r + v(s_1(\boldsymbol{\lambda}) - r). \quad (105)$$

In particular,

$$1 - u_1(y_1^1, y_1^2) \leq S(\boldsymbol{\lambda}), \quad 1 - u_2(y_1^1, y_1^2) \leq T(\boldsymbol{\lambda}). \quad (106)$$

Here the one-argument notation $T(\boldsymbol{\lambda})$ always denotes the scalar upper-bound function in (105). The four-argument expression used in (17) continues to denote the stationary payoff comparison. The six functions $s_i(\boldsymbol{\lambda}), h_i(\boldsymbol{\lambda}), S(\boldsymbol{\lambda}), T(\boldsymbol{\lambda})$ are nonnegative at the scalar values supplied by the execution, since each bounds a nonnegative quantity of the form $1 - u_i$.

Second-round comparisons at four specified strategies. The preceding six payoff bounds allow four instantiations of the dual inequalities of the second-round operations $\text{SimpleMixing}(y_1^1)$ and $\text{SimpleMixing}(y_1^2)$ to be written without additional payoff variables. Define

$$\begin{aligned} U_g &= \alpha s_1(\boldsymbol{\lambda}) + (1 - \alpha)h_2(\boldsymbol{\lambda}), & V_g &= (1 - \gamma)h_1(\boldsymbol{\lambda}) + \gamma s_2(\boldsymbol{\lambda}), \\ H_g &= \alpha S(\boldsymbol{\lambda}) + (1 - \alpha)T(\boldsymbol{\lambda}), & K_g &= (1 - \gamma)S(\boldsymbol{\lambda}) + \gamma T(\boldsymbol{\lambda}). \end{aligned} \quad (107)$$

Instantiate the dual inequality (83c) of $\text{SimpleMixing}(y_1^1)$ with $x^{2'} = y_s^2$ and y_1^2 ; instantiate the dual inequality (83d) of $\text{SimpleMixing}(y_1^2)$ with $x^{1'} = y_s^1$ and y_1^1 . Each deviation payoff is at most one, and each regret is at most one minus the actual payoff. The four bounds therefore imply

$$U_g \geq r, \quad V_g \geq r, \quad H_g \geq r, \quad K_g \geq r. \quad (108)$$

For example, instantiating the dual inequality of $\text{SimpleMixing}(y_1^1)$ with $x^{2'} = y_s^2$ gives an expression bounded above by $\alpha[1 - u_1(y_1^1, y_s^2)] + (1 - \alpha)[1 - u_2(y_1^1, y_s^2)]$, which is at most U_g . The lower bound r comes from $F_2^1 > r$ in (31).

Comparisons combining the two rounds. Instantiating two dual inequalities at the same arbitrary strategy can give a restriction that separate instantiations do not supply. We use the first-round operation $\text{SimpleMixing}(y_s^2)$ and the second-round operation $\text{SimpleMixing}(y_1^2)$ to bound a row best-response payoff at a mixture, and then instantiate the dual inequality of the second-round operation $\text{SimpleMixing}(y_1^1)$ at that mixture. Fix $x^{1'} \in \Delta_{n_1}$. Equations (83b) and (83d), the failures (31), and the regret definition give

$$\begin{aligned} r &\leq \xi[u_2(x^{1'}, y_1^2) - u_2(x^{1'}, y_s^2)] + (1 - \xi)[B_1(y_s^2) - u_1(x^{1'}, y_s^2)], \\ r &\leq \gamma[u_2(x^{1'}, y_2^2) - u_2(x^{1'}, y_1^2)] + (1 - \gamma)[B_1(y_1^2) - u_1(x^{1'}, y_1^2)]. \end{aligned} \quad (109)$$

Use $u_2(x^{1'}, y_s^2) \geq 0$, $B_1 \leq 1$, and $u_2(x^{1'}, y_2^2) \leq 1$, then rearrange:

$$\begin{aligned} \xi u_2(x^{1'}, y_1^2) &\geq r - (1 - \xi) + (1 - \xi)u_1(x^{1'}, y_s^2), \\ \gamma u_2(x^{1'}, y_1^2) &\leq 1 - r - (1 - \gamma)u_1(x^{1'}, y_1^2). \end{aligned} \quad (110)$$

Multiply the first inequality by the positive number γ/ξ and substitute its lower bound for the common column payoff into the second inequality. With v from (96), this gives

$$\forall x^{1'} \in \Delta_{n_1}, \quad (1 - \gamma)u_1(x^{1'}, y_1^2) + \gamma v u_1(x^{1'}, y_s^2) \leq 1 - r - \frac{\gamma}{\xi}[r - (1 - \xi)]. \quad (111)$$

Both occurrences of the eliminated payoff concern the same strategy and the same fixed dual solutions. This is why the universal comparison holds with the same ξ, γ .

Recall D_E and y_E^2 from (39). They normalize the coefficients on the left of (111):

$$D_E = 1 - \gamma + \gamma v > 0, \quad y_E^2 = \frac{(1 - \gamma)y_1^2 + \gamma v y_s^2}{D_E} \in \Delta_{n_2}. \quad (112)$$

This is a strategy chosen for the proof. It gives a valid instantiation of the dual inequality (83c) of **SimpleMixing**(y_1^1), whose comparison domain is the entire column simplex. Payoff linearity and (111) give

$$B_1(y_E^2) \leq \frac{1 - r - (\gamma/\xi)[r - (1 - \xi)]}{D_E}. \quad (113)$$

Instantiate the dual inequality (83c) of **SimpleMixing**(y_1^1) with $x^{2'} = y_E^2$. Bound the row-deviation payoff by (113) and the column best-response payoff by one. Expand each actual payoff using (112), and apply (99), (103), and (106). Multiplying by the positive denominator D_E gives

$$\begin{aligned} rD_E \leq & \alpha\{1 - r - (\gamma/\xi)[r - (1 - \xi)] - (1 - \gamma)[1 - S(\boldsymbol{\lambda})] - \gamma v[1 - s_1(\boldsymbol{\lambda})]\} \\ & + (1 - \alpha)[(1 - \gamma)T(\boldsymbol{\lambda}) + \gamma v h_2(\boldsymbol{\lambda})]. \end{aligned} \quad (114)$$

Grouping by the functions in (107) defines the left side of an additional necessary inequality:

$$E_{1,g} = (1 - \gamma)(H_g - r) + \gamma v(U_g - r) + \alpha\gamma(1 - r - rv) - \alpha r \geq 0. \quad (115)$$

For completeness, the other player's calculation starts by instantiating the dual inequalities (83a) of **SimpleMixing**(y_s^1) and (83c) of **SimpleMixing**(y_1^1) at the same $x^{2'} \in \Delta_{n_2}$. Eliminating their common row payoff after multiplying the first rearranged inequality by $\alpha/\eta > 0$ gives

$$\forall x^{2'} \in \Delta_{n_2}, \quad (1 - \alpha)u_2(y_1^1, x^{2'}) + \alpha u u_2(y_s^1, x^{2'}) \leq 1 - r - \frac{\alpha}{\eta}[r - (1 - \eta)]. \quad (116)$$

Define the row comparison strategy and its positive denominator by

$$D_{E,1} = 1 - \alpha + \alpha u > 0, \quad y_E^1 = \frac{(1 - \alpha)y_1^1 + \alpha u y_s^1}{D_{E,1}} \in \Delta_{n_1}. \quad (117)$$

The best-response payoff $B_2(y_E^1)$ is bounded by the right side of (116) divided by $D_{E,1}$. Instantiate the dual inequality (83d) of **SimpleMixing**(y_1^2) with $x^{1'} = y_E^1$, expand payoffs, and multiply by $D_{E,1}$:

$$\begin{aligned} rD_{E,1} \leq & \gamma\{1 - r - (\alpha/\eta)[r - (1 - \eta)] - (1 - \alpha)[1 - T(\boldsymbol{\lambda})] - \alpha u[1 - s_2(\boldsymbol{\lambda})]\} \\ & + (1 - \gamma)[(1 - \alpha)S(\boldsymbol{\lambda}) + \alpha u h_1(\boldsymbol{\lambda})]. \end{aligned} \quad (118)$$

Grouping these terms gives

$$E_{2,g} = (1 - \alpha)(K_g - r) + \alpha u(V_g - r) + \alpha\gamma(1 - r - ru) - \gamma r \geq 0. \quad (119)$$

Thus **SimpleMixing**(y_s^2) and **SimpleMixing**(y_1^2) jointly bound the row best-response payoff at the mixture y_E^2 , and **SimpleMixing**(y_s^1) and **SimpleMixing**(y_1^1) jointly bound the column best-response payoff at the mixture y_E^1 . Instantiating the dual inequalities of **SimpleMixing**(y_1^1) and **SimpleMixing**(y_1^2) at these respective mixtures yields $E_{1,g}, E_{2,g} \geq 0$, which connect both first-round weights with both second-round weights. The arithmetic regroupings and fixed strategy bindings in this appendix are checked by `quantified-interface-identities`² and `ideal-cap-provenance-ledger`³. Their inputs are the specified building-block instantiations; LP duality and the strategy semantics are supplied by the mathematical argument above. All checker paths below are relative to the repository root.

²Checker: `proof/bound/quantified/replay_quantified_identities.py`.

³Checker: `proof/bound/global/replay_ideal_cap_provenance_ledger.py`.

B Jensen gaps and finite mixing comparisons

We now analyze the `OptimalMixing` comparisons [19] over all stationary/dual mixtures, using Jensen's inequality for convex best-response payoffs [21]. The zero-error and failed-output assumptions are (28) and (27); the resulting comparisons are (31). The positive weights proved in (86) and (95) will justify every division below.

Joint bounds on endpoint regrets. The Jensen-gap identity (11) minimizes the weighted endpoint regrets of one comparison strategy. To bound the column Jensen gap, define its two regret functions on the comparison simplex:

$$\begin{aligned} \forall x^{2'} \in \Delta_{n_2}, \quad A_c(x^{2'}) &= f_2(x_s^1, x^{2'}) \geq 0, \\ B_c(x^{2'}) &= f_2(y_s^1, x^{2'}) \geq 0. \end{aligned} \tag{120}$$

These are the same regrets introduced in (44), now written as functions of the comparison strategy:

$$A = A_c(x^{2'}), \quad B = B_c(x^{2'}).$$

The subscript c identifies column-player regrets. The same comparison strategy occurs at the two row-strategy endpoints. Instantiate (17) at $(x_s^1, x^{2'})$, and use (87) and nonnegative payoffs:

$$\begin{aligned} g &\leq \rho[u_1(y_s^1, x^{2'}) - u_1(x_s^1, x^{2'})] + \sigma A_c(x^{2'}), \\ u_1(y_s^1, x^{2'}) &\geq g/\rho - (\sigma/\rho)A_c(x^{2'}). \end{aligned} \tag{121}$$

Instantiate the dual inequality (83a) of the first-round operation `SimpleMixing`(y_s^1) at the same $x^{2'}$, and use this lower bound and $u_1(y_s^1, x^{2'}) \leq 1$. Since $F_1^1 > r$, this gives

$$r \leq \eta[1 - g/\rho + (\sigma/\rho)A_c(x^{2'})] + (1 - \eta)B_c(x^{2'}). \tag{122}$$

Rearrangement yields a joint lower bound valid for every column strategy. For the row counterpart, define the functions

$$\begin{aligned} \forall x^{1'} \in \Delta_{n_1}, \quad A_r(x^{1'}) &= f_1(x^{1'}, x_s^2) \geq 0, \\ B_r(x^{1'}) &= f_1(x^{1'}, y_s^2) \geq 0. \end{aligned} \tag{123}$$

Instantiate (17) at $(x^{1'}, x_s^2)$. Then instantiate the dual inequality (83b) of the first-round operation `SimpleMixing`(y_s^2) at the same $x^{1'}$, and use the resulting column-payoff lower bound. Explicitly, the two instantiations give

$$\begin{aligned} g &\leq \rho A_r(x^{1'}) + \sigma[u_2(x^{1'}, y_s^2) - u_2(x^{1'}, x_s^2)], \\ u_2(x^{1'}, y_s^2) &\geq g/\sigma - (\rho/\sigma)A_r(x^{1'}), \\ r &\leq \xi[1 - g/\sigma + (\rho/\sigma)A_r(x^{1'})] + (1 - \xi)B_r(x^{1'}). \end{aligned} \tag{124}$$

The two conclusions and their right sides are

$$\begin{aligned} \forall x^{2'} \in \Delta_{n_2}, \quad \frac{\eta\sigma}{\rho}A_c(x^{2'}) + (1 - \eta)B_c(x^{2'}) &\geq \ell_c(\boldsymbol{\lambda}), \quad \ell_c(\boldsymbol{\lambda}) = r - \eta(1 - g/\rho), \\ \forall x^{1'} \in \Delta_{n_1}, \quad \frac{\xi\rho}{\sigma}A_r(x^{1'}) + (1 - \xi)B_r(x^{1'}) &\geq \ell_r(\boldsymbol{\lambda}), \quad \ell_r(\boldsymbol{\lambda}) = r - \xi(1 - g/\sigma). \end{aligned} \tag{125}$$

When a right side is positive, no comparison strategy has zero regret at both endpoints.

We next convert a joint lower bound into a lower bound on the weighted regrets in the Jensen-gap identity (11). The elementary calculation must work even when $\ell_c(\boldsymbol{\lambda})$ or $\ell_r(\boldsymbol{\lambda})$ is negative.

Lemma 11 (A weighted lower bound on two regrets). *Let $a, b \geq 0$, $c_1, c_2 > 0$, and $\ell \in \mathbb{R}$ satisfy $c_1 a + c_2 b \geq \ell$. For $t \in [0, 1]$, define $M(t) = \min\{(1-t)/c_1, t/c_2\}$. Then*

$$(1-t)a + tb \geq M(t)\ell.$$

Proof. Both $1-t-c_1M(t)$ and $t-c_2M(t)$ are nonnegative. Therefore

$$\begin{aligned} (1-t)a + tb - M(t)\ell &= [1-t-c_1M(t)]a + [t-c_2M(t)]b \\ &\quad + M(t)[c_1a + c_2b - \ell] \geq 0. \end{aligned} \tag{126}$$

The calculation uses $M(t) \geq 0$ and does not require $\ell \geq 0$. $\square$

For the column regrets, apply the lemma with $a = A_c(x^{2'})$, $b = B_c(x^{2'})$, $c_1 = \eta\sigma/\rho$, $c_2 = 1-\eta$, $\ell = \ell_c(\boldsymbol{\lambda})$, and $t = p$. For the row regrets, use $a = A_r(x^{1'})$, $b = B_r(x^{1'})$, $c_1 = \xi\rho/\sigma$, $c_2 = 1-\xi$, $\ell = \ell_r(\boldsymbol{\lambda})$, and $t = q$. Minimize over the comparison strategy using (11). This defines explicit Jensen-gap lower bounds:

$$\begin{aligned} \forall p \in [0, 1], \quad J_2(p) &\geq j_c(\boldsymbol{\lambda}, p), \quad j_c(\boldsymbol{\lambda}, p) = \ell_c(\boldsymbol{\lambda}) \min \left\{ \frac{(1-p)\rho}{\eta\sigma}, \frac{p}{1-\eta} \right\}, \\ \forall q \in [0, 1], \quad J_1(q) &\geq j_r(\boldsymbol{\lambda}, q), \quad j_r(\boldsymbol{\lambda}, q) = \ell_r(\boldsymbol{\lambda}) \min \left\{ \frac{(1-q)\sigma}{\xi\rho}, \frac{q}{1-\xi} \right\}. \end{aligned} \tag{127}$$

The functions j_c, j_r are lower bounds on the actual Jensen gaps J_2, J_1 . A negative lower bound is still valid. We use these signed formulas throughout the arithmetic reduction. The unit **full-BR-rectangle-bridge**⁴ reconstructs the joint-regret combinations and the signed weighted-regret identities used to obtain (127).

Upper bounds on both regrets throughout the rectangle. To turn the universal mixing comparison (31) into finitely many conditions, we need upper bounds that can be evaluated along specified mixing segments. The stationary best-response identities (87) and the payoff lower bounds (88) give the endpoint bounds

(p, q)	(0, 0)	(1, 0)	(0, 1)	(1, 1)
row-regret upper bound	g	0	1	$1 - g/\rho$
column-regret upper bound	g	1	0	$1 - g/\sigma$

(128)

For example, the row regret at (1, 0) is zero because y_s^1 is a best response to x_s^2 . At (1, 1) it is at most $1 - u_1(y_s^1, y_s^2) \leq 1 - g/\rho$. Insert (128) into the exact regret interpolation (10), and subtract the lower bounds (127). The resulting functions are

$$\begin{aligned} \forall p, q \in [0, 1], \quad f_1(\mathbf{x}(p, q)) &\leq R_g(p, q), \quad R_g(p, q) = (1-p)(1-q)g + q(1 - pg/\rho) - j_r(\boldsymbol{\lambda}, q), \\ \forall p, q \in [0, 1], \quad f_2(\mathbf{x}(p, q)) &\leq C_g(p, q), \quad C_g(p, q) = (1-p)(1-q)g + p(1 - qg/\sigma) - j_c(\boldsymbol{\lambda}, p). \end{aligned} \tag{129}$$

Thus R_g, C_g denote upper bounds on the row and column regrets; they are distinct from the payoff matrices R, C .

Each minimum in (127) changes formula where its two ratios are equal. Define the positive denominators and the corresponding mixing coefficients by

$$\begin{aligned} D_c &= \rho(1-\eta) + \sigma\eta > 0, \quad p_{\dagger} = \frac{\rho(1-\eta)}{D_c}, \\ D_r &= \sigma(1-\xi) + \rho\xi > 0, \quad q_{\dagger} = \frac{\sigma(1-\xi)}{D_r}. \end{aligned} \tag{130}$$

⁴Checker: `proof/bound/quantified/replay_fullbr_rectangle_bridge.py`.

Both coefficients lie strictly between zero and one. They are determined by the ratios even when $\ell_c(\boldsymbol{\lambda})$ or $\ell_r(\boldsymbol{\lambda})$ is zero. The two cuts partition $[0, 1]^2$ into four rectangles. On each rectangle R_g, C_g are bilinear, so they are affine along a segment with one coordinate fixed. There are twelve distinct rectangle edges: each of three vertical levels and three horizontal levels is split into two segments.

We will use the four segments joining the central cut point to the boundary. Name their endpoints now for later reference:

$$\begin{aligned} e_0 &= (p_{\dagger}, q_{\dagger}), & e_1 &= (p_{\dagger}, 0), & e_2 &= (p_{\dagger}, 1), \\ e_3 &= (0, q_{\dagger}), & e_4 &= (1, q_{\dagger}), & \mathbf{c}_j(\boldsymbol{\lambda}) &= (R_g(e_j), C_g(e_j)) \quad (j \in \{0, \dots, 4\}). \end{aligned} \quad (131)$$

Here e_j denotes a pair of mixing coefficients. Evaluation at e_j means evaluation at its two coordinates. The pair $\mathbf{c}_j(\boldsymbol{\lambda})$ consists of regret upper bounds. Omitting the other eight edge conditions enlarges the necessary system that the proof must exclude.

The finite condition on one segment. We eliminate the continuous segment parameter by identifying the possible minimizers of the maximum of two affine functions. For this calculation, let $\mathbf{a} = (R_0, C_0), \mathbf{b} = (R_1, C_1) \in \mathbb{R}^2$ be the two endpoint upper-bound pairs. Parameterize the segment by $t \in [0, 1]$ and define

$$\begin{aligned} L_1(t) &= R_0 + (R_1 - R_0)t, & L_2(t) &= C_0 + (C_1 - C_0)t, \\ D &= (R_1 - R_0) - (C_1 - C_0), & N &= C_0 - R_0, & W &= R_1 C_0 - R_0 C_1. \end{aligned} \quad (132)$$

The symbols $R_0, R_1, C_0, C_1, D, N, W$ are local to this affine calculation. If $D \neq 0$, the lines intersect at $t = N/D$ with common value W/D . No intersection condition is needed outside $[0, 1]$. When $D = 0$, endpoint conditions suffice. The maximum is affine on either side of an intersection, so its minimum occurs at an endpoint or an intersection in the segment. Define the segment predicate by

$$\mathcal{T}_r(\mathbf{a}, \mathbf{b}) \iff \begin{aligned} &\max\{R_0, C_0\} \geq r, & \max\{R_1, C_1\} &\geq r, \\ &\text{if } D \neq 0 \text{ and } 0 \leq N/D \leq 1, \text{ then } W/D \geq r. \end{aligned} \quad (133)$$

The ratios in the last condition are evaluated only when $D \neq 0$. Equivalently, the predicate says that $\min_{t \in [0, 1]} \max\{L_1(t), L_2(t)\} \geq r$.

Every actual profile on one of the twelve edges has maximum regret above r by (31). The upper bounds (129) therefore satisfy the strict affine minimax inequality there; continuity on the compact segment justifies taking its minimum. Weakening it to (133) gives a necessary condition. When clearing denominators, we must check their signs. For example, if $R_0 < r < R_1$ and $C_1 < r < C_0$, then $D > 0$, the intersection lies in the segment, and its non-strict condition is

$$(C_0 - r)(R_1 - r) - (R_0 - r)(C_1 - r) = W - rD \geq 0. \quad (134)$$

Thus the inequality is equivalent to $W/D \geq r$ in this case. A symbolic routine constructs endpoint conditions, intersection conditions, and sign-correct polynomial inequalities for all twelve edges; the independent unit `parametric-OM-linearizer`⁵ reconstructs 131 identities for these constructions. The four segment conditions (46) are expressed by this finite predicate as

$$\forall j \in [4], \quad \mathcal{T}_r(\mathbf{c}_0(\boldsymbol{\lambda}), \mathbf{c}_j(\boldsymbol{\lambda})). \quad (135)$$

This calculation eliminates one-dimensional affine upper-bound problems. It supplies necessary conditions for a failed execution on the selected segments.

⁵Checker: `proof/bound/quantified/replay_parametric_optimalmix.py`.

C Forgetting and the finite necessary system

We now eliminate the remaining game-dependent quantities from the comparisons proved above. Throughout, the zero-error assumption is (28), and the failed-output assumption is (27). The purpose is to obtain necessary conditions involving only r and five dual weights.

Consistent replacement of payoff and regret terms. An instantiated building-block clause still concerns strategies in the fixed game. Forgetting replaces each distinct payoff, regret, or best-response payoff term by a real variable, using the same variable at every occurrence of the same term, as in the LegoNE abstraction [15, 16]. For this example only, introduce the scalar names

$$\hat{t} = F_1^1, \quad \hat{a}_1 = u_1(y_1^1, x_s^2), \quad \hat{a}_0 = u_1(y_s^1, x_s^2), \quad \hat{b} = f_2(y_s^1, x_s^2). \quad (136)$$

These scalar names represent the displayed payoff and regret values. The instantiation of the dual inequality (83a) of `SimpleMixing`(y_s^1) with $x^{2'} = x_s^2$, the payoff range, and (87) become

$$\hat{t} \leq \eta(\hat{a}_1 - \hat{a}_0) + (1 - \eta)\hat{b}, \quad 0 \leq \hat{a}_0, \hat{a}_1, \hat{b} \leq 1, \quad \hat{a}_1 \leq \hat{a}_0. \quad (137)$$

The additional instantiation at a column best response to y_s^1 , recorded in (90), and the failure (31) give $r < \hat{t} \leq \min\{\eta, 1 - \eta\}$. Likewise, both occurrences of $u_2(x^{1'}, y_1^2)$ in (109) must be replaced by the same scalar for their cancellation to remain valid. Relations proved from the regret definition, payoff linearity, LP duality, and complementary slackness [21] remain conditions on the replacement variables.

We simplify further by eliminating many payoff variables through their lower bounds. The resulting regret upper bounds depend only on weights, r , and g . Every failed execution supplies values satisfying the resulting conditions; a solution of the scalar conditions need not be realizable by a game. Deleting a constraint, increasing an upper bound, or changing a scalar point must therefore be justified in the necessary direction. The unit `ideal-cap-provenance-ledger`³ checks the strategy types and fixed dual solutions, then reconstructs the nonnegative combinations, positive divisions, and payoff mixing identities used here.

Eliminating the stationary regret. The remaining game-dependent scalar is the stationary regret g . Setting it to r requires proof: the term $(1 - p)(1 - q)g$ in (129) increases with g , whereas other terms decrease. We first identify the admissible interval and then prove preservation of the selected inequalities while lowering g . The sign certificates cover $3/10 \leq r \leq 1/3$.

Instantiate the dual inequality (83a) of the first-round operation `SimpleMixing`(y_s^1) with $x^{2'} = y_s^2$ and the dual inequality (83b) of the first-round operation `SimpleMixing`(y_s^2) with $x^{1'} = y_s^1$. These instantiations use the stationary dual profile (y_s^1, y_s^2) . Use (88), bound the other deviation payoff by one, and use (31). The resulting two inequalities are

$$\begin{aligned} r &\leq \eta(1 - g/\rho) + (1 - \eta)(1 - g/\sigma), \\ r &\leq \xi(1 - g/\sigma) + (1 - \xi)(1 - g/\rho). \end{aligned} \quad (138)$$

With the positive D_c, D_r from (130), solving these inequalities bounds g from above. Weakening the strict weight and regret inequalities from (86), (95), and (31) gives the domain used in the projection:

$$\begin{aligned} 3/10 \leq r \leq 1/3, \quad r \leq \rho, \eta, \xi, \alpha, \gamma \leq 1 - r, \quad \sigma = 1 - \rho, \\ r \leq g \leq \min \left\{ \frac{\rho\sigma(1 - r)}{D_c}, \frac{\rho\sigma(1 - r)}{D_r} \right\}. \end{aligned} \quad (139)$$

The projection will preserve the nonnegative payoff upper bounds, the four comparisons from the second-round operations $\text{SimpleMixing}(y_1^1)$ and $\text{SimpleMixing}(y_1^2)$, the two comparisons obtained by instantiating their dual inequalities at mixtures, and the four selected segment conditions. For a general scalar point, record these premises as

$$\begin{aligned} s_1(\boldsymbol{\lambda}), s_2(\boldsymbol{\lambda}), h_1(\boldsymbol{\lambda}), h_2(\boldsymbol{\lambda}), S(\boldsymbol{\lambda}), T(\boldsymbol{\lambda}) &\geq 0, \\ U_g, V_g, H_g, K_g &\geq r, \quad E_{1,g}, E_{2,g} \geq 0, \\ \forall j \in [4], \quad \mathcal{T}_r(\mathbf{c}_0(\boldsymbol{\lambda}), \mathbf{c}_j(\boldsymbol{\lambda})). \end{aligned} \quad (140)$$

The functions and coefficient points are defined in (98), (102), (105), (107), (115), (119), and (131). The predicate is (133). A failed execution satisfies these premises by the payoff interpretations, the instantiations of the dual inequalities of $\text{SimpleMixing}(y_1^1)$ and $\text{SimpleMixing}(y_1^2)$, and (135).

Lemma 12 (Lowering the stationary regret). *Let a scalar point satisfy the domain (139) and the inequalities (140). Hold $r, \rho, \eta, \xi, \alpha, \gamma$ fixed and evaluate every function in (140) with $g = r$. All the inequalities in (140) continue to hold.*

Proof. We first treat the payoff upper bounds. Direct differentiation gives

$$\frac{\partial s_1(\boldsymbol{\lambda})}{\partial g} = -\frac{1}{\rho} - \frac{u}{\sigma} < 0, \quad \frac{\partial s_2(\boldsymbol{\lambda})}{\partial g} = -\frac{1}{\sigma} - \frac{v}{\rho} < 0. \quad (141)$$

Each $h_i(\boldsymbol{\lambda})$ is the minimum of one and an affine function decreasing in g , by (102). Thus lowering g cannot decrease any $s_i(\boldsymbol{\lambda})$ or $h_i(\boldsymbol{\lambda})$. The formulas (105), with $u, v > 0$, imply the same for $S(\boldsymbol{\lambda}), T(\boldsymbol{\lambda})$. The nonnegative coefficients in (107) then give the same conclusion for U_g, V_g, H_g, K_g . In (115) and (119), only these weighted bounds depend on g , and their coefficients are nonnegative. Therefore $E_{1,g}, E_{2,g}$ also cannot decrease. This proves preservation of the first two lines of (140).

We next treat the four segment predicates, beginning with their shared central endpoint $e_0 = (p_\dagger, q_\dagger)$. Differentiate (129), using the equality of the two ratios at each cut in (130):

$$\begin{aligned} \partial_g R_g(p_\dagger, q_\dagger) &= \frac{\xi}{D_r} [\rho(1 - p_\dagger) - 1] - \frac{p_\dagger q_\dagger}{\rho} < 0, \\ \partial_g C_g(p_\dagger, q_\dagger) &= \frac{\eta}{D_c} [\sigma(1 - q_\dagger) - 1] - \frac{p_\dagger q_\dagger}{\sigma} < 0. \end{aligned} \quad (142)$$

The signs follow from $0 < \rho, \sigma, p_\dagger, q_\dagger < 1$. At the two endpoints with a coefficient equal to one, the Jensen gap associated with that coefficient is zero and direct differentiation gives

$$\begin{aligned} \partial_g R_g(p_\dagger, 1) &= -p_\dagger / \rho < 0, \\ \partial_g C_g(p_\dagger, 1) &= -p_\dagger / \sigma - \eta p_\dagger / [\rho(1 - \eta)] < 0, \\ \partial_g R_g(1, q_\dagger) &= -q_\dagger / \rho - \xi q_\dagger / [\sigma(1 - \xi)] < 0, \\ \partial_g C_g(1, q_\dagger) &= -q_\dagger / \sigma < 0. \end{aligned} \quad (143)$$

Both upper-bound functions are affine along each selected segment. Their derivatives at an interior point are therefore convex combinations of their endpoint derivatives. Equations (142) and (143) prove that lowering g cannot decrease either bound on the segments from e_0 to e_2 and e_4 . Their predicates are preserved.

For the segment from $e_1 = (p_\dagger, 0)$ to e_0 , we must also consider an intersection of the two affine bounds. Recall $\mathcal{D}(g)$ from (48), which expresses the intersection condition:

$$\begin{aligned} \mathcal{D}(g) &= \frac{1}{q_\dagger} \{ [C_g(p_\dagger, 0) - r][R_g(p_\dagger, q_\dagger) - r] \\ &\quad - [R_g(p_\dagger, 0) - r][C_g(p_\dagger, q_\dagger) - r] \}. \end{aligned} \quad (144)$$

Division by $q_{\dagger} > 0$ preserves its sign. The endpoint ordering follows from (138), which are equivalent to the upper bounds on g in (139). To see the ordering explicitly, define two temporary functions for the affine row formula on this segment:

$$B_e(g) = (1 - p_{\dagger})g, \quad U_e(g) = 1 - p_{\dagger}g/\rho - \ell_r(\boldsymbol{\lambda})/(1 - \xi). \quad (145)$$

$B_e(g)$ is the row upper bound at e_1 ; $U_e(g)$ is its affine extrapolation to coefficient one. It is distinct from the weighted bound U_g in (107). For $0 \leq q \leq q_{\dagger}$, the formula is

$$R_g(p_{\dagger}, q) = (1 - q)B_e(g) + qU_e(g). \quad (146)$$

Direct subtraction and the first-round comparisons give

$$\begin{aligned} [U_e(g) - B_e(g)] - (1 - p_{\dagger})(g/\rho - g) &= \frac{1 - r - g((1 - \xi)/\rho + \xi/\sigma)}{1 - \xi} \geq 0, \\ C_g(p_{\dagger}, 0) - R_g(p_{\dagger}, 0) &= \frac{\rho(1 - r) - \eta g}{D_c} \geq 0. \end{aligned} \quad (147)$$

For the second sign, the first upper bound on g gives $\eta g \leq \eta \rho \sigma (1 - r)/D_c \leq \rho(1 - r)$, because $\eta \sigma \leq D_c$. Since $g > 0$, $\rho < 1$, and $p_{\dagger} < 1$, the first line implies $U_e(g) - B_e(g) > 0$. The row bound increases with q , whereas the column bound decreases:

$$\partial_q C_g(p_{\dagger}, q) = -(1 - p_{\dagger})g - p_{\dagger}g/\sigma < 0, \quad \partial_g C_g(p_{\dagger}, 0) = -\rho\eta/D_c < 0. \quad (148)$$

At e_1 the column bound is the larger one, so its maximum cannot decrease when g is lowered. The central endpoint is protected by (142). If the increasing row line and decreasing column line intersect inside the segment, their common value is at least r exactly when $\mathcal{D}(g) \geq 0$, by (133) and (134). We will prove $\partial_g \mathcal{D}(g) < 0$ throughout the admissible interval. This preserves the intersection condition as g decreases. An intersection can enter or leave the segment only through an endpoint, whose condition is already preserved. Continuity consequently preserves the full segment predicate.

The derivative sign reduces to a polynomial sign. Define the function $\mathcal{Q}(g)$ and the positive coefficient Z_Q by

$$\begin{aligned} \partial_g \mathcal{D}(g) &= -\frac{\mathcal{Q}(g)}{\sigma D_c^2 (1 - \xi)}, \\ Z_Q &= \rho \sigma \eta \xi + \sigma^2 \eta (1 - \xi) + \rho^2 (1 - \eta) \xi + 2 \rho \sigma (1 - \eta) (1 - \xi) > 0. \end{aligned} \quad (149)$$

The numerator can be written explicitly as

$$\begin{aligned} \mathcal{Q}(g) &= [\sigma(1 - \xi)(1 - \rho\eta) + D_c \xi][\rho(1 - r) - r D_c] \\ &\quad + \sigma D_c \rho \eta (1 - r) + r D_c (1 - \xi)[\sigma^2 \eta + \rho(1 - \eta)] - 2 \eta Z_Q g. \end{aligned} \quad (150)$$

The unit **g-identities**⁶ reconstructs this derivative identity and the endpoint and intersection identities above. This formula follows by expanding (144) using (146) and (129). It gives $\partial_g \mathcal{Q}(g) = -2\eta Z_Q < 0$. Thus it suffices to prove positivity at the largest admissible g . Define that endpoint, its feasibility margin, and a cleared numerator by

$$\begin{aligned} D_* &= \max\{D_c, D_r\}, & G_* &= \frac{\rho \sigma (1 - r)}{D_*}, \\ B_* &= \rho \sigma (1 - r) - r D_* \geq 0, & \mathcal{H}_* &= D_* \mathcal{Q}(G_*). \end{aligned} \quad (151)$$

⁶Checker: `proof/analytic_scalar_bound/cross_projection_symbolic_checks.py`.

The sign of $\mathcal{B}_*$ is exactly $G_* \geq r$, which follows from (139).

Before invoking the certificates, specify the four parameter regions and their unit-box coordinates. Split the weight domain at $\rho = 1/2$ and $\eta + \xi = 1$. The identity

$$D_c - D_r = (\rho - \sigma)(1 - \eta - \xi) \quad (152)$$

determines which denominator equals D_* in each region. Let $t_r, t_\rho, t_\eta, t_\xi \in [0, 1]$, and substitute

$$\begin{aligned} r &= 3/10 + t_r/30, & \eta &= r + (1 - 2r)t_\eta, \\ \rho &= \begin{cases} r + (1/2 - r)t_\rho, & \rho \leq 1/2, \\ 1/2 + (1/2 - r)t_\rho, & \rho \geq 1/2, \end{cases} & \xi &= \begin{cases} r + (1 - \eta - r)t_\xi, & \eta + \xi \leq 1, \\ 1 - \eta + (\eta - r)t_\xi, & \eta + \xi \geq 1. \end{cases} \end{aligned} \quad (153)$$

These substitutions cover all four regions, including their boundaries. After clearing positive denominators, the checker reconstructs the resulting polynomials and their rational Bernstein coefficients [31]. For degree at most d , the Bernstein basis on $[0, 1]$ satisfies

$$b_i^{(d)}(t) = \binom{d}{i} t^i (1 - t)^{d-i}, \quad i = 0, \dots, d, \quad b_i^{(d)}(t) \geq 0, \quad \sum_{i=0}^d b_i^{(d)}(t) = 1. \quad (154)$$

On a rational box $\prod_j [a_j, b_j]$, substitute $x_j = a_j + (b_j - a_j)t_j$ and expand in products of these basis functions. The polynomial's value is a convex combination of its coefficients, so their signs certify its sign throughout the box. The checker uses Python 3 rational arithmetic with `fractions.Fraction` [30] to verify these signs exactly.

On the two regions with $\rho \leq 1/2$, choose the positive rational certificate multiplier $m_Q = 12609/50000$. Subtracting $m_Q \mathcal{B}_*$ produces a polynomial whose strict positivity can be checked through its Bernstein coefficients; the known $\mathcal{B}_* \geq 0$ then recovers positivity of $\mathcal{H}_*$. On the other two regions the certificate proves positivity of $\mathcal{H}_*$ directly. The certified inequalities are

$$\begin{cases} \mathcal{H}_* - m_Q \mathcal{B}_* > 0, & m_Q = 12609/50000, \quad \rho \leq 1/2, \\ \mathcal{H}_* > 0, & \rho \geq 1/2. \end{cases} \quad (155)$$

All reconstructed Bernstein coefficients for these stated polynomials are strictly positive on the respective unit boxes. The unit `g-derivative-small`⁷ reconstructs the two $\rho \leq 1/2$ certificates, and `g-derivative-large`⁸ reconstructs the two $\rho \geq 1/2$ certificates. Both certificates imply $\mathcal{Q}(G_*) > 0$ because $D_* > 0$. Since $\mathcal{Q}$ decreases with g and $g \leq G_*$,

$$\mathcal{Q}(g) \geq \mathcal{Q}(G_*) > 0, \quad \partial_g \mathcal{D}(g) < 0. \quad (156)$$

This completes the segment from e_1 to e_0 .

The other coefficient-zero segment, from e_3 to e_0 , follows from the same identities under the explicit interchange

$$(\rho, \sigma, \eta, \xi, p, q, R_g, C_g) \longleftrightarrow (\sigma, \rho, \xi, \eta, q, p, C_g, R_g). \quad (157)$$

It exchanges D_c with D_r and the two first-round inequalities (138). The domain (139) is invariant under the interchange, so the preceding endpoint, determinant, and certificate arguments cover this segment as well. Together with the coefficient-one segments, all four predicates in (140) are preserved while g decreases to r . $\square$

⁷Checker: `proof/analytic_scalar_bound/replay_cross_derivative_certificate.py`.

⁸Checker: `proof/analytic_scalar_bound/replay_cross_derivative_large_mass_certificate.py`.

The projected upper-bound functions. Lemma 12 associates each failed execution with a scalar point at $g = r$. We now evaluate the arithmetic functions at this point. There is one further relaxation: replace d_1, d_2 in (102) by $(1 - \eta - r)/\eta$ and $(1 - \xi - r)/\xi$, respectively. Each replacement increases or leaves unchanged the corresponding h_i . With nonnegative u, v and dual weights, it also increases or leaves unchanged $S, T, U, V, H, K, E_1, E_2$. For compact formulas, define

$$\chi_1 = 1 - r - r/\rho, \quad \chi_2 = 1 - r - r/\sigma, \quad c = 1 - 2r. \quad (158)$$

The scalar c is a recurring coefficient. The symbols without an argument below are working names for the projected functions, including the stated relaxation of h_i :

$$\begin{aligned} s_1 &= \chi_1 + \chi_2 u, & s_2 &= \chi_2 + \chi_1 v, \\ h_1 &= \min\{1, 1 - r(1 + \sigma)/\rho + \sigma(1 - r)v/\rho\}, & h_2 &= \min\{1, 1 - r(1 + \rho)/\sigma + \rho(1 - r)u/\sigma\}, \\ S &= h_1 - r + u(s_2 - r), & T &= h_2 - r + v(s_1 - r). \end{aligned} \quad (159)$$

In terms of λ_r from (49), the relation to the original functions is

$$\forall i \in [2], \quad s_i = s_i(\lambda_r), \quad h_i \geq h_i(\lambda_r).$$

The larger h_i result from using the uncapped bounds in their affine formulas. The functions S, T are then recalculated using these relaxed h_i . For s_i, S, T , we use the displayed formulas directly. All six displayed upper bounds are nonnegative at the scalar point produced by the execution and the preceding projection. Later arithmetic transformations must establish the signs they need.

Use the same substitutions in the four weighted functions (107) and the two inequalities (115) and (119). Their projected working names and formulas are

$$\begin{aligned} U &= \alpha s_1 + (1 - \alpha)h_2 \geq r, & V &= (1 - \gamma)h_1 + \gamma s_2 \geq r, \\ H &= \alpha S + (1 - \alpha)T \geq r, & K &= (1 - \gamma)S + \gamma T \geq r, \\ E_1 &= (1 - \gamma)(H - r) + \gamma v(U - r) + \alpha\gamma(1 - r - rv) - \alpha r \geq 0, \\ E_2 &= (1 - \alpha)(K - r) + \alpha u(V - r) + \alpha\gamma(1 - r - ru) - \gamma r \geq 0. \end{aligned} \quad (160)$$

Each symbol denotes the displayed arithmetic function of r and the five weights.

Definition of the finite necessary system. The preceding projection removes g and preserves the selected necessary conditions. We can now specify their conjunction explicitly in five scalar variables. With e_j from (131), define

$$\mathbf{c}_j = \mathbf{c}_j(\lambda_r) = (R_r(e_j), C_r(e_j)) \quad (j \in \{0, \dots, 4\}), \quad \boldsymbol{\theta} = (\rho, \eta, \xi, \alpha, \gamma) \in \mathbb{R}^5. \quad (161)$$

Here R_r, C_r mean the functions R_g, C_g with $g = r$; the bold pairs record numerical regret upper bounds. For $3/10 \leq r \leq 1/3$, let $\Sigma_r(\boldsymbol{\theta})$ be the conjunction

$$\begin{aligned} \Sigma_r(\boldsymbol{\theta}) \iff & \begin{aligned} & r \leq \rho, \eta, \xi, \alpha, \gamma \leq 1 - r, \\ & s_1, s_2, h_1, h_2, S, T \geq 0, \\ & U, V, H, K \geq r, \quad E_1, E_2 \geq 0, \\ & \forall j \in [4], \quad \mathcal{T}_r(\mathbf{c}_0, \mathbf{c}_j). \end{aligned} \end{aligned} \quad (162)$$

The first line bounds the five weights. The second expresses nonnegativity of the six payoff upper bounds. The third records the four second-round comparisons and the two mixture comparisons.

The last line uses the finite affine segment predicate (133). All quantities are the functions defined in (96), (130), (159), (160), and (161), with $\sigma = 1 - \rho$.

The first-round inequalities used in the projection are already implied by this conjunction. To check that implication explicitly, subtract r from their projected upper bounds:

$$\begin{aligned}\eta(1 - r/\rho) + (1 - \eta)(1 - r/\sigma) - r &= \eta s_1, \\ \xi(1 - r/\sigma) + (1 - \xi)(1 - r/\rho) - r &= \xi s_2, \\ \eta h_1 + (1 - \eta)s_2 - r &= \eta S, \\ \xi h_2 + (1 - \xi)s_1 - r &= \xi T.\end{aligned}\tag{163}$$

Their nonnegative right sides give precisely the needed non-strict inequalities. Instantiating the dual inequality of the second-round operation `SimpleMixing`(y_1^1) with $x^{2'} = x_s^2$ gives the upper bound $\alpha d_1 + 1 - \alpha$, which is at least r because $d_1 \geq 0$ and $\alpha \leq 1 - r$. Instantiating the dual inequality of the second-round operation `SimpleMixing`(y_1^2) with $x^{1'} = x_s^1$ gives $\gamma d_2 + 1 - \gamma \geq r$ in the same way. At the respective first-round primal outputs x_1^2, x_1^1 , the exact best-response conclusions instead give the upper bounds $1 - \alpha$ and $1 - \gamma$, also at least r by the weight ranges. Thus these instantiations impose no additional condition on (162). Further mixture comparisons could strengthen the system, but are not part of its definition.

Every failed execution first yields (139) and (140). Lemma 12 and the subsequent increase of the payoff upper bounds then yield (162). This proves the implication

$$f(\mathbf{x}_{\text{out}}) > r \implies \exists \boldsymbol{\theta} \in \mathbb{R}^5, \Sigma_r(\boldsymbol{\theta}), \quad 3/10 \leq r \leq 1/3.\tag{164}$$

The system has fixed dimension, independent of the game. Replacing strict failure inequalities by non-strict necessary conditions and increasing upper bounds both enlarge the system. Consequently, infeasibility of Σ_r excludes a failed execution; realizability of every scalar point is not needed.

D The algebraic definition of the bound

This appendix describes how we compute a candidate bound and use the returned active comparisons to define it by exact polynomial equations. Its starting point is the finite necessary system (162):

$$\underset{r, \boldsymbol{\theta}}{\text{maximize}} \quad r \quad \text{subject to} \quad 3/10 \leq r \leq 1/3, \quad \Sigma_r(\boldsymbol{\theta}).\tag{165}$$

We minimize $-r$ subject to all the inequalities in (162), using SciPy's multi-start SLSQP routine [23, 24]. The segment minima are evaluated by the endpoint and intersection cases in (133). The computation is implemented by `find_bound.py`⁹. Its returned points and constraint residuals give a candidate threshold near 0.3095399654 and identify the active comparisons used in the following algebraic calculation. Exchanging the players if necessary, we describe the candidate in the orientation $S > r > T$. An inequality is called active when it holds with equality. Near this value, five inequalities appear active: the uncapped expression for h_1 equals one, $H = K = r$, one interior mixing comparison holds with equality, and $E_1 = 0$. Here H, K, E_1 are the expressions in (160). These observations motivate the equations below; they place no restriction on an arbitrary solution of Σ_r .

Recall $\sigma = 1 - \rho$ and the relative weights u, v from (96). Setting the uncapped formula for h_1 in (159) equal to one gives its change of formula:

$$v_0 = \frac{r(1 + \sigma)}{\sigma(1 - r)}, \quad v = v_0 \iff \xi = \frac{\sigma(1 - r)}{\sigma + r}, \quad h_1 = 1.\tag{166}$$

⁹Script: `tools/find_bound.py`. Parameters and reproduction instructions: `tools/README.md`.

At the numerical candidate $S > r > T$. Under this condition, substituting $H = K = r$ into their linear formulas in (160) determines both second-round weight variables:

$$A_0 = \frac{r - T}{S - T}, \quad \alpha = A_0, \quad \gamma = 1 - A_0. \quad (167)$$

The candidate also has $h_2 < 1$. For the following local calculation, we therefore fix $v = v_0$, use $h_1 = 1$, and substitute

$$\sigma = 1 - \rho, \quad u = \frac{1 - \eta}{\eta}, \quad \xi = \frac{\sigma(1 - r)}{\sigma + r}, \quad h_2 = 1 - \frac{r(1 + \rho)}{\sigma} + \frac{\rho(1 - r)}{\sigma}u. \quad (168)$$

The formulas for s_1, s_2, S, T remain those in (159), with these substitutions. This specifies which rational functions are used before defining their polynomial numerators. The global argument must later justify the use of this formula for h_2 at the points left by its reductions.

Substitute (167) into E_1 in (160). Define the quadratic Ψ with a real argument a ; evaluation at the balancing coefficient A_0 gives

$$\begin{aligned} \forall a \in \mathbb{R}, \quad \Psi(a) &= a(1 - a)K_0 + (1 - a)^2vh_2 - ra - r(1 - a)v, \\ K_0 &= 1 - r/\xi + vs_1, \quad E_1(A_0, 1 - A_0) = \Psi(A_0). \end{aligned} \quad (169)$$

The symbol K_0 denotes only the coefficient in this polynomial; it is distinct from the weighted bound K . Equation (169) expresses the necessary comparison $E_1 \geq 0$ in a short factored form. We have not yet proved that every possible scalar point can be moved to these weight values.

The observed interior mixing equality comes from the segment joining $(p_{\dagger}, 0)$ and $(p_{\dagger}, q_{\dagger})$. On (166), the two cut coefficients have the following denominators and values:

$$D_c = \rho + (1 - 2\rho)\eta, \quad J = \rho + 2\sigma r, \quad p_{\dagger} = \frac{\rho(1 - \eta)}{D_c}, \quad q_{\dagger} = \frac{r(2 - \rho)}{J}. \quad (170)$$

The expression D_c equals $\rho(1 - \eta) + \sigma\eta$ from the stationary-regret projection. The symbol J is a denominator, distinct from the Jensen gaps. Evaluating their lower bounds (127) at the cuts gives

$$J_C = j_c(\boldsymbol{\lambda}_r, p_{\dagger}) = \frac{r(\rho + \eta) - \rho\eta}{D_c}, \quad J_R = j_r(\boldsymbol{\lambda}_r, q_{\dagger}) = \frac{r(3 - 2\rho) - \sigma}{J}. \quad (171)$$

Thus J_C, J_R denote the Jensen-gap lower bounds evaluated at the two comparison coefficients. Substituting $g = r$ and the two endpoint coordinates into (129) gives

$$\begin{aligned} R_{\ell} &= (1 - p_{\dagger})r, & C_{\ell} &= (1 - p_{\dagger})r + p_{\dagger} - J_C, \\ R_k &= (1 - p_{\dagger})(1 - q_{\dagger})r + q_{\dagger}(1 - p_{\dagger}r/\rho) - J_R, \\ C_k &= (1 - p_{\dagger})(1 - q_{\dagger})r + p_{\dagger}(1 - q_{\dagger}r/\sigma) - J_C. \end{aligned} \quad (172)$$

The subscripts ℓ, k denote the outer endpoint and the central cut point, respectively; these are four scalar regret upper bounds.

We need polynomial equations with specified normalization to define the constant reproducibly. The intersection calculation (133), applied to (172), and the comparison (169) define their polynomial numerators by

$$\begin{aligned} (C_{\ell} - r)(R_k - r) - (R_{\ell} - r)(C_k - r) &= \frac{N_1(r, \rho, \eta)}{\sigma D_c^2 J}, \\ \Psi\left(\frac{r - T}{S - T}\right) &= \frac{N_2(r, \rho, \eta)}{D_2}, \\ D_2 &= \rho\eta\sigma(1 - r)[\rho\eta\sigma^2(1 - r)(S - T)]^2. \end{aligned} \quad (173)$$

Every quantity in these identities uses (168). On $3/10 \leq r \leq 1/3$, $r \leq \rho, \eta \leq 1 - r$, and $S > r > T$, the factors ρ, η, σ , $1 - r, D_c, J$, and $S - T$ are positive. Thus both displayed denominators are positive. Exact expansion shows that N_1 is quadratic and N_2 is cubic in η . The identities specify these polynomials without a large coefficient list.

The equations $N_1 = N_2 = 0$ suggest one further equation at an extremum of r . A gradient ∇N_i is the vector of partial derivatives in coordinates (r, ρ, η) . An intersection is regular here when the two gradients are linearly independent. At a regular constrained local extremum, the Lagrange multiplier condition [25] gives real multipliers ν_1, ν_2 such that

$$\nabla r = (1, 0, 0) = \nu_1 \nabla N_1 + \nu_2 \nabla N_2.$$

Taking the dot product with $\nabla N_1 \times \nabla N_2$ shows that the cross product's r -coordinate is zero, giving

$$N_1 = N_2 = 0, \quad H_{\text{root}} = (N_1)_\rho (N_2)_\eta - (N_1)_\eta (N_2)_\rho = 0. \quad (174)$$

A subscript on a polynomial denotes a partial derivative. The symbol H_{root} is distinct from the weighted bound H . Solving near the numerical candidate gives

$$(r_*, \rho_*, \eta_*) \approx (0.3095399654, 0.4799488361, 0.5461144474).$$

For an exact definition, take the rational center

$$\begin{aligned} c_r &= 0.30953996543144345279052787573817601463814080254235, \\ c_\rho &= 0.47994883614366958777809825446105296705426289894925, \quad \mathbf{c} = (c_r, c_\rho, c_\eta). \\ c_\eta &= 0.54611444737225673894884919253273990511184474731359, \end{aligned} \quad (175)$$

These terminating decimals denote exact rationals. Define (r_*, ρ_*, η_*) as the unique solution of (174) in the closed box

$$X = \{(r, \rho, \eta) : |r - c_r| \leq 10^{-30}, |\rho - c_\rho| \leq 10^{-30}, |\eta - c_\eta| \leq 10^{-30}\}. \quad (176)$$

The existence and uniqueness check below makes this definition well posed and also gives

$$0.30953996 < r_* < 0.30953997. \quad (177)$$

To explain exactly what this local check establishes, let $\mathcal{F} = (N_1, N_2, H_{\text{root}})$. Its Jacobian $D\mathcal{F}$ is the 3×3 matrix of partial derivatives, with one row for each component and columns for (r, ρ, η) . Let Y be an invertible rational matrix approximating the inverse Jacobian at $\mathbf{c}$. Rational interval arithmetic [26] computes lower and upper rational bounds containing all values of each entry on X ; write the resulting Jacobian enclosure as $D\mathcal{F}(X)$. The verification unit `natural-root`¹⁰ checks

$$\mathbf{c} - Y\mathcal{F}(\mathbf{c}) + [I - YD\mathcal{F}(X)](X - \mathbf{c}) \subset \text{int } X, \quad \sup_{\mathbf{z} \in X} \|I - YD\mathcal{F}(\mathbf{z})\|_\infty < 1. \quad (178)$$

The first expression is the interval enclosure commonly called the Krawczyk enclosure [27]. Here I is the identity matrix, $\text{int } X$ is the box interior, the vector norm $\|\mathbf{z}\|_\infty$ is the largest absolute coordinate, and the matrix norm is the largest sum of absolute entries in a row. The interval products enclose every possible value on X . Invertibility of Y makes zeros of $\mathcal{F}$ exactly the fixed points of $\mathbf{z} \mapsto \mathbf{z} - Y\mathcal{F}(\mathbf{z})$. The first condition maps the box into itself, and the second makes the map a contraction: distances are reduced by one uniform factor strictly below one. The contraction

¹⁰Checker: `proof/bound/critical/local_root_certificate.py`.

therefore has exactly one fixed point in X by the contraction mapping theorem [33]. This proves the local algebraic definition and the interval (177). Excluding feasible points elsewhere requires the separate global argument; the numerical search contributes no assumption to that argument. At this isolated root, the unit `scalar-root-fixture`¹¹ checks the selected scalar equality point: it verifies 58 strict signs and four active identities in the necessary relaxation. This check concerns scalar inequalities; an algorithmic tightness claim would additionally require a game realizing the point.

E Global reduction and exclusion

E.1 Exact sign checks and preliminary exclusions

The global proof needs polynomial signs throughout parameter regions. For this purpose we use the Bernstein representation defined in (154), following the polynomial range method of [31]. Its basis functions are nonnegative and sum to one, so a polynomial P with coefficients $b_0, \dots, b_d$ satisfies

$$\min_j b_j \leq P(t) \leq \max_j b_j \quad (0 \leq t \leq 1). \quad (179)$$

In several variables, products of the one-variable bases have the same properties. A rational affine substitution maps a rational box to a unit box, meaning a Cartesian product of intervals $[0, 1]$. Strictly negative coefficients then prove negativity throughout the box. If one coefficient list is insufficient, the certificate subdivides the box. The independent checker verifies that every subdivided box equals the union of its child boxes and that every terminal box is excluded by a polynomial sign or incompatible inequalities. This covers the whole region, including its boundaries. For expressions involving the isolated root, the checker uses rational interval enclosures [26] from (176).

It is enough to exclude solutions of Σ_r for $r_* < r \leq 1/3$. The interval (177) permits the slightly larger threshold range below. The weight bounds in (162) and the relative-weight definitions (96) give the following basic domain and coefficients:

$$\begin{aligned} r_0 &= \frac{30953996}{10^8}, & r_0 \leq r \leq 1/3, & & r \leq \rho \leq 1 - r, & & \sigma = 1 - \rho, \\ k &= \frac{r}{1 - r}, & k \leq u, v \leq 1/k, & & \chi_1 = 1 - r - r/\rho, & & \chi_2 = 1 - r - r/\sigma, & & c = 1 - 2r. \end{aligned} \quad (180)$$

The cutoff r_0 fixes the range of this proof. Throughout this domain, $\rho, \sigma, r, 1 - r, u, v$ and c are positive. The two clipped lower bounds corresponding to h_1, h_2 in (159) can be written as positive parts:

$$\begin{aligned} \kappa_1 &= \left[\frac{r(1 + \sigma) - \sigma(1 - r)v}{\rho} \right]_+ = 1 - h_1, \\ \kappa_2 &= \left[\frac{r(1 + \rho) - \rho(1 - r)u}{\sigma} \right]_+ = 1 - h_2. \end{aligned} \quad (181)$$

Here $[t]_+ = \max\{0, t\}$. Taking the positive part combines a derived payoff lower bound with the inherent bound that payoffs are nonnegative. It is equivalent to taking the minimum with one in the definitions of h_1, h_2 .

Two elementary coefficient bounds that will be used below are

$$\rho\sigma \leq 1/4, \quad \chi_1 - r, \chi_2 - r \leq \frac{1 - 4r + 2r^2}{1 - r} < 0. \quad (182)$$

¹¹Checker: `proof/bound/critical/local_signs.py`.

The first follows from $(\rho - \sigma)^2 \geq 0$ and $\rho + \sigma = 1$. For the second, use $\rho, \sigma \leq 1 - r$ in their definitions. Its numerator is negative at $r = 3/10$ and has derivative $4r - 4 < 0$ on the threshold interval.

For the odds coordinates u, v , the cut denominators have the following equivalent expressions:

$$D_{c,\text{odds}} = \rho u + \sigma = D_c/\eta, \quad D_{r,\text{odds}} = \sigma v + \rho = D_r/\xi. \quad (183)$$

Thus $p_{\dagger} = \rho u / D_{c,\text{odds}}$ and $q_{\dagger} = \sigma v / D_{r,\text{odds}}$ are the same cuts as in (130).

Before eliminating weights, we specify the two segment inequalities that later paths will use. Substituting the cut coefficients into (129) with $g = r$ gives the positive denominator and the central numerators

$$\begin{aligned} D_{\text{mix}} &= D_{c,\text{odds}} D_{r,\text{odds}} > 0, \\ N_R &= \sigma^2 \chi_2 + \rho(\sigma - 2r)u + \sigma^2 cv + \rho\sigma(\chi_1 - r)uv, \\ N_C &= \rho^2 \chi_1 + \sigma(\rho - 2r)v + \rho^2 cu + \rho\sigma(\chi_2 - r)uv, \\ R_r(p_{\dagger}, q_{\dagger}) - r &= N_R / D_{\text{mix}}, \quad C_r(p_{\dagger}, q_{\dagger}) - r = N_C / D_{\text{mix}}. \end{aligned} \quad (184)$$

Thus the signs of N_R, N_C specify which central regret upper bound is above the comparison threshold. Define

$$L = (\chi_1 + cu)N_R + ruN_C, \quad B_{\text{mix}} = (\chi_2 + cv)N_C + rvN_R. \quad (185)$$

These are positive multiples of the intersection determinants for the segments toward $(p_{\dagger}, 0)$ and $(0, q_{\dagger})$, respectively. The other endpoint differences are

$$\begin{aligned} R_r(p_{\dagger}, 0) - r &= -\frac{\rho ur}{\rho u + \sigma}, \quad C_r(p_{\dagger}, 0) - r = \frac{\rho(\chi_1 + cu)}{\rho u + \sigma}, \\ R_r(0, q_{\dagger}) - r &= \frac{\sigma(\chi_2 + cv)}{\sigma v + \rho}, \quad C_r(0, q_{\dagger}) - r = -\frac{\sigma vr}{\sigma v + \rho}. \end{aligned} \quad (186)$$

The segment predicates in (162) force $\chi_1 + cu, \chi_2 + cv \geq 0$: the other bound at each outer endpoint is strictly below r . If $N_C < 0$, the central comparison forces $N_R \geq 0$, and the intersection condition (133) gives $L \geq 0$. If $N_R, N_C \geq 0$, the definition of L and the endpoint coefficient signs give $L \geq 0$ directly. Both numerators negative would violate the central comparison. The remaining case $N_R < 0 \leq N_C$ gives $B_{\text{mix}} \geq 0$ from the second segment. Consequently, the initial necessary system implies the explicit guarded alternatives

$$\Sigma_r \implies \begin{aligned} &\chi_1 + cu \geq 0, \quad \chi_2 + cv \geq 0, \\ &(N_R \geq 0 \wedge L \geq 0) \quad \text{or} \quad (N_R < 0 \leq N_C \wedge B_{\text{mix}} \geq 0). \end{aligned} \quad (187)$$

These substitutions and determinant identities are reconstructed by the units **compact-identities**¹² and **mass-identities**¹³.

We next show that exactly one of S, T is above r . The inequalities $H, K \geq r$ in (160) imply $\max\{S, T\} \geq r$, because H, K are convex combinations of S, T . A preliminary uniform bound needed for this exclusion is

$$s_1 < r, \quad s_2 < r \quad \text{on the full domain (180)}. \quad (188)$$

Indeed, $s_2 = \chi_2 + \chi_1 v$ in (159) is affine in v , so its maximum on $[k, 1/k]$ occurs at an endpoint. Cauchy–Schwarz, applied to the two terms with denominators σ, ρ , gives

$$\frac{1}{\sigma} + \frac{v}{\rho} \geq (1 + \sqrt{v})^2. \quad (189)$$

¹²Checker: `proof/bound/simplify/verify_compact_identities.py`.

¹³Checker: `proof/bound/global/check_reduction_identities.py`.

At $v = k$, substitute (189) into s_2 and use $r \geq 3/10$, $k \geq 3/7 > (13/20)^2$ to obtain

$$s_2 \leq 1 - k - 2r\sqrt{k} \leq 1 - 3/7 - 39/100 = 127/700 < r.$$

At $v = 1/k$, the same substitution gives $s_2 \leq 1/r - 2 - 2\sqrt{r(1-r)} < r$. To verify the last inequality, both sides of $1/r - 2 - r < 2\sqrt{r(1-r)}$ are positive on this interval; squaring and multiplying by r^2 gives $5r^4 + 2r^2 - 4r + 1 < 0$. This polynomial is negative at r_0 , and its derivative $20r^3 + 4r - 4$ is at most $-52/27$ for $r \leq 1/3$. The proof for s_1 exchanges ρ, σ and u, v . In particular, (188) does not assume $\rho \leq 1/2$.

To exclude $S, T \geq r$ simultaneously, first exchange all player labels, if necessary, so that $\rho \leq 1/2$. This temporary labeling does not assume an order between S, T . Under $S \geq r$, we will prove $u < 99/100$ before using a separate implication valid over the full range of ρ . Define the uncapped expression for h_1 and the affine expression for S obtained from it by

$$\begin{aligned} u_c &= 99/100, & \bar{h}_1(v) &= 1 - r(1 + \sigma)/\rho + \sigma(1 - r)v/\rho, \\ S_0(v) &= \bar{h}_1(v) - r, & S_u(v) &= s_2 - r < 0, \\ S^{\text{raw}}(u, v) &= S_0(v) + uS_u(v). \end{aligned} \tag{190}$$

The superscript denotes replacement of h_1 by its uncapped formula. Since $h_1 \leq \bar{h}_1$, equation (159) gives $S \leq S^{\text{raw}}$. Its slope in v at u_c is positive:

$$\begin{aligned} A_v &= \frac{\sigma(1-r)}{\rho} + u_c\chi_1 = \frac{(1-u_c)\sigma(1-r) + u_cc}{\rho} > 0, \\ v_s &= \frac{r - S_0(0) - u_cS_u(0)}{A_v}, & S^{\text{raw}}(u_c, v_s) &= r. \end{aligned} \tag{191}$$

Thus v_s is the value at which this uncapped expression equals r . If $u \geq u_c$ and $S \geq r$, the negative u -slope in (190) gives

$$S^{\text{raw}}(u_c, v) \geq S^{\text{raw}}(u, v) \geq S \geq r, \quad v \geq v_s. \tag{192}$$

We must obtain the left-segment condition with its original guards before contradicting it. Write the central column numerator as

$$\begin{aligned} N_C &= C_b(v) + uC_s(v), \\ C_b(v) &= \rho^2\chi_1 + \sigma(\rho - 2r)v, & C_s(v) &= \rho^2c + \rho\sigma(\chi_2 - r)v, \\ (N_C)_v &= \sigma(\rho - 2r) + \rho\sigma(\chi_2 - r)u < 0. \end{aligned} \tag{193}$$

The subscripts b, s distinguish the constant and u -slope of the affine expression. The derivative sign follows from $\rho \leq 1/2 < 2r$, $u > 0$, and (182). If $C_s(v) \leq 0$, equations (192) and (193) imply $N_C(u, v) \leq N_C(u_c, v) \leq N_C(u_c, v_s)$.

If $C_s(v) > 0$, use $S \geq r$ to bound u before substitution into N_C . On $v \leq v_0$, $h_1 = \bar{h}_1$ and $u \leq (S_0 - r)/(-S_u)$; on $v \geq v_0$, $h_1 = 1$ and $u \leq c/(-S_u)$. The denominator is positive by (188). Consequently

$$\begin{aligned} P_{\text{raw}} &= (-S_u)C_b + C_s(S_0 - r), & N_C &\leq P_{\text{raw}}/(-S_u) \quad (v \leq v_0), \\ P_{\text{clip}} &= (-S_u)C_b + cC_s, & N_C &\leq P_{\text{clip}}/(-S_u) \quad (v \geq v_0). \end{aligned} \tag{194}$$

These inequalities follow by replacing u in (193) by its respective upper bound. Both numerator polynomials have degree at most two in v .

To state their sign checks exactly, for a quadratic polynomial P on an interval $[a, b]$ with $a \leq b$, define

$$\begin{aligned}\beta_0(P; a, b) &= P(a), \\ \beta_1(P; a, b) &= P(a) + (b - a)P'(a)/2, \\ \beta_2(P; a, b) &= P(b), \\ P(a + (b - a)t) &= \beta_0(P; a, b)(1 - t)^2 + 2\beta_1(P; a, b)t(1 - t) + \beta_2(P; a, b)t^2.\end{aligned}\tag{195}$$

For $0 \leq t \leq 1$, negative coefficients imply a negative polynomial by (179). Unit `canonical-u-cap`¹⁴ reconstructs the following seven sign checks:

$$\begin{aligned}N_C(u_c, v_s) &< 0, \\ \beta_j(P_{\text{raw}}; v_s, v_0) &< 0 \quad (j = 0, 1, 2; \ v_s \leq v_0), \\ \beta_j(P_{\text{clip}}; v_0, 1/k) &< 0 \quad (j = 0, 1, 2).\end{aligned}\tag{196}$$

Their exact parameter domain is $619/2000 \leq r \leq 1/3$, $r \leq \rho \leq 1/2$, covered by

$$r = 619/2000 + (1/3 - 619/2000)t_r, \quad \rho = r + (1/2 - r)t_\rho, \quad 0 \leq t_r, t_\rho \leq 1.\tag{197}$$

It covers the required domain because $619/2000 < r_0$. The denominators cleared in these checks are positive products of $\rho, \sigma, r, 1 - r$ and the factor $J_{\text{CN}} = 100\rho A_v > 0$. There are 352 strictly negative rational Bernstein coefficients. The interval $[v_0, 1/k]$ is nonempty on this domain. If $v_s > v_0$, no uncapped- h_1 point can satisfy $v \geq v_s$, so that case needs no first interval check. Equations (194) and (196), together with the $C_s \leq 0$ case, prove $N_C < 0$. The original segment alternative (187) now gives $L \geq 0$.

To contradict that inequality, expand L about (u_c, v_s) :

$$\begin{aligned}L(u_c + d_u, v_s + d_v) &= b_{00} + b_{10}d_u + b_{01}d_v + b_{20}d_u^2 + b_{11}d_u d_v + b_{21}d_u^2 d_v, \\ d_u = u - u_c &\geq 0, \quad d_v = v - v_s \geq 0.\end{aligned}\tag{198}$$

Here b_{ij} is the coefficient of $d_u^i d_v^j$, equivalently the corresponding derivative of L at (u_c, v_s) divided by $i!j!$. The same verification unit reconstructs these six coefficients and proves each strictly negative on (197). Every monomial is nonnegative and $b_{00} < 0$, so (198) gives $L < 0$. We have proved

$$\Sigma_r, \quad r_0 \leq r \leq 1/3, \quad \rho \leq 1/2, \quad S \geq r \implies u < 99/100.\tag{199}$$

The next implication supplies the contradiction if $T \geq r$ also. It uses the explicit capped formulas (159) and holds on the full stationary-weight range:

$$\begin{aligned}r_0 \leq r \leq 1/3, \quad r \leq \rho \leq 1 - r, \quad \sigma = 1 - \rho, \quad 0 \leq u \leq 1, \quad v \geq 0, \\ S \geq r \implies T < r.\end{aligned}\tag{200}$$

In particular, it requires no second-round weight comparison. Define the other uncapped expression and replace both capped formulas in S, T by their uncapped expressions:

$$\begin{aligned}\bar{h}_2(u) &= 1 - r(1 + \rho)/\sigma + \rho(1 - r)u/\sigma, \\ \bar{S}(u, v) &= \bar{h}_1(v) - r + u(s_2 - r), \\ \bar{T}(u, v) &= \bar{h}_2(u) - r + v(s_1 - r).\end{aligned}\tag{201}$$

¹⁴Checker: `proof/bound/simplify/replay-relaxed-source-knot-u-cap.py`.

Thus $S \leq \bar{S}$ and $T \leq \bar{T}$. Subscripts u, v on these functions denote partial derivatives. Their slopes in v have the endpoint values

$$\begin{aligned} \bar{S}_v \Big|_{u=0} &= \sigma(1-r)/\rho > 0, & \bar{S}_v \Big|_{u=1} &= c/\rho > 0, \\ \bar{T}_v \Big|_{u=0} &= \chi_1 - r < 0, & \bar{T}_v \Big|_{u=1} &= \chi_1 + \chi_2 - r < 0. \end{aligned} \quad (202)$$

The slopes are affine in u , so these signs hold for $0 \leq u \leq 1$. The first negative sign uses (182); the second uses $\rho\sigma \leq 1/4$ to give $\chi_1 + \chi_2 - r = 2 - 3r - r/(\rho\sigma) \leq 2 - 7r < 0$. These arguments do not need bounds on v .

Because $\bar{S}_v > 0$, the equation $\bar{S}(u, v) = r$ has the unique solution

$$v_b(u) = \frac{r - \bar{S}(u, 0)}{\bar{S}_v(u)}. \quad (203)$$

Its numerator and denominator are affine in u , so the derivative of their ratio has constant sign. To control $\bar{T}$ along this boundary, the unit **full-mass-both-high**¹⁵ checks

$$\begin{aligned} 0 < v_b(0), v_b(1) < 7/3, & \quad \bar{T}(1, v_b(1)) < r, \\ \mathcal{J}_{\text{cap}} = \bar{T}_u \bar{S}_v - \bar{T}_v \bar{S}_u > 0 & \quad (0 \leq u \leq 1, \ 0 \leq v \leq 7/3). \end{aligned} \quad (204)$$

The determinant $\mathcal{J}_{\text{cap}}$ combines the two rows of partial derivatives of $\bar{T}, \bar{S}$. Its positive denominator is $\rho^2 \sigma^2$; the endpoint expressions also have positive rational denominators. The exact coefficient-check domain is $r \in [r_0, 1/3]$, $\rho \in [r, 1-r]$, $u \in [0, 1]$, $v \in [0, 7/3]$. Since v_b stays between its endpoint values, this domain contains the whole boundary curve. If $S \geq r$, then $\bar{S} \geq r$ and (203) imply $v \geq v_b(u)$. Using the negative slope (202) and differentiating $\bar{S}(u, v_b(u)) = r$ gives

$$\begin{aligned} T &\leq \bar{T}(u, v) \leq \bar{T}(u, v_b(u)), \\ \frac{d}{du} \bar{T}(u, v_b(u)) &= \frac{\mathcal{J}_{\text{cap}}}{\bar{S}_v} > 0. \end{aligned} \quad (205)$$

Hence $T \leq \bar{T}(1, v_b(1)) < r$, proving (200), even for $v > 7/3$.

If $S, T \geq r$ in the temporary labeling with $\rho \leq 1/2$, equation (199) makes (200) applicable and contradicts $T \geq r$. The impossibility of both inequalities is invariant under exchanging the players. Together with $\max\{S, T\} \geq r$, this proves

$$\Sigma_r, \quad r_0 \leq r \leq 1/3 \quad \implies \quad (S > r > T) \text{ or } (T > r > S). \quad (206)$$

For strictness, $S = r > T$ would give $H = \alpha r + (1 - \alpha)T < r$, because $\alpha \leq 1 - r < 1$; the case $T = r > S$ similarly contradicts $K \geq r$.

E.2 Eliminating the second-round weights

The purpose of this reduction is to eliminate α, γ while preserving enough necessary inequalities for the later paths. By (206), exchange all player labels, if necessary, to obtain $S > r > T$. This exchange acts simultaneously:

$$(\rho, \sigma, \eta, \xi, \alpha, \gamma, u, v, s_1, s_2, h_1, h_2, S, T) \longleftrightarrow (\sigma, \rho, \xi, \eta, \gamma, \alpha, v, u, s_2, s_1, h_2, h_1, T, S). \quad (207)$$

¹⁵Checker: `proof/bound/simplify/replay-fullmass-both-high.py`.

The stationary row weight after this exchange can exceed $1/2$. We therefore use the full domain (180) throughout.

Define the two distances from r and their ratio by

$$m = S - r > 0, \quad n = r - T > 0, \quad z = \frac{n}{m} > 0, \quad A_0 = \frac{n}{m+n} = \frac{r-T}{S-T}. \quad (208)$$

This is the same coefficient A_0 used in (167); it balances S, T at r . Substitute (208) into H, K in (160). The results are

$$H - r = \alpha(m+n) - n, \quad K - r = m - \gamma(m+n), \quad \alpha \geq A_0, \quad \gamma \leq 1 - A_0, \quad 0 < A_0 \leq 1 - r. \quad (209)$$

The two weight inequalities use $H, K \geq r$ and $m+n > 0$; the upper bound on A_0 also uses $\alpha \leq 1 - r$ from (162). They do not yet justify changing the weights, because $E_1 \geq 0$ must survive.

To lower α while preserving $E_1 \geq 0$, we need its α -derivative to be negative throughout the allowed interval for γ . Two signs will imply this:

$$M = (1-r)S - (1-2r)T - rh_2/\xi < 0, \quad h_2 > \xi. \quad (210)$$

Their hypotheses differ. The sign $M < 0$ needs (180) and $T \geq 0$. The sign $h_2 > \xi$ also uses the initial nonnegativity $s_1, s_2, h_1, h_2, S, T \geq 0$, $S \geq r$, and the original $H \geq r$ with $\alpha \leq 1 - r$. These are conditions of (162) and (160), before any later path drops them. We explain the checks before using the two signs to move the weights.

First suppose $h_2 = \bar{h}_2$, using its uncapped formula in (201). For either formula of h_1 , the functions T, M are affine in u . Write

$$\begin{aligned} T &= t_0 + t_1 u, & M &= M_0 + M_1 u, \\ t_0 &= \bar{h}_2(0) - r + v(\chi_1 - r), & t_1 &= \rho(1-r)/\sigma + v\chi_2 > 0, \\ M_1 &= (1-r)(s_2 - r) - (1-2r)t_1 - r(1+v)\rho(1-r)/\sigma < 0. \end{aligned} \quad (211)$$

Here M_0 is the constant coefficient obtained by formally evaluating this affine formula at $u = 0$. To justify $t_1 > 0$, if $\chi_2 \geq 0$ both terms are nonnegative and the first is positive. If $\chi_2 < 0$, $v \leq 1/k$ gives $t_1 \geq \rho(1-r)/\sigma + \chi_2/k = (1-r)c/r > 0$. The sign of M_1 then follows from (188), $c > 0$, and $1/\xi = 1 + v$ in (96). Using $T \geq 0$ in (211) gives the exact elimination

$$M = M_0 - \frac{M_1 t_0}{t_1} + \frac{M_1}{t_1} T \leq M_0 - \frac{M_1 t_0}{t_1}. \quad (212)$$

It remains to check the sign of the right-hand side after clearing its positive denominators. The unit **mass-M-small**¹⁶ checks $619/2000 \leq r \leq 1/3$, $r \leq \rho \leq 1/2$; the unit **mass-M**¹⁷ adds $r_0 \leq r \leq 1/3$, $1/2 \leq \rho \leq 1 - r$. In both cases the h_1 intervals are $k \leq v \leq v_0$ and $v_0 \leq v \leq 1/k$. They are in this order because $v_0 \leq 1/k$ is equivalent to $r^2 \leq \sigma c$, which follows from $\sigma \geq r$ and $c \geq r$. The smaller- ρ checks have strictly negative Bernstein coefficients. In each of the two larger- ρ checks, three terms are grouped into a strictly negative quadratic; all other coefficients are negative. The grouped coefficients b_0, b_1, b_2 satisfy $b_0, b_2 < 0$ and $60b_0b_2 - 36b_1^2 > 0$, proving negativity of $b_0(1-t)^2 + 6b_1t(1-t) + 15b_2t^2$. Thus (212) proves $M < 0$ on the uncapped- h_2 branch.

On the other branch $h_2 = 1$, the definition of the cap implies $u \geq u_C$, where

$$u_C = \frac{r(1+\rho)}{\rho(1-r)}, \quad M \leq (1-r)[h_1 - r + u_C(s_2 - r)] - r(1+v). \quad (213)$$

¹⁶Checker: `proof/bound/global/replay_m_chart_certificate.py`.

¹⁷Checker: `proof/bound/global/replay_m_chart_full_mass_certificate.py`.

Indeed, $S = h_1 - r + u(s_2 - r)$ decreases with u by (188), and the term $-(1 - 2r)T$ is nonpositive because $T \geq 0$. The right-hand side of (213) is affine in v on each formula of h_1 . The unit `mass-M`¹⁷ checks its three endpoint values at $v = k, v_0, 1/k$ are negative on $r_0 \leq r \leq 1/3$, $r \leq \rho \leq 1 - r$. This completes $M < 0$ on both h_2 branches.

For the other derivative sign, the unit `mass-h_2-xi`¹⁸ excludes $h_2 \leq \xi$ under the hypotheses stated for (210). Its sufficient contradiction is the following conditional sign:

$$\begin{aligned} s_1, s_2, h_1, h_2, S, T \geq 0, \quad S \geq r, \quad h_2 \leq \xi \\ \implies Z_\xi = (1 - r)S + (1 + r)T - r < 0, \end{aligned} \quad (214)$$

on (180). To specify the substitutions in this check, put $x = \xi$ and regard h_2 temporarily as a coordinate. Since $h_2 \leq x \leq 1 - r < 1$, its uncapped formula applies:

$$\begin{aligned} u &= \frac{\sigma h_2 - \sigma + r(1 + \rho)}{\rho(1 - r)}, \quad v = \frac{1 - x}{x}, \\ s_1 &= c_0 + c_1 h_2, \quad c_0 = \chi_1 - \frac{\chi_2[\sigma - r(1 + \rho)]}{\rho(1 - r)}, \quad c_1 = \frac{\sigma(1 - r) - r}{\rho(1 - r)}. \end{aligned} \quad (215)$$

The weight ranges give $c_1 < 1$, $c_0 < r$, and $\partial_{h_2} T = 1 + v c_1 > 0$. For $c_1 < 1$, use $(1 - 2\rho)(1 - r) < r$, a consequence of (182). For $\rho \leq 1/2$, $\chi_2 \geq 0$, $\sigma - r(1 + \rho) \geq 0$, and $\chi_1 \leq 1 - 3r < r$ give $c_0 < r$. For $\rho \geq 1/2$, $\chi_1 \leq 19/70$. If the product subtracted in c_0 is negative, then $0 < \chi_2 \leq 1/10$, $0 < r(1 + \rho) - \sigma < 1/20$, and $\rho(1 - r) \geq 1/3$; hence $c_0 \leq 19/70 + 3/200 = 401/1400 < 3/10 \leq r$. If that product is nonnegative, $c_0 \leq \chi_1 < r$ directly. Finally, $c_1 \geq 0$ when $\rho \leq 1/2$; otherwise $c_1 \geq -2r^2/(1 - r)$ and $v \leq 1/k$ give $1 + v c_1 \geq 1 - 2r > 0$.

Increasing h_2 to x therefore gives the upper bound

$$T \leq \hat{T}(x) = x - r/x + (1 - x)(c_0 + c_1 x)/x, \quad \hat{T}'(x) = 1 - c_1 + (r - c_0)/x^2 > 0. \quad (216)$$

If $h_1 = 1$, the switch formula (166) gives $x \leq \theta = \sigma(1 - r)/(\sigma + r)$. The first two sign checks in `mass-h_2-xi`¹⁸ certify $\hat{T}(\theta) < 0$, contradicting $T \geq 0$. Their rational boxes are $r \in [3/10, 1/3]$, $\rho \in [3/10, 1/2]$ and $r \in [3/10, 1/3]$, $\rho \in [1/2, 7/10]$. The numerators have positive Bernstein coefficients and their denominators are negative.

Thus h_1 is uncapped. Equations (159) and (188) give $S < h_1 - r$, so $S \geq r$ implies $h_1 > 2r$. Solving these two strict bounds on the uncapped formula for h_1 gives

$$\theta < x < \theta_2, \quad \theta = \frac{\sigma(1 - r)}{\sigma + r}, \quad \theta_2 = \frac{\sigma(1 - r)}{1 + r - 2\rho(1 - r)}. \quad (217)$$

The denominator of θ_2 is positive on the stated domain. After (215), differentiation at fixed x gives

$$\partial_{h_2} Z_\xi = (\sigma/\rho)(s_2 - r) + (1 + r)(1 + v c_1) > 0.$$

For $\rho \leq 1/2$, $s_2 \geq 0$ and $c_1 \geq 0$ give a lower bound $1 + r - \sigma r/\rho \geq 2r > 0$. For $\rho \geq 1/2$, use $\sigma/\rho \leq 1$, $s_2 \geq 0$, and $v c_1 \geq -2r$ to get $1 - 2r - 2r^2 \geq 1/9 > 0$. Thus substituting $h_2 = x$ increases Z_ξ . The remaining two sign checks substitute $x = \theta + (\theta_2 - \theta)t$, $0 \leq t \leq 1$, and prove that even this increased expression is negative. Their exact boxes are $r \in [4/13, 1/3]$, $\rho \in [3/10, 1/2]$, $t \in [0, 1]$ and $r \in [619/2000, 1/3]$, $\rho \in [1/2, 7/10]$, $t \in [0, 1]$. After clearing positive denominators, the numerators of $-Z_\xi$ have positive Bernstein coefficients. These four boxes cover all weights needed for (214).

¹⁸Checker: `proof/bound/simplify/replay-h_2-xi-certificates.py`.

In our ordered case $S > r > T$, the original $H \geq r$ and $\alpha \leq 1 - r$ imply $r \leq H \leq (1 - r)S + rT$. Together with $T \geq 0$, this gives $Z_\xi \geq T \geq 0$, contradicting (214). Hence $h_2 > \xi$, completing (210).

Now differentiate the particular expression E_1 in (160). Using $1/\xi = 1 + v$ gives

$$\begin{aligned}\partial_\alpha E_1 &= (1 - \gamma)(S - T) + \gamma(1 + T - h_2/\xi) - r, \\ \partial_\alpha E_1|_{\gamma=r} &= M < 0, \\ \partial_\alpha E_1|_{\gamma=1-A_0} &= \frac{(S - r)(1 - h_2/\xi) + T(T - r)}{S - T} < 0.\end{aligned}\tag{218}$$

The second endpoint is negative because $S - r > 0$, $S - T > 0$, $0 \leq T < r$, and $h_2 > \xi$. The derivative is affine in γ , so it is negative throughout $r \leq \gamma \leq 1 - A_0$. Lowering α to A_0 therefore cannot decrease E_1 . It cannot decrease $U = \alpha s_1 + (1 - \alpha)h_2$ either: $s_1 < r$ by (188), and $h_2 > \xi \geq r$ by (210).

After this move E_1 is affine in γ , with

$$E_1(A_0, 0) = -A_0 r < 0.\tag{219}$$

Its nonnegativity at the present $\gamma > 0$ therefore forces its γ -slope to be positive. Raising γ to $1 - A_0$ preserves $E_1 \geq 0$ and does not affect U . We have justified $\alpha = A_0$, $\gamma = 1 - A_0$ by necessary comparisons, without assuming the observed numerical equalities.

The inequalities needed next are $U(A_0) \geq r$, $A_0 \leq 1 - r$, and $E_1(A_0, 1 - A_0) \geq 0$. Their positive rescalings are

$$\begin{aligned}G &= (m + n)[U(A_0) - r] = m(h_2 - r) - n(r - s_1) \geq 0, \\ B_{\text{mass}} &= (m + n)[(1 - r) - A_0] = (1 - r)m - rn \geq 0, \\ E &= \frac{E_1(A_0, 1 - A_0)}{(1 - A_0)^2} = v(h_2 - r) + z[c + v(s_1 - 2r)] - rz^2 \geq 0.\end{aligned}\tag{220}$$

Thus B_{mass} is exactly the upper coefficient bound $A_0 \leq 1 - r$, while B_{mix} is the distinct segment determinant in (185). At the initial point, changing only α, γ leaves all caps and segment guards unchanged. For the subsequent paths we use the following necessary domain, with the functions still defined by (159):

$$\begin{aligned}r_0 \leq r \leq 1/3, \quad r \leq \rho \leq 1 - r, \quad k \leq u, v \leq 1/k, \quad \sigma = 1 - \rho, \quad k = r/(1 - r), \\ m = S - r > 0, \quad n = r - T > 0, \quad T \geq 0, \quad s_2 \geq 0, \quad (\rho \leq 1/2 \implies s_1 \geq 0), \\ G \geq 0, \quad B_{\text{mass}} \geq 0, \quad E \geq 0, \\ (N_R \geq 0 \wedge L \geq 0) \quad \text{or} \quad (N_R < 0 \leq N_C \wedge B_{\text{mix}} \geq 0).\end{aligned}\tag{221}$$

The original point also satisfies $s_1 \geq 0$ for $\rho \geq 1/2$, but the paths on that half use the weaker system without it. The guarded alternatives specify how a segment inequality is chosen at the start; each later path will state which guards it needs throughout the move. We have dropped E_2 and the remaining weighted comparisons. We have also dropped the lower bound $A_0 \geq r$, which has not been proved for this reduction. The second-round weights have therefore disappeared, leaving only (r, ρ, u, v) .

We next prove $u < 1$, so that h_2 always uses its uncapped formula in this reduced problem. For $\rho \leq 1/2$, assume $u \geq u_c$. The column-numerator checks (196), using $S > r$, force $N_C < 0$. The opposite guarded alternative in (221) is then impossible, and the first gives $L \geq 0$. The negative coefficients (198) contradict this inequality. Thus $u < u_c < 1$ on this half.

For $\rho \geq 1/2$, replace both capped expressions by (201) and define

$$\begin{aligned} B_{\text{mass}}^{\text{raw}} &= (1-r)\bar{S} + r\bar{T} - r = \beta_0^{\text{mass}} + \beta_1^{\text{mass}}v, \\ L &= \ell_0 + \ell_1v. \end{aligned} \tag{222}$$

The four coefficients are the constant and linear coefficients in v , and depend on (r, ρ, u) . Since $S \leq \bar{S}$ and $T \leq \bar{T}$, $B_{\text{mass}} \leq B_{\text{mass}}^{\text{raw}}$. The unit **fixed-v-and-u1**¹⁹ certifies, on $619/2000 \leq r \leq 1/3$, $1/2 \leq \rho \leq 1-r$, $1 \leq u \leq 1/k$,

$$\beta_1^{\text{mass}} > 0, \quad \ell_1 < 0, \quad \Omega = \beta_0^{\text{mass}}\ell_1 - \beta_1^{\text{mass}}\ell_0 > 0. \tag{223}$$

If $L \geq 0$, its affine formula gives $v \leq -\ell_0/\ell_1$. Substitution into the increasing affine expression in (222) yields

$$B_{\text{mass}} \leq B_{\text{mass}}^{\text{raw}} \leq \Omega/\ell_1 < 0,$$

contradicting (221).

The same unit supplies the explicit alternative for $B_{\text{mix}} \geq 0$. Write $B_{\text{mix}} = b_0^{\text{mix}} + b_1^{\text{mix}}v + b_2^{\text{mix}}v^2$ by collecting powers of v . On the same (r, ρ, u) domain, it certifies

$$\begin{aligned} Q_0 &= (\beta_1^{\text{mass}})^2 b_0^{\text{mix}} - \beta_0^{\text{mass}} \beta_1^{\text{mass}} b_1^{\text{mix}} + (\beta_0^{\text{mass}})^2 b_2^{\text{mix}} < 0, \\ Q_1 &= \beta_1^{\text{mass}} b_1^{\text{mix}} - 2\beta_0^{\text{mass}} b_2^{\text{mix}} < 0, \\ Q_2 &= b_2^{\text{mix}} < 0. \end{aligned} \tag{224}$$

The nonnegative coefficient bound in (221) implies $B_{\text{mass}}^{\text{raw}} \geq 0$, hence $v \geq v_W = -\beta_0^{\text{mass}}/\beta_1^{\text{mass}}$. Putting $d = v - v_W \geq 0$ gives the exact identity

$$B_{\text{mix}}(v_W + d) = \frac{Q_0}{(\beta_1^{\text{mass}})^2} + \frac{Q_1}{\beta_1^{\text{mass}}}d + Q_2d^2 < 0.$$

This contradicts the opposite segment inequality. Neither argument uses a sign of N_C , so both are valid after the weight change. We have proved $u < 1$ throughout (221).

Finally, reaching the cap $h_2 = 1$ would require

$$u \geq \frac{r(1+\rho)}{\rho(1-r)} \geq \frac{r(2-r)}{(1-r)^2} > 1. \tag{225}$$

The middle inequality uses $\rho \leq 1-r$; the last is equivalent to $4r - 2r^2 - 1 > 0$, which is already positive at $r = 3/10$ and increases for $r \leq 1/3$. Since $u < 1$, h_2 is uncapped. Only the change of formula for h_1 at $v = v_0$ remains. The next paths show how every remaining point reaches that boundary or a boundary system that is excluded directly.

E.3 Paths to the switch boundary

We now show that a point satisfying the reduced necessary conditions can be moved to $v = v_0$, or to one of two boundaries excluded below. The moves vary scalar variables and preserve the specified inequalities, producing new points in the necessary system. Throughout, r and ρ are fixed. With m, n, z, A_0 defined in (208), the conditions used by the paths are

$$\begin{aligned} r_0 \leq r \leq 1/3, \quad r \leq \rho \leq 1-r, \quad k \leq u < 1, \quad k \leq v \leq 1/k, \quad S > r > T, \\ m = S - r > 0, \quad n = r - T > 0, \quad z = n/m, \quad A_0 = z/(1+z), \quad T, s_2 \geq 0, \\ G \geq 0, \quad B_{\text{mass}} \geq 0, \quad E \geq 0, \quad L \geq 0 \text{ or } (N_R < 0 \leq N_C \text{ and } B_{\text{mix}} \geq 0). \end{aligned} \tag{226}$$

¹⁹Checker: `proof/bound/global/replay_preclip_left_projection_signs.py`.

The three inequalities involving G, B_{mass}, E are (220); the determinant polynomials are defined in (185). When $\rho \leq 1/2$, we also use $s_1 \geq 0$. No such lower bound is imposed for $\rho > 1/2$. For differentiating E , it is convenient to define

$$F = E/z, \quad R_s = r - s_1 > 0, \quad A_h = h_2 - r > 0. \quad (227)$$

Since $z > 0$, F and E have the same sign. The symbol F here is distinct from the candidate regrets F_j^i . The preliminary bound $s_1 < r$ gives $R_s > 0$, and $G = m(A_h - zR_s) \geq 0$ gives $A_h \geq zR_s > 0$.

The switch value lies within the allowed range of v . Indeed, $v_0/k = (1+\sigma)/\sigma > 1$, and $v_0 \leq 1/k$ is equivalent to $r^2(1+\sigma) \leq \sigma(1-r)^2$. The latter follows from $\sigma \geq r$ and $r \leq 1/3$. Thus moving toward v_0 cannot leave the range in (226) through an endpoint of v .

First suppose $v \geq v_0$, so the definition of h_1 in (159) gives $h_1 = 1$. We keep u fixed and decrease v to v_0 . On the domain (226) with $v \geq v_0$, the unit `postclip-envelope`²⁰ gives

$$z_v > 0, \quad F = 0 \implies F_v < 0. \quad (228)$$

A subscript denotes a partial derivative with the other variables fixed. At a first crossing from $F \geq 0$ to $F < 0$ as v decreases, the directional derivative would be nonpositive, whereas (228) gives $-F_v > 0$. Hence $E \geq 0$ is preserved, including when the initial point satisfies $E = 0$. The variables s_1, h_2 are constant during this move, and

$$G = m(A_h - zR_s), \quad B_{\text{mass}} = m[(1-r) - rz]. \quad (229)$$

Since z decreases, both inequalities in (229) remain valid while $m > 0$. On the same domain, using $T \geq 0$ and $u < 1$, the unit `postclip-knots`²¹ gives the determinant signs

$$L_v < 0, \quad (B_{\text{mix}})_v < 0, \quad (N_R)_v \geq 0, \quad (N_C)_v < 0. \quad (230)$$

Decreasing v therefore preserves $L \geq 0$ in the first alternative of (226). In the second alternative it preserves $B_{\text{mix}} \geq 0$ and $N_R < 0 \leq N_C$. The first alternative uses $L \geq 0$ alone and does not require the initial signs of N_R, N_C to persist.

The unit `postclip-orientation`²² gives $T < r$ and checks the endpoint identities on this domain. Since $T_v = s_1 - r < 0$, decreasing v increases T and preserves $T \geq 0$. The inequality $s_2 \geq 0$ is also preserved. If $\chi_1 \leq 0$, decreasing v increases s_2 . If $\chi_1 > 0$, then $\rho > k$ and the minimum of s_2 during the move is attained at v_0 ; the identity

$$\rho\sigma(1-r) s_2|_{v=v_0} = r(1-3r) + (1-r)(\rho-k)(1-r-\rho) \geq 0 \quad (231)$$

establishes that minimum's nonnegativity. At a possible limiting point, $n = 0$ would mean $T = r$, contrary to the cap bound. If $m = 0$ while $n > 0$, then $z = n/m$ would diverge, contrary to its decrease in (228). The denominator $m = S - r$ therefore remains positive throughout this move, including its endpoint. The endpoint satisfies $S > r > T$, $0 < A_0 \leq 1 - r$, $h_2 > r$, $E \geq 0$, and one determinant alternative in (226).

Now suppose $v < v_0$, so h_1 equals the affine expression inside the minimum in (159). We keep v fixed and increase u until a determinant reaches zero. The unit `fixed-v-and-u`¹⁹ reconstructs the source and derivative signs used in this move. Here $S_u = s_2 - r < 0$, $T_u = \rho(1-r)/\sigma + v\chi_2$, and

$$z_u = \frac{-T_u + z(r - s_2)}{m}, \quad T_u R_s - A_h(r - s_2) > 0, \quad \partial_u(A_h/R_s) > 0. \quad (232)$$

²⁰Checker: `proof/bound/global/replay_postclip_crossing_certificate.py`.

²¹Checker: `proof/bound/global/replay_postclip_knot_direction.py`.

²²Checker: `proof/bound/global/replay_postclip_orientation_certificate.py`.

The inequality $G \geq 0$ in (229) is $z \leq A_h/R_s$. Substituting it into (232) gives $z_u < 0$; the increasing upper bound A_h/R_s then preserves $G \geq 0$. For $E \geq 0$, differentiation of $F = E/z$ gives

$$F_u = v(\chi_2 + (h_2)_u/z) - (r + vA_h/z^2)z_u, \quad (h_2)_u R_s + \chi_2 A_h > 0. \quad (233)$$

When $\chi_2 \geq 0$, the first bracket is positive because $(h_2)_u = \rho(1-r)/\sigma > 0$. When $\chi_2 < 0$, the positive quantities A_h, R_s, z and $z \leq A_h/R_s$ give

$$\chi_2 + (h_2)_u/z \geq [\chi_2 A_h + (h_2)_u R_s]/A_h > 0.$$

Both terms in (233) are thus positive, so $F_u > 0$ preserves $E \geq 0$. The decrease of z also preserves $B_{\text{mass}} \geq 0$ through (229). These sign checks cover the full range $r \leq \rho \leq 1-r$ and allow the affine expression for s_1 to become negative when $\rho > 1/2$.

During this move, s_2 is constant and the checked sign $T_u > 0$ preserves $T, s_2 \geq 0$. For $\rho \leq 1/2$, $\chi_2 \geq 0$ also preserves $s_1 \geq 0$. In the alternative $N_R < 0 \leq N_C$, the checked sign $(N_R)_u < 0$ preserves $N_R < 0$. If N_C reached zero, then (185) would give $B_{\text{mix}} = rvN_R < 0$, contradicting the determinant inequality. The previously proved implication $S \geq r, u \leq 1 \Rightarrow T < r$ excludes $n = 0$. A limit with $m = 0$ and $n > 0$ would contradict the decrease of z . Finally, the bounds using B_{mass} and the determinant at $u = 1$ exclude that endpoint, as established in the weight reduction. Thus the move reaches $L = 0$ or $B_{\text{mix}} = 0$ before any other condition can fail.

For $\rho \leq 1/2$, the unit **canonical-alignment**²³ gives $S \geq r, u \leq 1, v < v_0 \Rightarrow N_C < 0$ on the range in (226); the first segment condition therefore gives $L \geq 0$. We choose a contact approached from $L > 0$ as u increases, so $L_u \leq 0$ there. If the initial point already has $L = 0$ and $L_u \leq 0$, it is used immediately. If instead $L = 0$ and $L_u > 0$, the move first enters $L > 0$ and continues to the next contact. We now justify the additional bounds needed to continue along $L = 0$. They are proved before the continuation checker is applied.

First write the polynomial L , which is affine in v , as

$$L = \ell_0 + \ell_1 v, \quad \ell_0 = \sigma\chi_2(\sigma + \rho u)(\chi_1 + cu), \quad \ell_1 = L_v. \quad (234)$$

The inequalities $\chi_2 \geq 0$ and $s_1 \geq 0$ give $\chi_1 + cu = s_1 + (\rho r/\sigma)u > 0$. Consequently $\ell_0 > 0$, except at $r = 1/3, \rho = \sigma = 1/2$. At that corner $L = -u^2 v/18 < 0$, so no contact is possible. At a contact with $v > 0$,

$$\ell_1 = -\ell_0/v < 0, \quad d_L = -\rho L_v = \rho \ell_0/v > 0, \quad v = \rho \ell_0/d_L. \quad (235)$$

This proves that division in the affine graph is valid before any additional bound on z is used.

To obtain $A_0 > r$, define

$$W_2 = rS + (1-r)T - r, \quad A_0 - r = -W_2/(S-T). \quad (236)$$

On the present domain, the checked bound $T_u \geq r$ and $s_2 \geq 0$ imply

$$(W_2)_u = r(s_2 - r) + (1-r)T_u \geq rs_2 + r(1-2r) > 0.$$

The unit **canonical-A_0-stop**²⁴ gives $W_2(9/10, k) < 0$ and $W_2(9/10, v_0) < 0$. Since W_2 is affine in v , it is negative at $u = 9/10$ throughout $k \leq v \leq v_0$. Thus $W_2 \geq 0$ requires $u > 9/10$. Writing $W_2 = w_0 + w_1 v$, where w_0, w_1 are its constant and linear coefficients, the checks on $9/10 < u < 1$ give

$$w_0 < 0, \quad \ell_1 < 0, \quad \Omega_2 = w_0 \ell_1 - w_1 \ell_0 > 0. \quad (237)$$

²³Checker: `proof/bound/preclip_canonical_alignment.py`.

²⁴Checker: `proof/bound/simplify/replay-A_0-mass-stop-exclusion.py`.

If $W_2 \geq 0$, then $w_1 > 0$ and $Lw_1 = \ell_1 W_2 - \Omega_2 < 0$, contradicting $L = 0$. By (236), this proves

$$A_0 > r, \quad z > k. \quad (238)$$

The equivalence uses $A_0 = z/(1+z)$ and $k = r/(1-r)$.

To bound z from above, use the quadratic obtained from E in (220):

$$\mathcal{E}(t) = vA_h + [c - v(r + R_s)]t - rt^2 \quad (t \in \mathbb{R}), \quad E = \mathcal{E}(z). \quad (239)$$

On the graph in (235), the unit **canonical-E1**²⁵ proves

$$\mathcal{P}_{\text{arc}} = \rho^2 \sigma^2 d_L [-\mathcal{E}(1) - 2G] > 0. \quad (240)$$

The identity defines $\mathcal{P}_{\text{arc}}$ after graph substitution; its clearing factor is positive. Its 600 rational Bernstein coefficients are checked on $r \in [619/2000, 1/3]$, $\rho \in [3/10, 1/2]$, $u \in [2/5, 1]$, a box containing the present domain because $r \geq r_0 > 619/2000$, $\rho \geq r > 3/10$, and $u \geq k > 2/5$. Of the 600 coefficients, 582 are individually positive. The remaining terms form six quadratic groups. For each group with coefficients b_2, b_3, b_4 , the checker verifies $b_2, b_4 > 0$ and $3b_2b_4 - 2b_3^2 > 0$. For its unit coordinate $w \in [0, 1]$, the factor

$$6b_2(1-w)^2 + 4b_3w(1-w) + b_4w^2$$

is positive because its matrix determinant is $2(3b_2b_4 - 2b_3^2) > 0$. The other Bernstein factors are nonnegative. This proves (240) without requiring each individual coefficient to be positive. Since $G \geq 0$, it follows that $\mathcal{E}(1) < 0$. The identity $\mathcal{E}(z) = z\mathcal{E}(1) + (1-z)(vA_h + rz)$ would then give $E < 0$ for $z \geq 1$. Hence

$$z < 1, \quad B_{\text{mass}} = m[(1-r) - rz] > mc > 0. \quad (241)$$

The second inequality shows that the mass condition is automatic on this part of $L = 0$.

One further bound excludes small v . On the branch $v < v_0$, $S_u = s_2 - r < 0$ and $S_v > 0$, so $u \geq k$ and $v \leq 4/5$ would imply $S \leq S|_{u=k, v=4/5}$. The unit **canonical-boundary-stops**²⁶ checks the fifteen coefficients proving $r - S|_{u=k, v=4/5} > 0$. Consequently,

$$S \geq r \implies v > 4/5. \quad (242)$$

We can now continue the contact along $L = 0$ toward v_0 . The preceding calculations justify all conditions of the 5,098-leaf continuation certificate:

$$\begin{aligned} r_0 \leq r \leq 1/3, \quad r \leq \rho \leq 1/2, \quad L = 0, \quad L_u \leq 0, \quad d_L > 0, \\ m, n > 0, \quad T, s_1, s_2 \geq 0, \quad G, E \geq 0, \quad k \leq u < 1, \quad k < z < 1, \quad 4/5 < v < v_0, \quad k \leq v \leq 1/k. \end{aligned} \quad (243)$$

On this domain, the unit **canonical-upper-arc**²⁷ gives the exact interval and Bernstein checks

$$L_u < 0, \quad L_v < 0, \quad u > k, \quad s_1 > 0, \quad s_2 > 0, \quad E = 0 \implies E_v - E_u L_v / L_u > 0. \quad (244)$$

The implicit-function theorem [33] gives a curve $L = 0$ with increasing v and $du/dv = -L_v/L_u < 0$. At $E = 0$, the last sign in (244) is precisely $dE/dv > 0$ and excludes exit through $E = 0$.

²⁵Checker: `proof/bound/simplify/replay-arc-e1-source-budget.py`.

²⁶Checker: `proof/bound/simplify/replay-preclip-stop-events.py`.

²⁷Checker: `proof/bound/kink/replay-preclip_bernstein_interval.py`.

For completeness, we check the other possible exits. The calculations establishing (238), (241), and (242) apply at every point satisfying the same inequalities on $L = 0$. If $G = 0$, then $A_h = zR_s$ and

$$E = z[c - r(v + z)] < 0,$$

because $v + z > 4/5 + k > c/r$ for $r \geq r_0$, contradicting $E \geq 0$. If $T = 0$, then $z = r/(S - r) < 1$ gives $S > 2r$, whereas the separate cap check in **canonical-boundary-stops**²⁶ at $T = 0$ gives $S < 2r$. The strict signs in (244) exclude $u = k$, $s_1 = 0$, $s_2 = 0$, $L_u = 0$, and $L_v = 0$. The implication $S \geq r$, $u \leq 1 \Rightarrow T < r$ excludes $n = 0$, and $z < 1$ then excludes $m = 0$. Equation (241) excludes $B_{\text{mass}} = 0$. The domain bounds and these strict exclusions prevent a limiting stop before v_0 ; the continuation therefore reaches $v = v_0$.

For $\rho \geq 1/2$, we instead exclude the two contact boundaries directly. Their common conditions are

$$\begin{aligned} r_0 \leq r \leq 1/3, \quad 1/2 \leq \rho \leq 1 - r, \quad k \leq u < 1, \quad k \leq v \leq v_0, \\ S > r > T, \quad T, s_2 \geq 0, \quad G, B_{\text{mass}}, E \geq 0. \end{aligned} \quad (245)$$

The unit **high-rho-left**²⁸ adds $L = 0$ to (245) and needs no signs on N_R, N_C . The unit **high-rho-below**²⁹ adds $B_{\text{mix}} = 0$ and $N_R < 0 \leq N_C$. Both domains allow $s_1 < 0$ and $A_0 < r$. The two exclusion certificates have 19,771 and 4,945 leaves; the continuation certificate has 5,098 leaves. Each leaf is checked by exact rational interval or Bernstein bounds. Substitution in an affine graph such as $L = 0$ is made only after its linear coefficient has a certified nonzero sign. A box containing a possible zero coefficient is split or checked without division. The independent replayers check all leaf inequalities and the complete binary partitions, including these division conditions. Together, the paths and the two exclusions cover every point satisfying (226).

E.4 Exclusion on the switch boundary

We now exclude the endpoints $v = v_0$ produced by the preceding paths when $r > r_*$. The proof must exclude simultaneous inequalities: the nonexistence of a common equality root alone would not suffice. We use only the following conditions on this boundary:

$$\begin{aligned} r_0 \leq r \leq 1/3, \quad r \leq \rho \leq 1 - r, \quad v = v_0, \quad 1/2 < \eta \leq 1 - r, \\ S > r > T, \quad 0 < A_0 \leq 1 - r, \quad h_2 > r, \quad E \geq 0, \end{aligned} \quad (246)$$

together with one determinant alternative from (226). The condition $\eta > 1/2$ follows from $u < 1$. Thus h_2 equals its affine expression in (159), as in the definition of N_2 . Neither $E_2 \geq 0$ nor $s_1, s_2 \geq 0$ is required here.

The polynomial definitions in (173) and the determinant definitions in (185) agree on this boundary. For the second determinant, define N_B by

$$[q_{\dagger}(1 - r) - J_R](C_k - r) + q_{\dagger}r(R_k - r) = \frac{N_B(r, \rho, \eta)}{\sigma D_c J^2}. \quad (247)$$

The left side is the intersection determinant for the segment toward $(0, q_{\dagger})$; D_c, J, J_R, C_k, R_k were defined in the boundary polynomial calculation. Exact substitution gives

$$\begin{aligned} N_1 &= \rho \sigma \eta^2 (1 - r) L, \\ N_B &= \sigma^2 \eta (1 - r)^2 B_{\text{mix}}, \\ \Psi(A_0) &= (1 - A_0)^2 E = N_2 / D_2. \end{aligned} \quad (248)$$

²⁸Checker: `proof/bound/global/replay_preclip_left_interval.py`.

²⁹Checker: `proof/bound/global/replay_preclip_below_interval.py`.

Every factor multiplying a determinant, and D_2 , is positive under (246). Hence $L \geq 0$ gives $N_1 \geq 0$, $B_{\text{mix}} \geq 0$ gives $N_B \geq 0$, and $E \geq 0$ gives $N_2 \geq 0$.

The alternative requiring N_B can occur only when $\rho > 1/2$. To see this on $r \leq \rho \leq 1/2$, write $T = T_0 + T_1/\eta$ on $v = v_0$, where T_0, T_1 are its constant and reciprocal coefficients. Define $\eta_T = T_1/(r - T_0)$ and the polynomial $\widehat{C} = \sigma D_c J(C_k - r)$. The unit **full-rho-kink-face**³⁰ checks the boundary polynomial normalizations in (173) and (247), and gives the following signs on this smaller-weight domain:

$$T_1 > 0, \quad r - T_0 > 0, \quad \partial_\eta \widehat{C} < 0, \quad \widehat{C}(\eta_T) < 0. \quad (249)$$

Since $T < r$ implies $\eta > \eta_T$, these inequalities give $C_k < r$, or $N_C < 0$ by (184). The alternative $N_R < 0 \leq N_C$ is therefore impossible there.

Both determinant alternatives will use the same sign argument for $\Psi(A_0)$ as η increases. With its cap coefficients fixed, the quadratic Ψ satisfies $\Psi(0) = v(h_2 - r) > 0$ and $\Psi(1) = -r < 0$. It has a unique zero in $(0, 1)$, and $\Psi_a < 0$ at that zero. Holding a fixed while varying η gives

$$\partial_\eta \Psi(a)|_a = -\frac{(1-a)v}{\eta^2} \left[a\chi_2 + (1-a)\frac{\rho(1-r)}{\sigma} \right] < 0 \quad (0 \leq a \leq 1-r). \quad (250)$$

The bracket is affine in a and has positive endpoint values $\rho(1-r)/\sigma$ and $(1-r)(1-2r)$. Consequently, on any interval satisfying the conditions in (246) other than $E \geq 0$, and also $(A_0)_\eta > 0$, a zero of $\Psi(A_0)$ satisfies

$$\frac{d}{d\eta} \Psi(A_0) = \partial_\eta \Psi(a)|_{a=A_0} + \Psi_a(A_0)(A_0)_\eta < 0. \quad (251)$$

A negative value therefore cannot become nonnegative as η increases on that interval. Below we check the interval conditions separately for N_1 and N_B .

We first treat $N_1 \geq 0$. Write

$$N_1 = a_2\eta^2 + b_2\eta + c_2, \quad \Delta = (b_2^2 - 4a_2c_2)/\sigma^2, \quad (252)$$

where a_2, b_2, c_2 are functions of (r, ρ) . The quadratic is strictly concave throughout the physical domain. Indeed, expansion gives

$$a_2 = r[(1-r)\sigma^3(1-2\sigma) - r^2\rho^2(1+\sigma)].$$

For $\rho \leq 1/2$, its first term is nonpositive and its second is negative. For $\rho \geq 1/2$, the bounds $\sigma^3(1-2\sigma) \leq 1/32$ and $\rho^2(1+\sigma) \geq 1/4$ give $a_2/r \leq (1-r)/32 - r^2/4 < 0$ for $r \geq 3/10$.

Before using a root of N_1 , the unit **full-rho-kink-face**³⁰ excludes two larger parts of its domain:

$$\begin{aligned} r_0 \leq r \leq 1/3, \quad r \leq \rho \leq 21/50 &\implies b_2^2 - 4a_2c_2 < 0, \\ 8/25 \leq r \leq 1/3, \quad r \leq \rho \leq 1/2, \quad r \leq \eta \leq 1-r &\implies N_1 < 0, \\ 8/25 \leq r \leq 1/3, \quad 1/2 \leq \rho \leq 1-r &\implies \\ (N_1)_\eta(1-r) > 0, \quad N_1(1-r) < 0. \end{aligned} \quad (253)$$

The first line excludes real roots. The second directly contradicts $N_1 \geq 0$. For the third, concavity gives $(N_1)_\eta(\eta) \geq (N_1)_\eta(1-r) > 0$ when $\eta \leq 1-r$; thus $N_1(\eta) \leq N_1(1-r) < 0$ there as well.

³⁰Checker: `proof/bound/critical/replay_full_rho_kink_face.py`.

The remaining parameter pairs lie in the rational rectangles

$$\mathcal{C}_- = [r_0, 8/25] \times [21/50, 1/2], \quad \mathcal{C}_+ = [r_0, 8/25] \times [1/2, 1 - r_0], \quad (254)$$

with coordinates (r, ρ) ; $r \leq \rho \leq 1 - r$ is imposed for a physical point. On both rectangles the checks give $N_1(1/2) < 0$ and $(N_1)_\eta(1/2) > 0$. Therefore $N_1 \geq 0$ requires $\Delta \geq 0$ and

$$\eta \geq \eta_-(r, \rho) > 1/2, \quad \eta_-(r, \rho) = \frac{-b_2 + \sigma\sqrt{\Delta}}{2a_2}. \quad (255)$$

The displayed expression is the smaller root because $a_2 < 0$; when $\Delta = 0$, the two roots coincide.

The derivative checks of **full-rho-kink-face**³⁰ on (254) give

$$S_\eta > 0, \quad T_\eta < 0, \quad S|_{\eta=1/2} > r > T|_{\eta=1/2}, \quad (A_0)_\eta > 0. \quad (256)$$

The last sign holds throughout $[\eta_-, \eta]$ for a proposed feasible endpoint η . For the lower rectangle, positivity of $(A_0)_\eta$ uses $\Delta \geq 0$. For the upper rectangle, it uses $N_1(1 - r) \geq 0$, which follows from $N_1(\eta) \geq 0$ and the checked positive derivative $(N_1)_\eta(1 - r)$. Throughout $[\eta_-, \eta]$, the first three signs give $S > r > T$. The increasing A_0 stays positive and at most its final value $1 - r$. The explicit formula for h_2 gives $(h_2)_\eta < 0$, so h_2 stays above its final value r . These facts justify applying (251) on the whole interval.

It remains to establish a negative starting value at η_- . The resultant [28], which vanishes whenever N_1, N_2 share a root in η , satisfies the exact identity

$$\text{Res}_\eta(N_1, N_2) = \sigma^6 \rho^2 r^2 R_{\text{resultant}}(r, \rho). \quad (257)$$

This defines the polynomial factor $R_{\text{resultant}}$; **full-rho-kink-face**³⁰ reconstructs the resultant identity. At the isolated solution (r_*, ρ_*, η_*) of (174) in (176), exact algebra gives

$$R_{\text{resultant}}(r_*, \rho) = (\rho - \rho_*)^2 W(\rho), \quad \deg W = 18, \quad \det \frac{\partial(N_1, N_2)}{\partial(r, \eta)}(r_*, \rho_*, \eta_*) \neq 0. \quad (258)$$

The determinant permits solving locally for (r, η) as functions of ρ . The equation $H_{\text{root}} = 0$ then gives $r'(\rho_*) = 0$. Differentiating the resultant along this curve proves that its ρ derivative also vanishes there, yielding the double factor. This determinant differs from the 3×3 Jacobian used for root isolation. The unit **full-rho-kink-face**³⁰ reconstructs all nineteen Bernstein coefficients of W on $[r_*, 1 - r_*]$ and checks that they are strictly negative. It also checks the exact signs $\partial_r R_{\text{resultant}} < 0$ on both rectangles in (254). Hence

$$(r, \rho) \in \mathcal{C}_- \cup \mathcal{C}_+, \quad r > r_*, \quad r \leq \rho \leq 1 - r \quad \implies \quad R_{\text{resultant}}(r, \rho) < 0. \quad (259)$$

Indeed, such a ρ lies in $[r_*, 1 - r_*]$, and increasing r from r_* strictly decreases the resultant factor. Thus N_1, N_2 cannot have a common root in this domain.

To determine the sign, use the exact reference check in **full-rho-kink-face**³⁰:

$$0.55278962 < \eta_-(r_*, 1/2) < 0.55278965, \quad N_2(r_*, 1/2, \eta_-) < 0. \quad (260)$$

The signs $\Delta_\rho > 0$ and $\Delta_r < 0$ on (254) connect every real smaller-root point to this reference. In the lower rectangle, first increase ρ to $1/2$, then decrease r to r_* . Both moves increase Δ . In the upper rectangle, first decrease r to r_* , then decrease ρ to $1/2$. The last move stays in $\Delta > 0$ because $\Delta(r_*, \rho) \geq \Delta(r_*, 1/2) > 0$ for $\rho \geq 1/2$. The paths obey $r \leq \rho \leq 1 - r$. Their portions at $r > r_*$ have

nonzero resultant by (259); the portion at $r = r_*$ has $\rho \geq 1/2 > \rho_*$ and hence nonzero resultant by (258). Continuity of η_- and N_2 therefore propagates the negative reference sign to every such point. Points with $\Delta = 0$, where the two roots coincide, follow by continuity; $N_2 = 0$ there would also contradict the nonzero resultant. Thus (248) gives $\Psi(A_0) < 0$ at η_- . Equations (256) and (251) exclude a nonnegative value at any larger feasible η . This excludes the simultaneous conditions $N_1 \geq 0, N_2 \geq 0$.

We now treat $N_B \geq 0$ on $1/2 \leq \rho \leq 1 - r$. The polynomial in (247) is affine in η . On $r_0 \leq r \leq 1/3, 1/2 \leq \rho \leq 1 - r$, **full-rho-kink-face**³⁰ gives

$$(N_B)_\eta > 0, \quad N_B(1/2) < 0, \quad \eta_B = -\frac{N_B(r, \rho, 0)}{(N_B)_\eta} > 1/2, \quad N_B \geq 0 \implies \eta \geq \eta_B. \quad (261)$$

The zero η_B is uniquely determined by this affine equation. Define $\mathcal{G}_B$ by the exact substitution identity

$$N_2(r, \rho, \eta_B) = \frac{\sigma^3 r \mathcal{G}_B(r, \rho)}{[(N_B)_\eta]^3}. \quad (262)$$

The denominator is positive. To check $\mathcal{G}_B$ on its full physical triangle, substitute $r = r_0 + (1/3 - r_0)s$ and $\rho = 1/2 + (1/2 - r)t$, where $s, t \in [0, 1]$. After clearing positive rational constants, **full-rho-kink-face**³⁰ reconstructs all 612 Bernstein coefficients and checks that they are strictly negative. Hence

$$N_2(r, \rho, \eta_B) < 0 \quad (r_0 \leq r \leq 1/3, 1/2 \leq \rho \leq 1 - r). \quad (263)$$

The interval conditions for this alternative must be checked on the whole triangle, since (254) covers less. The checks of **full-rho-kink-face**³⁰ on this triangle give

$$\begin{aligned} S_\eta > 0, \quad T_\eta < 0, \quad S|_{\eta=1/2} \geq r > T|_{\eta=1/2}, \\ N_B(1 - r) \geq 0 \implies (A_0)_\eta > 0 \text{ for every } \eta \in [\eta_B, 1 - r]. \end{aligned} \quad (264)$$

A feasible endpoint has $\eta_B \leq \eta \leq 1 - r$, so the increasing N_B gives $N_B(1 - r) \geq 0$. The equality $S|_{\eta=1/2} = r$ occurs only at $(r, \rho) = (1/3, 2/3)$; $\eta_B > 1/2$ then ensures $S > r$ on the interval used. As in the first alternative, A_0 increases toward a final value at most $1 - r$, and h_2 decreases toward a final value greater than r . Thus all hypotheses of (251) hold from η_B to the proposed endpoint. The starting sign (263) therefore excludes $N_2 \geq 0$ at that endpoint. No additional candidate algebraic root is needed.

The two alternatives arising from the paths in (226) are now both excluded. The alternative $L \geq 0$ gives $N_1 \geq 0$ and $N_2 \geq 0$, which the first switch argument excludes. The other alternative has $B_{\text{mix}} \geq 0$ and $N_R < 0 \leq N_C$; (249) places it in the larger-weight domain, where the second switch argument excludes $N_B, N_2 \geq 0$. These conclusions use exactly the conditions preserved by the paths. Combining them with the two direct contact exclusions proves

$$r_* < r \leq 1/3 \implies \nexists \theta, \Sigma_r(\theta). \quad (265)$$

The proof uses computation for three stated purposes. Numerical optimization suggests the boundary and the active equations. Symbolic calculation constructs the specified nonnegative combinations, cut points, determinant numerators, and resultant identities. Exact checks establish the signs, root enclosure, and regional coverage required by the mathematical argument. The repository¹ records the inputs and dependencies of 29 verification units. Some units recompute fixed algebraic checks; dedicated independent replays reconstruct certificate identities, coefficient signs, or all leaves and partitions of boundary trees. Their mathematical premises are the building-block properties, LP duality, payoff linearity, maximum attainment, and universal instantiation with the specified strategy types. These checks certify the arithmetic of the stated proof; they are not a completed formalization in a theorem-prover kernel. None is a step of Algorithm 2.

F Transfer of the approximation errors

We now prove that the actual errors δ_f and τ enter the regret bound once each. This establishes the implication in Lemma 10 using the same necessary system (162) as in the ideal analysis. The error contracts for the stationary-point and mixing operations come from [12, 13, 18, 19]. Suppose an actual execution satisfies

$$f(\mathbf{x}_{\text{out}}) > r + \delta_f + \tau, \quad r_* < r < 1/3. \quad (266)$$

Output selection makes every candidate exceed this threshold. Substituting any $p, q \in [0, 1]$ into (24) subtracts τ once. The error-free endpoint comparison at $p = q = 0$ also gives $g \geq f(\mathbf{x}_o)$. Hence

$$\forall p, q \in [0, 1], \quad f(\mathbf{x}(p, q)) > r + \delta_f, \quad g > r + \delta_f + \tau. \quad (267)$$

Primal optimality of $\text{SimpleMixing}(y_s^1)$, $\text{SimpleMixing}(y_s^2)$, $\text{SimpleMixing}(y_1^1)$, and $\text{SimpleMixing}(y_1^2)$ gives the same strict $r + \delta_f$ lower bound throughout each call's domain with its input strategy fixed. All four LPs are solved exactly, so their duality and complementary-slackness properties add no error.

To use stationarity, substitute (x_s^1, y_s^2) and (y_s^1, x_s^2) into (17). With $\lambda_{\text{payoff}}, \mu_{\text{payoff}}$ the payoff differences in (84), this yields

$$G_s = g - \delta_f > r, \quad \rho \lambda_{\text{payoff}} \geq G_s, \quad \sigma \mu_{\text{payoff}} \geq G_s, \quad r < \rho < 1 - r. \quad (268)$$

The last interval follows because $\lambda_{\text{payoff}}, \mu_{\text{payoff}} \leq 1$ and $\rho + \sigma = 1$. In particular, both stationary weights are positive. The notation G_s denotes this single error-adjusted stationary value; it differs from the polynomial G in (220). The error-adjusted lower-bound values are

$$\lambda_0 = G_s / \rho, \quad \mu_0 = G_s / \sigma. \quad (269)$$

They reduce to (89) when $\delta_f = 0$. Equation (5) gives the actual errors

$$\begin{aligned} \tau_1 &= B_1(x_s^2) - u_1(y_s^1, x_s^2), \\ \tau_2 &= B_2(x_s^1) - u_2(x_s^1, y_s^2), \end{aligned} \quad 0 \leq \tau_1, \tau_2 \leq \delta_f. \quad (270)$$

The dual inequality (83a) for $\text{SimpleMixing}(y_s^1)$ is exact. Instantiating it at a column best response to y_s^1 and at x_s^2 , respectively, gives

$$F_1^1 \leq \eta, \quad F_1^1 \leq 1 - \eta + \eta \tau_1 \leq 1 - \eta + \eta \delta_f. \quad (271)$$

Together with $F_1^1 > r + \delta_f$, these inequalities give $\eta > r$ and $r < 1 - \eta - \delta_f + \eta \tau_1 \leq 1 - \eta$. Thus $r < \eta < 1 - r$. The corresponding instantiations of the dual inequality for $\text{SimpleMixing}(y_s^2)$ give $r < \xi < 1 - r$. Positive first-round weights permit the exact best-response conclusion (15) at x_1^2, x_1^1 . Instantiating the dual inequality for $\text{SimpleMixing}(y_1^1)$ at x_1^2 and at a column best response to y_1^1 gives $F_2^1 \leq 1 - \alpha$ and $F_2^1 \leq \alpha$; the corresponding instantiations for $\text{SimpleMixing}(y_1^2)$ are symmetric. We therefore have

$$r < \rho, \eta, \xi, \alpha, \gamma < 1 - r. \quad (272)$$

The row regret of y_1^1 , the dual strategy returned by $\text{SimpleMixing}(y_s^1)$, against x_s^2 must be bounded before taking the minimum with one. Set $d = B_1(x_s^2) - u_1(y_1^1, x_s^2) \geq 0$. Instantiate the dual inequality (83a) for $\text{SimpleMixing}(y_s^1)$ at x_s^2 . This gives

$$\begin{aligned} r + \delta_f &< F_1^1 \leq \eta(\tau_1 - d) + (1 - \eta), \\ \eta d &< 1 - \eta - r - \delta_f + \eta \tau_1 \leq 1 - \eta - r. \end{aligned} \quad (273)$$

Since normalized payoffs also give $d \leq 1$, we obtain the same $d \leq d_1 = \min\{1, (1 - \eta - r)/\eta\}$ as in the ideal argument. The symmetric instantiation of the dual inequality for `SimpleMixing`(y_s^2) at x_s^1 gives the ideal bound d_2 for the column regret. In each case the error is canceled before the minimum is taken; an additional subtraction from the capped quantity is unnecessary. The unit `near-BR-delta-ledger`³¹ reconstructs this cancellation identity and both cases of the minimum with one.

The payoff bounds can now be evaluated at

$$\lambda_s = (r, G_s, \rho, \eta, \xi). \quad (274)$$

For example, instantiate the dual inequality (83a) for `SimpleMixing`(y_s^1) at y_s^2 and use the stationary payoff lower bounds in (268):

$$\begin{aligned} r + \delta_f &< F_1^1 \leq \eta[u_1(y_1^1, y_s^2) - G_s/\rho] + (1 - \eta)(1 - G_s/\sigma), \\ 1 - u_1(y_1^1, y_s^2) &< s_1(\lambda_s) - \delta_f/\eta \leq s_1(\lambda_s). \end{aligned} \quad (275)$$

Here $s_1(\lambda_s)$ is the function in (98) evaluated at the tuple (274). Instantiating the dual inequality for `SimpleMixing`(y_s^2) at y_s^1 gives $s_2(\lambda_s)$. For the cross payoff, substitute (y_1^1, x_s^2) into (17). Using (273) gives

$$G_s \leq \rho(d - \tau_1) + \sigma[u_2(y_1^1, y_s^2) - u_2(y_1^1, x_s^2)], \quad u_2(y_1^1, y_s^2) \geq G_s/\sigma - (\rho/\sigma)d_1.$$

This is the lower bound giving $h_2(\lambda_s)$ in (102); the symmetric substitution gives $h_1(\lambda_s)$. Instantiating the dual inequalities for `SimpleMixing`(y_s^1) at y_1^2 and for `SimpleMixing`(y_s^2) at y_1^1 gives $S(\lambda_s), T(\lambda_s)$, respectively, by (105), using $F_1^i > r$ and these payoff bounds. There is no new stationary error in either instantiation.

The joint regret comparisons also have favorable error signs. For arbitrary $x^{1'} \in \Delta_{n_1}$ and $x^{2'} \in \Delta_{n_2}$, define

$$\begin{aligned} \ell_1^s &= B_1(x_s^2) - u_1(x^{1'}, x_s^2), & \ell_1^y &= B_1(y_s^2) - u_1(x^{1'}, y_s^2), \\ \ell_2^s &= B_2(x_s^1) - u_2(x_s^1, x^{2'}), & \ell_2^y &= B_2(y_s^1) - u_2(y_s^1, x^{2'}). \end{aligned} \quad (276)$$

All four quantities are nonnegative. The column comparison follows by adding η/ρ times stationarity instantiated at $(x_s^1, x^{2'})$ to the dual inequality for `SimpleMixing`(y_s^1) instantiated at the same $x^{2'}$, and bounding $u_1(y_1^1, x^{2'}) - u_1(x_s^1, x^{2'}) \leq 1$. The corresponding calculation using the dual inequality for `SimpleMixing`(y_s^2) is symmetric. The resulting inequalities are

$$\begin{aligned} \frac{\eta\sigma}{\rho}\ell_2^s + (1 - \eta)\ell_2^y &\geq r - \eta(1 - G_s/\rho) + \frac{\eta\sigma}{\rho}\tau_2, \\ \frac{\xi\rho}{\sigma}\ell_1^s + (1 - \xi)\ell_1^y &\geq r - \xi(1 - G_s/\sigma) + \frac{\xi\rho}{\sigma}\tau_1. \end{aligned} \quad (277)$$

The last term on each right side is nonnegative, so omitting it weakens the lower bound and gives exactly the ideal joint regret comparison evaluated at $g = G_s$. Applying the same minimum argument as in (127) therefore bounds the actual Jensen gaps from below by $j_r(\lambda_s, q), j_c(\lambda_s, p)$.

For the rectangle regrets, use (10) with $g = G_s + \delta_f$. The endpoint regrets against the stationary opponent are τ_1, τ_2 , while (268) gives the opposite endpoint bounds $1 - G_s/\rho, 1 - G_s/\sigma$. The error contributions satisfy, for every $p, q \in [0, 1]$,

$$\begin{aligned} \delta_f(1 - p)(1 - q) + \tau_1 p(1 - q) &\leq \delta_f(1 - q) \leq \delta_f, \\ \delta_f(1 - p)(1 - q) + \tau_2(1 - p)q &\leq \delta_f(1 - p) \leq \delta_f. \end{aligned} \quad (278)$$

³¹Checker: `proof/bound/critical/replay_delta_f_lift_ledger.py`.

Together with (277) and the functions in (129), this gives

$$\begin{aligned}\forall p, q \in [0, 1], \quad f_1(\mathbf{x}(p, q)) &\leq R_{G_s}(p, q) + \delta_f(1 - q), \\ \forall p, q \in [0, 1], \quad f_2(\mathbf{x}(p, q)) &\leq C_{G_s}(p, q) + \delta_f(1 - p).\end{aligned}\tag{279}$$

Here R_{G_s}, C_{G_s} are the functions R_g, C_g evaluated at $g = G_s$. If their maximum were at most r at any (p, q) , (278)–(279) would give $f(\mathbf{x}(p, q)) \leq r + \delta_f$, contradicting (267). Thus all four segment predicates (133) hold with the common parameter G_s .

The remaining ideal necessary inequalities follow from the same payoff bounds and the exact **SimpleMixing** dual comparisons. In particular, all six bounds $s_i(\boldsymbol{\lambda}_s), h_i(\boldsymbol{\lambda}_s), S(\boldsymbol{\lambda}_s), T(\boldsymbol{\lambda}_s)$ are nonnegative, $U_{G_s}, V_{G_s}, H_{G_s}, K_{G_s} \geq r$, and $E_{1,G_s}, E_{2,G_s} \geq 0$. The shared-envelope calculation (111)–(115) uses only these bounds and the same actual α, γ ; it introduces no further stationary error. Instantiating the dual inequalities for **SimpleMixing**(y_s^1) at y_s^2 and for **SimpleMixing**(y_s^2) at y_s^1 also gives the parameter range

$$r < G_s \leq \min \left\{ \frac{\rho\sigma(1-r)}{D_c}, \frac{\rho\sigma(1-r)}{D_r} \right\}.\tag{280}$$

Equations (272) and (280) meet the hypotheses (139). Since $(r_*, 1/3) \subset [3/10, 1/3]$, the proved projection of the common parameter from G_s to r preserves the four segment predicates and the other necessary inequalities. At this projected point, replacing d_i by the uncapped bounds used in (159) increases the corresponding upper bounds. This gives the same bare functions and necessary system Σ_r as in the ideal derivation.

The independent unit **near-BR-delta-ledger**³¹ checks the specified 34 nonnegative combinations, seven identities, the two minimum cases in (273), and the source substitutions with common weights. The mathematical substitutions above and those exact checks establish

$$f(\mathbf{x}_{\text{out}}) > r + \delta_f + \tau \implies \exists \boldsymbol{\theta}, \Sigma_r(\boldsymbol{\theta}), \quad r_* < r < 1/3.\tag{281}$$

Equation (265) contradicts its conclusion. If $f(\mathbf{x}_{\text{out}}) > r_* + \delta_f + \tau$, choose r strictly between r_* and $\min\{f(\mathbf{x}_{\text{out}}) - \delta_f - \tau, 1/3\}$. This would satisfy (266), so

$$f(\mathbf{x}_{\text{out}}) \leq r_* + \delta_f + \tau.\tag{282}$$

Taking $\delta_f = \tau = \delta/2$ proves Theorem 1; the polynomial running time follows from the fixed composition in Section 4.4. If $\delta > 1$, normalized regrets are at most one and any profile suffices. A scalar point at r_* has been checked, but no game realizing it has been established. The conclusion is an upper bound on the algorithm's regret, without a tightness claim.